ETBIZ (Exam ETZ-210)

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ETBIZ (Exam ETZ-210)

Part Number: CNX0480
Course Edition: 2.0

Acknowledgements

PROJECT TEAM

Author	*Content Editor*
Jason Nufryk Pamela J. Taylor	Michelle Farney

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About This Course

In the past few years, the business world has undergone significant changes in response to emerging technologies like the Internet of Things (IoT), data science, and artificial intelligence (AI). These emerging technologies are not just another set of tools or processes for the business to consider—they are truly disruptive forces. Even organizations that resist adopting them will feel their impact. If the organization wants to thrive and survive in this transforming business landscape, it will need to harness the power of emerging technologies.

This course is designed to help business professionals conquer and move beyond the basics of emerging technologies to apply IoT, data science, and AI concepts for the benefit of the business. It will give you the essential knowledge of these emerging technologies that you'll need to steer the business forward.

Course Description

Target Student

This course is primarily designed for business leaders, consultants, product and project managers, and other decision makers who are interested in growing the business by leveraging the power of emerging technologies like IoT, data science, and AI. Other individuals who wish to explore basic emerging-technology concepts are also candidates for this course.

Course Prerequisites

To ensure your success in this course, you should have a foundational knowledge of business processes and general business concepts. You should also have at least a basic understanding of information technology resources and systems, including networks, computers, and other electronic devices used in the enterprise.

Course Objectives

In this course, you will identify ways in which emerging technologies can bring significant value to the business. You will:

- Plan an IoT implementation.
- Manage an IoT prototyping and development project throughout the development lifecycle.
- Explain the fundamentals of data science.
- Identify functions of data science for business.
- Implement business requirements for data science.
- Describe AI fundamentals.
- Identify the functions of AI in business.

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- Implement business requirements for AI.

The CHOICE Home Screen

Logon and access information for your CHOICE environment will be provided with your class experience. The CHOICE platform is your entry point to the CHOICE learning experience, of which this course manual is only one part.

On the CHOICE Home screen, you can access the CHOICE Course screens for your specific courses. Visit the CHOICE Course screen both during and after class to make use of the world of support and instructional resources that make up the CHOICE experience.

Each CHOICE Course screen will give you access to the following resources:

- **Classroom**: A link to your training provider's classroom environment.
- **eBook**: An interactive electronic version of the printed book for your course.
- **Files**: Any course files available to download.
- **Checklists**: Step-by-step procedures and general guidelines you can use as a reference during and after class.
- **Assessment**: A course assessment for your self-assessment of the course content.
- Social media resources that enable you to collaborate with others in the learning community using professional communications sites such as LinkedIn or microblogging tools such as Twitter.

Depending on the nature of your course and the components chosen by your learning provider, the CHOICE Course screen may also include access to elements such as:

- LogicalLABS, a virtual technical environment for your course.
- Various partner resources related to the courseware.
- Related certifications or credentials.
- A link to your training provider's website.
- Notices from the CHOICE administrator.
- Newsletters and other communications from your learning provider.
- Mentoring services.

Visit your CHOICE Home screen often to connect, communicate, and extend your learning experience!

How to Use This Book

As You Learn

This book is divided into lessons and topics, covering a subject or a set of related subjects. In most cases, lessons are arranged in order of increasing proficiency.

The results-oriented topics include relevant and supporting information you need to master the content. Each topic has various types of activities designed to enable you to solidify your understanding of the informational material presented in the course. Information is provided for reference and reflection to facilitate understanding and practice.

Data files for various activities as well as other supporting files for the course are available by download from the CHOICE Course screen. In addition to sample data for the course exercises, the course files may contain media components to enhance your learning and additional reference materials for use both during and after the course.

Checklists of procedures and guidelines can be used during class and as after-class references when you're back on the job and need to refresh your understanding.

At the back of the book, you will find a glossary of the definitions of the terms and concepts used throughout the course. You will also find an index to assist in locating information within the instructional components of the book. In many electronic versions of the book, you can click links on key words in the content to move to the associated glossary definition, and on page references in

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the index to move to that term in the content. To return to the previous location in the document after clicking a link, use the appropriate functionality in your PDF viewing software.

As You Review

Any method of instruction is only as effective as the time and effort you, the student, are willing to invest in it. In addition, some of the information that you learn in class may not be important to you immediately, but it may become important later. For this reason, we encourage you to spend some time reviewing the content of the course after your time in the classroom.

As a Reference

The organization and layout of this book make it an easy-to-use resource for future reference. Taking advantage of the glossary, index, and table of contents, you can use this book as a first source of definitions, background information, and summaries.

Course Icons

Watch throughout the material for the following visual cues.

Icon	*Description*
	A **Note** provides additional information, guidance, or hints about a topic or task.
	A **Caution** note makes you aware of places where you need to be particularly careful with your actions, settings, or decisions so that you can be sure to get the desired results of an activity or task.
	**Checklists** provide job aids you can use after class as a reference to perform skills back on the job. Access checklists from your CHOICE Course screen.
	**Social** notes remind you to check your CHOICE Course screen for opportunities to interact with the CHOICE community using social media.

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1 Planning an IoT Implementation

Lesson Time: 1 hour, 55 minutes

Lesson Introduction

Before you can successfully plan and implement an Internet of Things (IoT) solution, you must understand the various factors that will drive your decisions, such as identifying what IoT can do for your organization, and the various business and technical challenges you'll need to address.

In this lesson, you will examine the wide range of possibilities that IoT offers, and you will develop an understanding of the components and design decisions that go into an IoT project by creating a simple IoT device from parts.

Lesson Objectives

In this lesson, you will:

- Define IoT and its components.
- Describe common elements within the IoT ecosystem.
- Identify benefits and challenges of IoT.

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TOPIC A

Define IoT

A good place to start with an IoT project is to determine what you intend to accomplish, the various components you'll need to bring together to meet your requirements, and to identify the benefits and challenges that you'll encounter on such a project.

Dawn of a New Day

Consider the following scenario as you use IoT technologies to start your day.

- **6:34 A.M.**—Your alarm clock has been monitoring your sleep cycles, and awakens you at the optimal time, when you'll feel completely rested.
- **6:52 A.M.**—Your refrigerator door shows a recommended breakfast, with nutritional information based on your food preferences, weight goals, and what is in your kitchen.
- **6:53 A.M.**—Your smart speaker informs you of all the meetings and appointments you have today. Traffic is congested on your normal route, so the speaker tells you that your autonomous car has prepared an alternate route that will save you 10 minutes but take 2 minutes longer than your normal route.
- **7:12 A.M.**—Your electric car warmed up its interior just in time for you to get in. A speaker in the car informs you that your work colleague has signed the contracts, and will have them delivered to you before your first meeting.
- **8:50 A.M.**—Your smartwatch reminds you of your 9:00 A.M. meeting.
- **8:51 A.M.**—A delivery drone enters your office, bearing a package containing signed contracts for the meeting.

Not too long ago, this scenario may have seemed to be science-fiction hype, but each of the technologies described is available today through the ***Internet of Things (IoT)***.

IoT

IoT refers to the ability to connect everyday things to the cloud, which leads to many possibilities for collecting, sharing, and interpreting data, and control of remote devices. IoT's potential applications are widespread, and have the potential to increase global food production, promote safety and security, improve the quality of life in our cities, create new markets and opportunities for business, and improve our conservation of natural resources.

Kevin Ashton, who coined the expression "Internet of Things" in 1999, wrote, "If we had computers that knew everything there was to know about things—using data they gathered without any help from us—we would be able to track and count everything, and greatly reduce waste, loss and cost. We would know when things needed replacing, repairing or recalling, and whether they were fresh or past their best. We need to empower computers with their own means of gathering information, so they can see, hear and smell the world for themselves, in all its random glory. Radio-frequency identification (RFID) and sensor technology enable computers to observe, identify and understand the world—without the limitations of human-entered data."

IoT Ingredients

Many different technologies come together to provide the ingredients for the IoT. These include:

- **IoT devices** that are:
  - **Sensing**: Able to measure the world around them, such as local temperature, humidity, or water level, or number of cars passing by.

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- **Active**: Able to take local action as necessary, such as turning off a light, opening a valve, or sounding an alarm.
- **Connected**: Able to share data with other systems in the cloud.
- **Smart**: Able to:
  - Perform initial processing and filtering of sensor data.
  - Determine whether immediate (real-time) action should be taken in response to locally acquired data.
  - Take action based on remote instructions from the cloud.
- **The cloud**, where powerful cloud applications can:
  - Collect data from IoT devices.
  - Combine IoT device data with other data sources.
  - Perform data analytics to reveal trends, identify problems, predict the future.
- **Client applications** that enable users to:
  - Access and view data processed in the cloud.
  - Issue commands to remote IoT devices.



Note: Smart devices alone are not IoT. Connected devices alone are not IoT. The cloud alone is not IoT. IoT becomes possible when all three of these aspects are combined, enabling IoT devices and IoT client applications to work together, data to be stored and analyzed, and cybersecurity to be maintained.

IoT Devices

IoT devices (the ***things*** in the Internet of Things) take many forms, providing many possibilities for automation and data collection. IoT devices may be geared toward home and consumer use, industry, agriculture, or various other domains. Some devices are geared toward *remote sensing*, while others are geared toward *process automation* or *remote control*.

Devices may be intended to be seen, manipulated, and handled by users (like a fitness watch), or they may be intended to function completely behind the scenes, out of sight, with very little direct contact by end users, such as a basement flooding monitor or weather sensors. Devices may be very specialized toward supporting a specific feature, or they may be much more general in purpose.

Examples of IoT Devices

The diversity of IoT devices makes it difficult to articulate an all-encompassing description of the typical device. Examples may provide a better sense of the scope encompassed by the phrase *IoT device*.

- **Agricultural sensor**: Based on measurements of soil quality and geographic location, controls the application of chemicals and amendments to the soil in real time.
- **Air quality monitor**: Monitors air quality in a room and sends alerts via a smartphone app, providing useful data for people with allergies or who otherwise have a need to monitor air quality in their living or working environment.
- **Augmented-reality (AR) headset**: Enables the wearer to look around and see the local view augmented with data from the Internet, which could be useful in tourism and travel, real-estate sales, museums, and many similar situations.
- **Connected universal remote control**: Televisions and home audio have long been controllable through remote controls, but with many other devices in the smart home now accessible through the home network, a universal remote control can control much more than entertainment devices. Likewise, devices like smart speakers can also be used to control functions that previously were the domain of the universal remote.
- **Dash button**: A single-purpose connected button. The concept is named after a device sold by Amazon that automatically places an order for a consumable product like detergent when

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pressed. The concept has been expanded to perform tasks such as starting a car, opening a garage door, and enabling users to create their own custom tasks, such as sending an email or activating a remote buzzer.

- **Home-systems monitor**: Monitor home systems such as heating, cooling, and drainage to alert the user when there are problems, when systems need to be tuned or maintained, and so forth.
- **Industrial control**: Sensors and actuators in production equipment enable operations staff to monitor and control production and implement process automation.
- **Remote-controlled mood lighting**: Enables users to define various lighting schemes involving numerous lights throughout their home. With a single command and through pre-programmed scripts, lights can be set in a combination of light intensities and colors.
- **Scientific instrument**: Scientists can monitor and control instruments in remote locations, including places that people can't easily reach, such as the poles, the ocean, other planets, within living organisms, and so forth.
- **Smart battery**: Smart batteries can transform your conventional devices into IoT devices. For example, a smart battery in a smoke or carbon monoxide alarm could warn you through text messages and other notifications when the power is getting low. The smart battery could also immediately notify you if the alarm sounds and you're not at home to hear it.
- **Smart door lock**: Enables users to use an electronic device such as a smartphone or fob as a key, eliminating having to fumble for a key. Also enables an owner to lock and unlock a door from a remote location using a smartphone app (to enable someone without a physical key to enter under the owner's control, for example). The device can monitor and log activity at the door, activating cameras and alerting the owner through a notification on their smartphone that someone is at the door.
- **Smart home appliance** (refrigerator, range, coffee maker, etc.): Smart appliances can notify you if there is a problem or maintenance need (to change a water filter in a refrigerator, for example, or to warn that spoiling food has been detected), help you monitor energy consumption, and provide remote access to the device. For example, from the grocery store, you could use a camera in the refrigerator to check if you're out of milk.
- **Smart speaker**: A cloud-connected speaker that responds to a wide range of voice commands, enabling the user to control music streaming, command home-automation devices (such as smart thermostats and smart outlets), communicate with other smart speakers (like an intercom), order products online, and perform web searches using voice and audio.
- **Smart thermostat**: Monitors and controls home heating, cooling, and airflow functions, enabling remote control and data logging through other devices, automatically sensing when users are at home and adjusting the home environment accordingly, logging use of heating and cooling systems, and suggesting ways to reduce energy costs.
- **Traffic monitoring**: Traffic patterns on roads, highways, sidewalks, and hallways can be monitored to provide data related to safety, civil engineering, and timing of traffic.
- **Wireless breath analyzer**: People tend to think of breath analyzers that detect levels of alcohol consumption, and certainly there are IoT applications for such devices, but breath analyzers can also be used to monitor various health-related conditions, such as dangerous ketone levels in diabetes patients, airway inflammation in asthma patients, and many other potential health problems.
- **Wearable device**: Wearable devices can monitor for conditions such as heart irregularities, low blood oxygen levels, and so forth, to provide users with early warning of impending medical problems. They can also be used to monitor babies, patients, pets, and others who require a caretaker. Caretakers can be alerted if their charge leaves a particular area, and the device can provide data on the wearer's health and well-being.
- **Weather sensor**: Sensors mounted outdoors can monitor local weather conditions, such as temperature, humidity, and air pressure. The user can read and analyze their weather data on a mobile or desktop application. Using a ***crowdsourcing*** approach, data from weather stations owned by many different individuals can be aggregated to provide detailed weather data on a global scale, as done by Weather Underground (**https://www.wunderground.com**).

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Figure 1-1: IoT devices take many forms.

IoT in Industry

Many of the general concepts of IoT, such as the ability to measure and control processes from afar, have been used in manufacturing for decades, and, in fact, manufacturers have been quick to adopt various new IoT technologies into their own process control applications. The following is a brief history of process automation in industry.

Time Period	*Description*
1970s	• ***Programmable logic controllers (PLCs)*** were integrated into manufacturing facilities, enabling process automation and robotics, which transformed the way many products were manufactured. PLCs were revolutionary because they were relatively simple to program, optimized for process automation, and very reliable. • Early ***machine-to-machine (M2M)*** data communication capabilities were developed based on telephone protocols, enabling computers to communicate with manufacturing equipment. This provided limited ***command and control (C2)*** capabilities from remote locations, in ways that resemble IoT capabilities of today. • The first iteration of ***supervisory control and data acquisition (SCADA)***, an industrial control-system architecture typically used in industrial, energy, and utility companies is introduced, starting as a system for mainframe computers with limited interoperability and networking capabilities.

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Time Period	Description
1980s	• The advent of inexpensive personal computers (PCs) and client–server networking (using Ethernet, for example) made it possible for desktop computers in the production office to interface with PLCs on the manufacturing floor. • SCADA is expanded to interface with PCs, local area networks (LANs), and control through PC applications. Interfacing between SCADA systems from different vendors was not supported.
1990s	• Following the advent of the World Wide Web, PLCs were enabled to connect to the Internet, providing a relatively simple way to expand the connection between office and manufacturing systems on an enterprise level, beyond a single location or campus. • SCADA evolved to use an open architecture and communication protocols. Over common network protocols such as Ethernet, systems from different vendors were able to communicate. • ***Open Platform Communications (OPC)***, based on Microsoft standards such as ***Object Linking and Embedding (OLE)*** and ***Component Object Model (COM)***, was created to provide open standards and specifications for communication of real-time plant data between control devices from different manufacturers.
2000s	• The development of cloud technologies and ubiquitous connected computing devices (smartphones and tablets) provided a means for those monitoring and managing manufacturing processes to become more mobile and connected than ever before.
2010s	• Inexpensive and pervasive web technologies, combined with low-cost microcontrollers, sensors, and high-performing batteries, resulted in new capabilities for data acquisition and process automation on a huge scale. • In terms of spending, industry has been the primary driver of IoT, investing more in IoT technologies than any other sector.

Industry Evolution

Industry 4.0 refers to the state of process automation, data collection, analysis, and communication in today's "smart factories." Modern industry is described as being in the fourth industrial revolution.

- **Industry 1.0**: Hydro and steam-powered mechanization of production.
- **Industry 2.0**: Mass production through assembly lines, and the introduction of electrical power.
- **Industry 3.0**: Automation of individual manufacturing processes through computers and robotics.
- **Industry 4.0**: Massive automation and interconnection of global manufacturing systems, using:
  - **Sensors** to collect extensive process data.
  - **Software** to accurately model physical processes based on that data.
  - **Smart production machines** that can interpret data, and adjust operations in real time.
  - **Global networking** to enable extensive cooperation and coordination among the various machines, devices, sensors, and people involved.

Because of the way in which Industry 4.0 uses software (*cyber* systems) to model production processes (*physical* systems), the technologies used in this approach are often called ***cyber-physical systems (CPS)***.

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Note: Some people refer to Industry 4.0 as the ***Industrial Internet of Things (IIOT)***. However, one might argue that Industry 4.0 preceded IoT, and has produced concepts and ideas that have subsequently been adopted by IoT, so it is not accurate to consider Industry 4.0 as a subset or derivative of IoT.

Enabling Technologies

Various developments in technology provide the basis for IoT.

Enabling Factor	*Description*
Miniaturization	Significant processing power is available in a small package, with relatively powerful processors embedded in compact devices such as smartphones, smartwatches and other wearable devices, home appliances, and other everyday things.
Connectivity	The ability to wirelessly connect devices, even those in remote locations, through a wide variety of technologies such as cellular data, Wi-Fi, Bluetooth®, RFID, and others, enable them to share data and respond to remote control.
Advanced power sources and power management	The ability to operate on very small amounts of electrical energy, using rechargeable batteries or harvesting energy from the surrounding environment, enable devices to function in mobile applications or remote locations, away from the power grid.
Inexpensive processors, sensors, and actuators	Relatively inexpensive components enable the proliferation of small, capable devices that can measure their local environment, process data, and respond accordingly.
Cloud-based processing	The ability to delegate data collection and analysis tasks to more powerful computers in the cloud enable IoT devices to remain compact and operate on minimal power.
Ubiquitous computing	More than ever before, people are often just an arm's length away from a computing device, whether it be a smartphone, tablet, desktop computer, wearable device, smart appliance, or some other device that provides a user interface through which users can receive notifications and run applications to control or pull information from remote devices and services.

Industry 4.0 Promise

Enabling new capabilities can help businesses add value for their customers. We can now see (and measure) not only what is happening in real time, but also even predict, with a high degree of accuracy, and respond (or actuate) in a prescriptive way. This is a powerful new capability that we did not have in the past.

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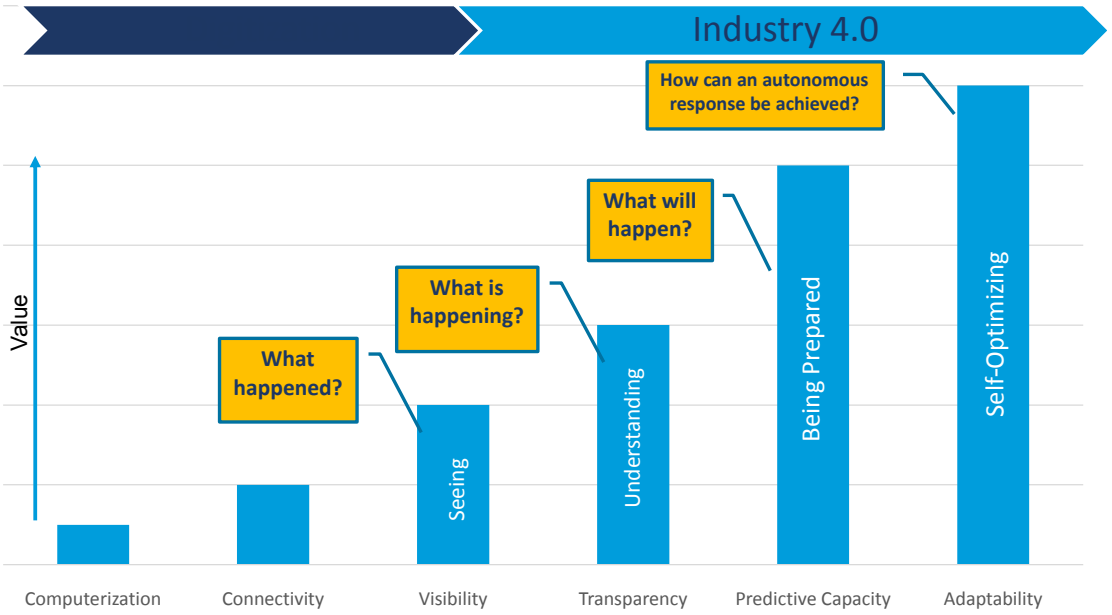


Figure 1-2: The promise of Industry 4.0.

An excellent article that describes this phenomenon is *Industry 4.0 and the digital twin*, which you can find at **https://www2.deloitte.com/insights/us/en/focus/industry-4-0/digital-twin-technology-smart-factory.html**.

As manufacturing processes become increasingly digital, the digital twin is now within reach.

By providing companies with a complete digital footprint of products, the digital twin enables companies to detect physical issues sooner, predict outcomes more accurately, and build better products.

Four Ps of Planned Maintenance

Early maintenance practices were *reactive*. When a component in the production system failed, causing production systems to come to a halt, the necessary repairs and maintenance would be performed, and production would then resume. This approach is sometimes called ***run to failure (RTF)***.

Such unplanned downtime is costly, so manufacturers have long looked for ways to anticipate problems and prevent them by replacing components at a convenient time, before they fail. Listed in the following are four methods for planned maintenance that have been developed over the years to provide improved proactive approaches.

- ***Preventive maintenance*** was an early take on planned maintenance. It is driven by recommendations of the manufacturer who produces the component. Based on the estimated mean time between failure (MTBF) for a particular component, companies attempt to replace components before they are expected to go bad. This provides an advantage over reactive maintenance, as it helps to avoid many instances of downtime.
- ***Predictive maintenance*** relies on the ability of highly trained maintenance technicians to routinely inspect components for performance factors such as vibration, overheating, and so forth to predict when they are nearing their end of life. Components are monitored while they

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are running, and fairly reliable indicators are used to detect when components are about to fail, so maintenance can be planned for a time when there will be minimal disruption to production.

- ***Proactive maintenance*** builds upon predictive maintenance, looking to also control the problems that lead to wear and tear. Remediations identified through this process include such things as training operators in best practices and identifying sources for better components, lubricants, and so forth.
- ***Prescriptive maintenance*** takes these approaches even further by using Industry 4.0 tools to automate measurement and analysis functions, so the system itself can spot the need for maintenance and prescribe mitigations or repair, even prioritizing those tasks among other tasks that need to be performed on the system.

Three Key IoT Business Strategy Steps to Follow

Before you dive into IoT solutions, business leaders need to solve for these three focus areas, *in this order*:

1. **Value**: What new or incremental value does your idea provide to your customers?
2. **Insights**: What information will you need to be able to provide to support this new or incremental value?
3. **Data**: How is IoT a possible solution for you to obtain the data needed to enable these new insights?

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ACTIVITY 1-1
Understanding Value, Insights, and Data

Data File

C:\CNX0480Data\Planning an IoT Implementation\Understanding Value, Insights, and Data.docx

Scenario

You will evaluate an IoT system to monitor the environment within agricultural greenhouses operated by Greene Organix. Greene Organix produces fresh roses for the northeast U.S. regional distributors florist market. Your goal is to see if you can identify some key areas of value that we must provide in order to deliver the best quality and freshest roses we can produce to our regional distributors.

The IoT system will collect data, analyze that data, and control the greenhouse environment as necessary to ensure all plants are subjected to optimal growing conditions. In this activity, you will identify the value of producing high-quality products, what insights are needed to support that value, and what data is required. Finally, you will determine the capability of IoT to accomplish these objectives.

In the spaces provided, write your ideas regarding requirements for the IoT devices' key areas of value for this scenario. (If you'd prefer to type your ideas in a Word document, a worksheet is provided in the aforementioned data file location.)

---

1. A) Identify new/incremental value: What elements of value would be important to produce year-round, high-quality fresh flowers for Greene Organix's distributors?

2. B) Understand insights needed to support value: What information may be needed to drive down production costs so you can continue to deliver best-in-class flowers at improved competitive prices?

3. C) Data: What data do you need to collect to support these insights?

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4. D) Identify data-collection capabilities you need to include in your IoT solution.

   Sensors: What useful information might an IoT device measure in this environment?

   Actuators: What actions might an IoT device need to perform in this environment?

   Other sources: What external data might be of use to support the insights?

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TOPIC B

IoT Infrastructure

The IoT ecosystem is typically complex. In reality, it is many systems operating under a broader ecosystem. In the previous topic, you explored the diversity of IoT devices. The IoT ecosystem can be just as diverse; however, there are common elements within the ecosystem, regardless of options.

IoT Infrastructure

Just as IoT devices are quite diverse, the infrastructure in which they operate may vary significantly from one system to another. However, common elements of IoT infrastructure include:

- **IoT devices**, including hardware and software that directly interacts with the world and measures local data (called ***telemetry***, meaning "measurements from far away").



Note: Borrowing terminology from industrial automation, people sometimes refer to IoT devices as *field devices*.

- **A network connection** using wireless or wired communication.
- **An IoT gateway** that serves as an intermediary between local networks of IoT devices and the cloud, enabling local IoT devices to communicate with each other, and to an external network. The IoT gateway might be a discrete piece of hardware, or its functionality might be built into one of the IoT devices in the local network, such as a thermostat in a home network, or an industrial machine in a factory.
- **The edge**, defined by various points where traffic leaves (or enters) the local network; for example, a router providing access to an Internet carrier. A router that conveys network traffic between the local area network and the wider network beyond the control of the local organization is described as an ***edge*** device.
- **Global network**, which provides the long-haul connection to the cloud.
- **Cloud services** that collect data from devices, combine it with data from other devices and data sources, and analyze resulting datasets, which can become quite large when combined from hundreds or thousands of IoT devices. Massive data-processing capabilities, known as ***big data***, may be necessary, requiring the use of ***artificial intelligence (AI)*** and ***machine learning (ML)*** strategies and technologies.
- **User interface** that provides a user with the ability to view and analyze IoT data, issue commands and control IoT devices, and perform related tasks. The user interface may be provided through a mobile app, web application, or desktop application.

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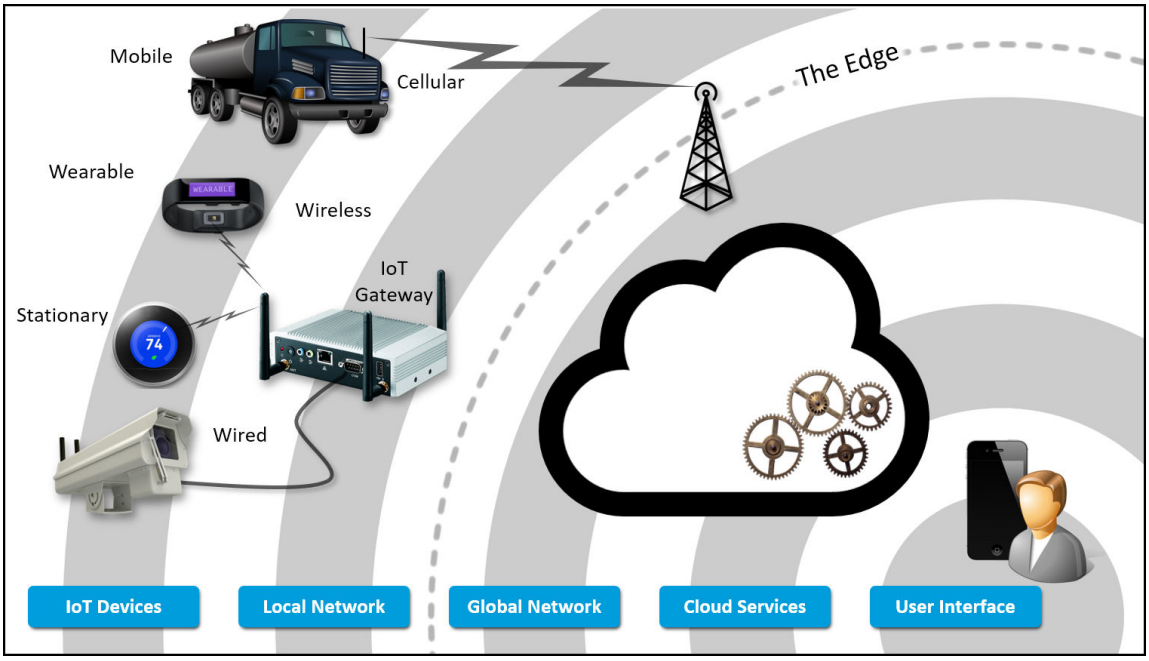


Figure 1-3: General components of IoT architecture.

IoT Gateway

An IoT gateway might perform tasks such as the following:

- **Support local communication and messaging** on a local network that is optimized or well suited for IoT devices with limited data-processing and communication capabilities.
- **Aggregate and store data from IoT devices temporarily** to take advantage of off-hour data rates or to deal with intermittent connections.
- **Forward data acquired from IoT devices to cloud services**, providing the increased security, routing capabilities, and electrical power requirements that communication across the Internet requires.
- **Perform local data analysis** where a fast turnaround is required, and sending data to the cloud for analysis would take too long.
- **Prepare data for cloud services** by filtering, compressing, or performing other processing tasks on it.
- **Translate between IPv6 and IPv4.**
- **Serve as a local cache for software updates** sent to IoT devices.
- **Receive device commands from the cloud** and send to IoT devices.

A wide variety of devices may be used as an IoT gateway, and they vary by industry. For example, many consumer-oriented IoT products (smart thermostats, smart speakers, and so forth) include built-in IoT gateways, although they're marketed by other names, such as *smart-home hubs*.

For business implementations, IoT gateways may be constructed by installing the necessary software on an industrial or rack-mounted computer, or by using products specifically marketed as IoT gateways. Such systems are commonly sold by companies that traditionally have sold business computers and networking devices.

Cloud Services

Cloud services provide a central point where massive amounts of data from IoT devices and other sources can be stored and analyzed, and where other systems and solutions can be integrated, such as enterprise applications, cloud services on other platforms, and so forth.

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Cloud services may operate on cloud platforms such as Amazon Web Services™, Microsoft Azure®, Google Cloud™, Red Hat® OpenShift®, etc. They may also be hosted within an enterprise's own data center using commercial, open-source, and custom-developed software.

Cloud services can be scaled as needed to meet increased demand (e.g., more IoT devices are connected or more users need to access the system), and to support an increasing variety of different services.

User Interface

IoT client applications provide users with access to view IoT data and issue commands.

For consumer IoT, client applications often include smartphone, tablet, or desktop computer apps that enable users to control remote devices, or to view data they have accumulated, such as exercise statistics from a smartwatch or fitness band.

In business applications, ***business intelligence (BI)*** applications may include ***data dashboards*** and similar visualization tools that enable users to view collected business data in graphic form and possibly issue commands to manage business processes.

Various other user interfaces may be provided as well. For example, administrative applications enable administrators to set up and configure the various IoT devices, IoT gateways, and other systems supporting the IoT infrastructure.

Software Stacks

Some software capabilities are typically required by all hardware involved in IoT. Clearly, cybersecurity is a requirement for any piece of hardware involved in collecting, processing, and presenting sensitive data. Furthermore, since much of the software involved in IoT is not running directly on the user's computer, remote management capabilities are required to set up, configure, and maintain IoT devices, gateways, and cloud applications. However, the requirements for the different types of hardware used in IoT are generally very different based on the role played.

This figure and the following table describe the typical layers of software (***stacks***) used by IoT devices, IoT gateways, and cloud services that support IoT.

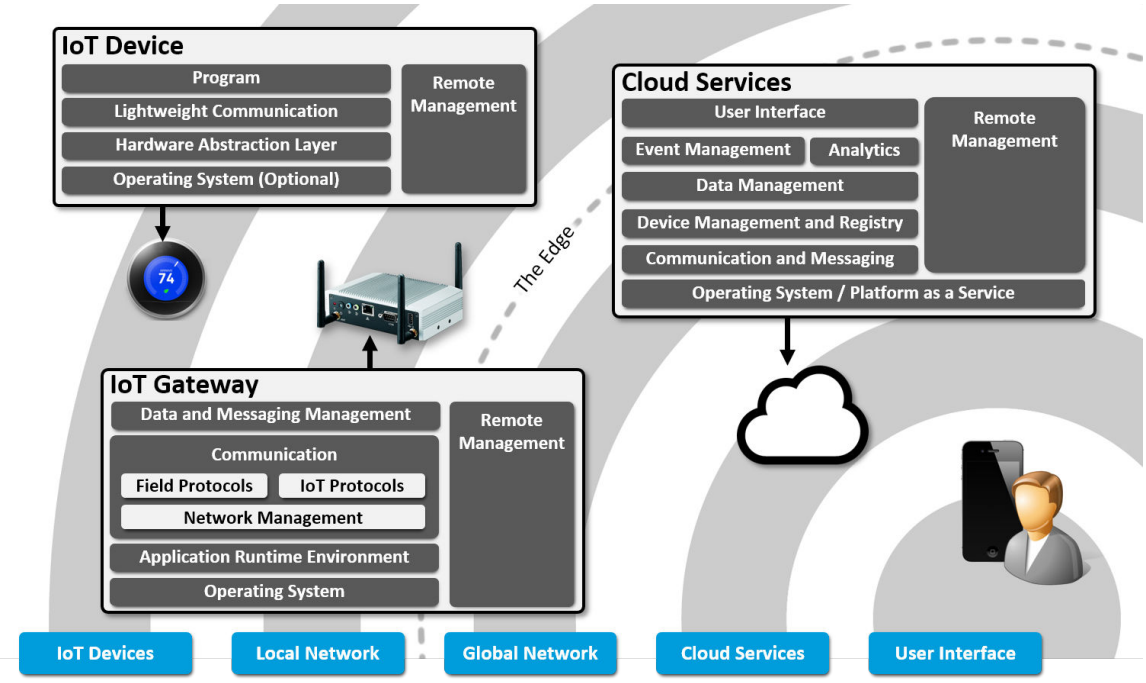


Figure 1-4: Types of software required based on role within the infrastructure.

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IoT Hardware	*Software Layers*
IoT devices	IoT devices are often ***constrained devices***, meaning they are purposely designed to function with little processing power, memory, energy requirements, physical size, and so forth. So, the software on these devices must be very lean and oriented to the required tasks. **Operating System**: A very lightweight operating system is used, which may be customized to support only essential tasks. If possible, no operating system may be used at all. **Hardware Abstraction Layer (HAL)**: Programmers may be provided with a programming interface that can perform operations on the device hardware using relatively simple code, without having to directly program the hardware. **Communication**: Capabilities include protocols the device needs to communicate with other devices and the IoT gateway. To conserve power, unused radios are typically disabled, and lightweight communication protocols optimized for constrained devices may be used to minimize the processing, memory, and power required. Encryption capabilities may be limited in these devices. **Program**: Typically, a single program that starts running when the device is powered on provides the functionality of the device.
IoT gateways	**Operating System**: IoT gateways generally have much more processing power than IoT devices, so they can run conventional operating systems, such as Linux® or Windows®. **Application Runtime Environment**: IoT gateways may perform many different tasks to collect data and perform initial processing before forwarding data on to cloud services. Typically, this is accomplished using programs written for Node.js, Python®, Java, or a similar application runtime environment, which must be installed on the IoT gateway. **Communication**: The role of a gateway in general is to serve as the go-between across two different types of networks. In the case of IoT gateways, these are the local network of IoT devices and the global network (the Internet), typically running TCP/IP. So IoT gateways must be able to support two types of communication, using lightweight protocols with local IoT devices and normal protocols (such as TCP/IP with TLS transport layer security) with the network at large. **Data and Messaging Management**: Because IoT gateways serve as a communication hub, they must provide the ability to manage communications between IoT devices and external network connections.

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IoT Hardware	*Software Layers*
Cloud services	The cloud services that support IoT typically perform heavy processing and analytics on data from IoT devices, and provide a central point for interfacing IoT with end users and enterprise applications. **Operating System/Platform as a Service**: Cloud services typically run on virtual machines that can be scaled up or down as needed to meet demand, so operating systems that can function in this way, such as customized versions of Linux and Windows, are used. **Communication and Messaging**: Cloud services must manage two-way communication and messaging with all devices, gateways, and clients they interact with, and must ensure that communications and transactions occur in proper sequence, even when dealing with huge numbers of devices scattered throughout the world. **Device Management and Registry**: IoT cloud services must be able to identify devices and gateways they communicate with, and provide a means to manage those devices and securely provide them with software updates. **Data Management**: The volume of aggregated data may become massive in scale. IoT cloud services must provide a way to manage this data, and handle it in ways that are economical, efficient, and secure. **Event Management**: Event management software enables the system to detect when certain conditions exist, log them, and perhaps trigger a response. This supports a variety of other capabilities, such as security monitoring, data management, and so forth. **Analytics**: Analytics software enables IoT data to be processed in light of business rules to provide insights that might not be readily apparent from just visually scanning the data. Analytics programs use advanced data processing techniques and artificial intelligence to reveal significant patterns of data. **User Interface**: IoT cloud services can present information in highly visual and structured ways, such as charts, reports, and dashboards that can be viewed from a web browser or client application.

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ACTIVITY 1-2
Selecting an IoT Infrastructure

Data File

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Scenario

Now that we have some exposure to the complexity within the IoT infrastructure, let's have a little fun again with our Greene Organix project. Automation will require that we both monitor (sense) important physical elements within the greenhouse, analyze the data we are collecting, and respond with control actions (actuate) to maintain optimal operating conditions within the greenhouses. Let's look at what a typical system would require from an IoT infrastructure and also consider some of the real-world constraints that may limit our options.

The system will:

- Collect data.
- Analyze the data.
- Control the greenhouse environment to ensure all plants have optimal growing conditions.
- Report data as needed.

Identify the basic IoT infrastructure you will need to include in your IoT solution for:

- Data collection and control
- Local connectivity
- Remote connectivity
- Remote data ingestion, analytics, backend applications
- Operational constraints

In the spaces provided, write your ideas regarding requirements for the IoT infrastructure you would use in this scenario. (If you'd prefer to type your ideas in a Word document, a worksheet is provided in the aforementioned data file location.)

---

1. A) What would be a typical IoT infrastructure setup for our greenhouses?

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2. B) What are some constraints we may be dealing with within our greenhouse operation?

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TOPIC C

Business Benefits and Challenges

Industry has long benefited from process automation. With the advent of IoT, other business processes are now being automated on a greater scale than ever before. However, while IoT brings new capabilities that can benefit business and society as a whole, it also brings new challenges.

Business Benefits

Applied effectively, IoT can enable organizations and businesses in many different sectors to transform how they operate, improving performance by providing abundant, accurate data that can help them make more informed decisions, reduce the need for "feet on the ground," and provide higher levels of situational awareness than they could manage previously.

IoT technologies have the potential to help businesses:

- Improve competitive position by providing information that leads to improvements to products, processes, and services.
- Anticipate and avoid expensive, time-consuming, and hazardous problems before they occur, reducing operational costs and the harmful impact of operations on the natural environment, while increasing safety and improving customer experience.
- Identify possibilities for new and enhanced revenue streams and markets by revealing previously unidentified problems that can be solved and areas in which products and services can be improved.
- Reduce operational costs and natural resources usage by identifying sources of waste and inefficiency.
- Reduce transportation and lodging expenses, and potentially reduce the impact on the environment by enabling some tasks to be performed from remote locations that previously had to be performed on site.
- Reduce waste and defects and increase productivity, agility, and product quality by improving product testing and manufacturing telemetry.
- Provide better tracking of logistics and transportation.

Business Considerations

Organizations fall into the trap of building IoT solutions that capture lots of new data but put little thought into how this new data will add business value. Focus first on what value your company or your organization provides and how to improve that value. If you focus on the data first, the result may end up being that your incremental costs exceed your incremental value created.

There are three items to consider when trying to determine when your organization is evaluating an IoT implementation.

- *What incremental value are you trying to capture?*
- *What new information (insights) will you need to be able to deliver on this new value?*
- *What data will you need to capture to help facilitate and support this new information?*

Business Challenges

Although many businesses may benefit from IoT, it also presents numerous challenges. For example, many organizations have invested heavily in existing infrastructure. Implementing IoT systems and adapting business processes to use them can be disruptive. Provisioning and tracking of additional devices, potentially numbering in the thousands for a single organization, will be

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challenging for some organizations. Significant cost may be associated with just the hardware and software, not to mention setup, configuration, and interfacing the new components with legacy systems. Data networks will have to be reconfigured to add new network nodes for IoT devices, gateways, analytics tools, and support for IoT protocols. There may be some disruption to current systems as all these changes are implemented.

Security, privacy, and safety concerns may be associated with the new types of data being collected and the systems used to secure the data. The organization may be exposed to new standards and regulations associated with the data they are capturing. For some situations, the cost and disadvantages of IoT automation may outweigh benefits, and it may be more beneficial to utilize people or traditional technology.

Traditionally, ***information technology (IT)*** and ***operational technology (OT)*** have been managed by separate groups within organizations. IT typically managed the company's information systems, geared toward collecting, storing, and sending data, whereas OT focused on measuring and controlling production and operations. With IoT, the distinctions between these two groups is blurring, with IT becoming more involved in measurement and control, and OT providing expertise in that area, while looking to take greater advantage of the information systems supported by IT.

Of course, IoT will drive change, and there may be some resistance within the organization. Some jobs will change, as more information can be collected automatically, and new skillsets will be required. To ensure a good transition, it is helpful to get buy-in and participation from all levels of the organization—particularly from those groups given the task of implementing and supporting the new technology.

Technical Challenges

A variety of factors make it challenging to design, develop, implement, and maintain an IoT system.

Factor	*Description*
Device constraints	The constraints on IoT devices can make it challenging to provide robust features and capabilities. Those who design IoT devices and systems must manage the tradeoff between low power consumption and features the device needs to support, such as data processing, communication with other devices, and consistent, reliable data sensing.
Data communication	IoT devices often use wireless (rather than cabled) connections to support remote or mobile use, and may have an unreliable, low-bandwidth, or intermittent connection, and their limited memory and storage may make it challenging to move large amounts of data quickly. The lack of standards and emergence of new standards makes it difficult to select technologies that will last over time. Organizations are working toward open standards, but the process takes time. Vendors are quite competitive in this new technology domain and trying to establish technology advantages over their competitors. While single-vendor, proprietary solutions are often quite good, they lead to interoperability issues, data silos, and security problems when products from different vendors are introduced into the mix.

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Factor	Description
Interoperability	Integrating products from different vendors into a unified solution for the customer can be challenging. Industrial systems have traditionally been highly customized and based heavily on legacy systems that use proprietary communication and messaging protocols. It is difficult to connect industrial systems between organizations using products from a different vendor, and to interface systems with new technologies that were not part of the original system. In the consumer space, IoT products have forced customers into assembling systems like home automation using only products from a particular product line, or else risk having systems that are awkward to use or unable to support all of the product's features. Even when interoperability standards are supported, the solution is often clumsy, involving various proprietary apps and complicated processes to link the devices. For example, to set up a smart outlet from one vendor to respond to voice controls through a smart speaker by another vendor, the user may have to download a smartphone app for the smart outlet, an app for the smart speaker, and numerous steps within both applications to link the devices. If the user buys another smart outlet from a different vendor, yet another app may be required, as well as a different complicated linking operation.
Real-time operations	Determining where to perform processing within the IoT infrastructure is a primary concern for the IoT designer. Some IoT devices must be able to measure and detect certain events within a fraction of a second, and take local action immediately. This type of processing is called a real-time operation.
Security and privacy	The Internet of Things brings new security challenges. IoT devices are often located in remote or public locations, where it may be more challenging to provide *physical* protection from attackers. They may be located on the edges of network infrastructure and exposed to Internet traffic, where it may be more challenging to provide *logical* protections through firewalls and other forms of network security. It may be easy to use radio devices to jam communication on low-power wireless devices. Encryption is a common approach to protecting data when it is in transit across data networks or at rest in storage. But in many cases, encryption requires more capability than constrained devices can spare. So devices may have minimal or no encryption to protect the data they send and receive. IoT has introduced concerns about privacy and data provenance. It may be unclear who owns data collected by IoT processes. Consumers may be skeptical about IoT products and concerned they won't be able to control who can use their data.
Usability	Because of the limited storage and data communication capabilities on an IoT device, it may be challenging to provide security and functionality updates. In some cases, high security may be at odds with usability. Tasks such as pairing, configuration, and authentication may require multiple forms of authentication and multiple devices to complete, causing some frustration for users.

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Factor	Description
Updates	It may be challenging to provide security and functionality updates to IoT devices. They may have an unreliable, low-bandwidth, or intermittent connection, and their limited memory and storage may make it difficult for them to manage updates.

Designing systems around constrained devices can be challenging, and when security requirements are added to the mix, the challenge is even greater.

Security Challenges

There is a famous Twitter joke about IoT that is attributed to web developer Tim Kadlec (@tkadlec): "The S in IoT stands for security." Think about that for a second as you say, "Wait, there is no S in IoT." That is exactly the point of Tim's statement. IoT manufacturers have been slow to build more needed security in their respective products.

Beecham Research's IoT Security Threat Map displays the full set of threat and vulnerability analyses that Beecham uses to help its clients shape their strategies.

This downloadable summary includes many of the top five features from each of those analyses.

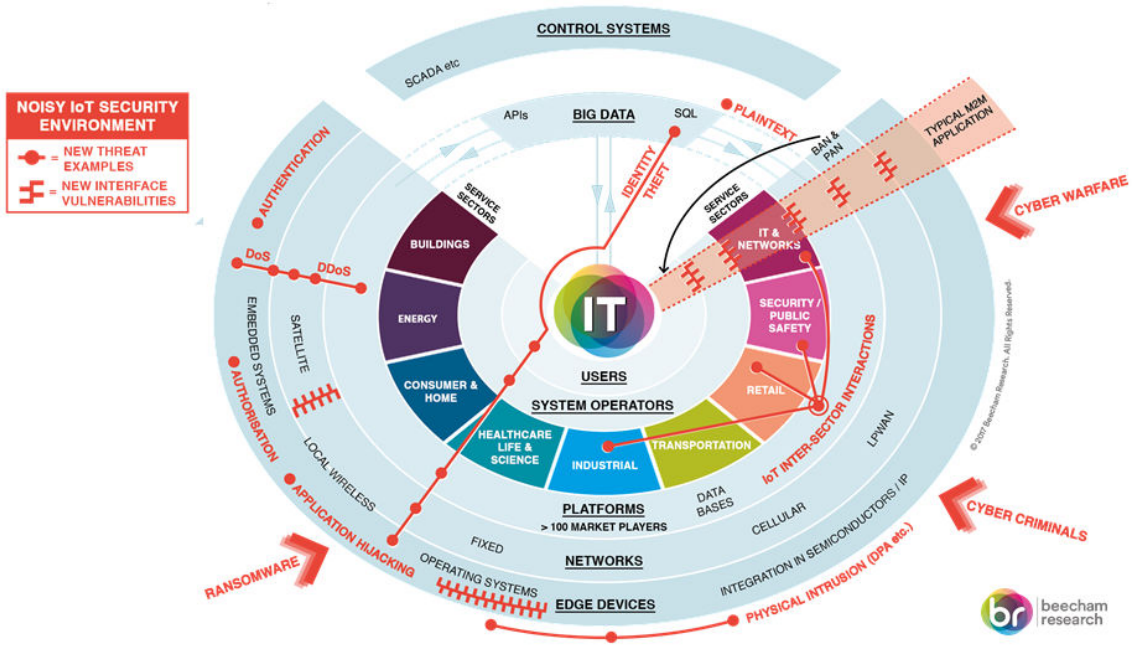


Figure 1-5: IoT Security Threat Map.

Traditional M2M (machine-to-machine) applications are typically very focused, using specific edge devices, a single network and custom platform, making it relatively easy for security professionals to secure to the acceptable level. IoT cuts across different sectors and embraces multiple devices and networks—from satellite to cellular—along with a growing number of IoT platforms and big data systems, which present threats on many different levels and fronts. Wherever there is a new interface between devices, networks, platforms, and users, there is the potential for a new weak link.

IoT Contributes to the Threat Landscape

According to the IDC, in 2016, there were 11 billion devices connected to the Internet, with an expansion to 30 billion expected by 2020. IoT contributes to the threat landscape primarily because of all the new devices that are added daily. IoT is the primary possible culprit in an additional 20 billion devices.

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The Dyn Attack

The 2016 Dyn cyberattack was a series of three distributed denial-of-service (DDoS) attacks that were launched on October 21, 2016. The attack targeted systems operated by former Domain Name System (DNS) provider Dyn. The attack left millions of users in Europe and North America without access to major Internet platforms and services. Although the groups Anonymous and New World Hackers, among others, claimed responsibility for the attack, no definitive evidence was provided.

As a DNS provider, Dyn provided end users with the service of mapping an Internet domain name to its corresponding IP address. The DDoS attack was caused by bogus DNS lookup requests from tens of millions of IP addresses. It is believed that the requests were executed through a botnet consisting of millions of IoT devices that had been infected with the Mirai malware.

Why IoT Is Lacking Fundamental Security

Three reasons why IoT is lacking fundamental security are: lack of industry knowledge, rapid time to market, and lack of developer knowledge.

- **Lack of industry knowledge**: Products are being added with IP interfaces that existed previously in a network-less world. Companies such as Cisco, Microsoft, and HPE have decades of security experience. Most of the IoT startups do not. This lack of security knowledge results in products that repeat the same problems of the previous two decades. IoT startups need to learn from industry's past mistakes.
- **Rapid time to market**: The age-old argument between time to market vs. securing the customer's environment. Startups are focused on deploying products and features with a velocity that catches the eye of investors and new customers.
- **Lack of developer knowledge**: IoT is lacking fundamental security in that the developers do not understand security. A perfect example of this is the Mirai botnet, and specifically the XM cameras that made up a large portion of that botnet.

The solution to this problem is to build security into everything that is created on the IoT, by using a secure development lifecycle (SDL). Also, many embedded security chipmakers and cryptography specialists are taking the matter of embedded security for IoT into their own hands and building security at the hardware-chip level.

Societal Impact

As with technologies that have preceded it, IoT has immense potential to disrupt and change the way things are done. Our history shows that some changes brought about through technology may benefit society, while others may cause harm. And different people may be affected differently. There may also be unintended side effects—new problems may be created while solving old problems. Technologists should consider the potential impact of IoT solutions they design and implement on their organization, people, and the environment.

For example, as machines gain new abilities to sense the world around them, process the data they take in, and take action based on that data, it will be possible for jobs previously performed by people to be performed by machines. There are clear societal benefits to having machines perform tasks that are dangerous, risky, or mundane. And, in some cases, machines may perform those tasks more efficiently and effectively than people. However, eliminating someone's job is disruptive—not only to that individual, but it may also have a significant negative impact on an entire family that can affect them for years.

IoT technologies may reduce manufacturing waste and the use of natural resources, and may significantly help to reduce pollution of the environment. But there are significant environmental costs for using those technologies, such as the mining operations needed to obtain raw materials for lithium batteries and other components; the energy, natural resources, and toxic chemicals used in their manufacture; and challenges associated with properly recycling or disposing of IoT hardware once it becomes damaged, obsolete, or worn out. These factors should all be included in evaluations of the benefits of IoT.

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Organizational Skills Impact

IT staff may need to develop new skills or combine efforts with OT groups to leverage the skills and experience of both groups. New skillsets for IT might include:

- Embedded devices.
- Sensors and actuators.
- Real-time data processing and networking.
- Edge computing.
- New methods of data processing and analysis.

The key is to effectively merge industry domain knowledge with the new technical knowledge required (OT and IT) to accomplish the new goals.

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ACTIVITY 1-3
Identifying Potential Challenges of IoT

Data File

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Scenario

An IoT project can provide a variety of business and technical challenges. In this activity, you will consider the challenges your organization might face in an IoT implementation.

Write your ideas in the spaces provided. (If you'd prefer to type your ideas in a Word document, a worksheet is provided in the aforementioned data file location.)

---

1. **A) Challenges: What challenges might your organization encounter in implementing an IoT solution?**

2. **B) Countermeasures and remediations: How might you address the challenges you identified?**

Case Study: Smart Outlet

A smart outlet has attributes you might find in a typical consumer-oriented IoT device. Following setup and configuration, a user can plug in a normal electrical appliance such as a light or fan, and can switch the appliance on or off through a smartphone app or by issuing a voice command through a smart speaker such as Amazon Echo or Google Home. The outlet might also perform other tasks, such as monitoring and logging electrical flow (when the switch was on, how much current it drew, and safety monitoring for overheating and overcurrent). The device uses Wi-Fi to connect with the applications that collect its usage data and enable the user (through other devices) to switch it on and off.

IoT devices like this can be purchased for less than $25 USD, but even a relatively inexpensive and limited device like this would have attributes such as:

- **Local controls and user interface**: Controls may be provided directly on the device to enable the operator to set it up, turn it on and off, and perform other tasks.
- **Remote controls and user interface**: A remote user interface may be provided through a smartphone app, desktop application, or through verbal commands issued to a smart speaker. A relay in the device enables it to switch power on or off under remote control. (Other types of IoT devices must include motors, servos, valves, and other types of actuators to local action to take place using remote control.) Some local controls, such as buttons or switches, may be

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provided locally for setup and configuration of the device, or as an alternative to remote controls, but local controls are often quite limited.

- **Remote sensing**: A heat sensor and electrical current sensor in the device measure local information that can be locally processed (for example, automatically switching the device off and sending a notification to the remote user if an overcurrent or overheating condition occurs). Data produced locally can be sent to a remote system for logging and analysis.
- **Connected**: Wi-Fi, Bluetooth®, Bluetooth Low Energy®, Zigbee®, or other data communication capabilities enable the device to communicate locally with other devices or with cloud-based applications over the Internet.
- **Smart**: Processing capabilities embedded in the device enable it to process local inputs and outputs (switch, LED), remote inputs and outputs (commands from a remote user, sending data logs to a server or gateway in the cloud). Some (typically small) amount of data storage might be present on the device, where it can be stored until it can be uploaded to a remote system for logging and analysis.

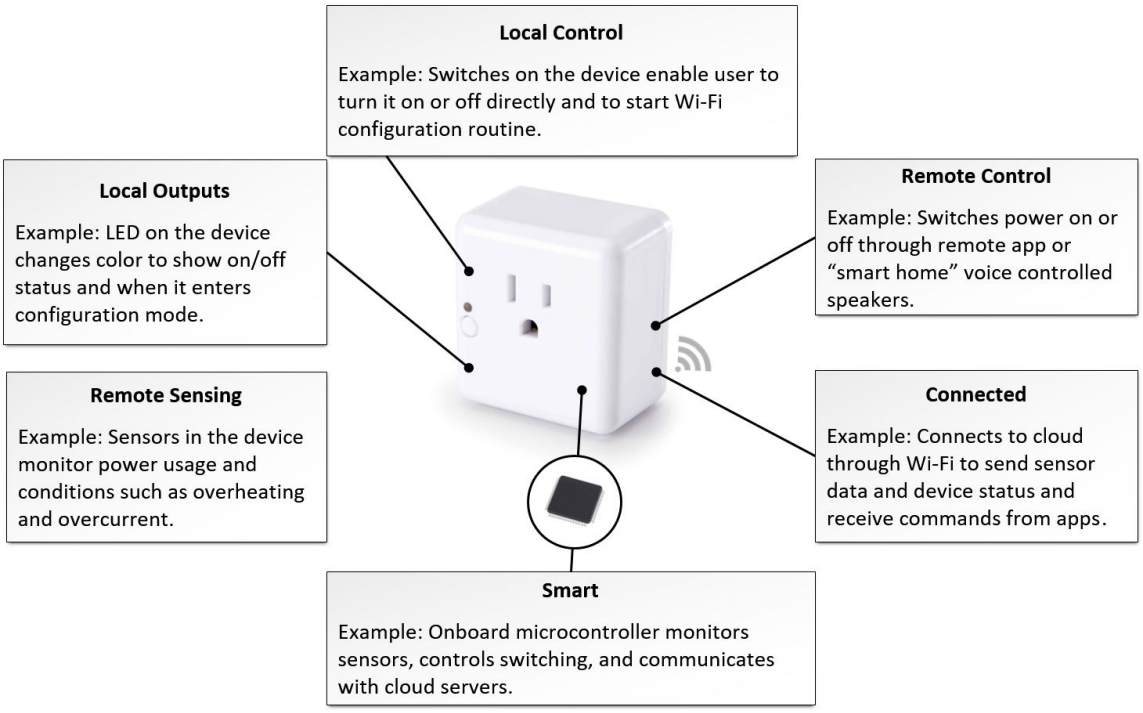


Figure 1-6: Common features of IoT devices, as exemplified in a smart outlet.

Industrial Controllers

Various systems and devices are used for industrial control, as described in the following table.

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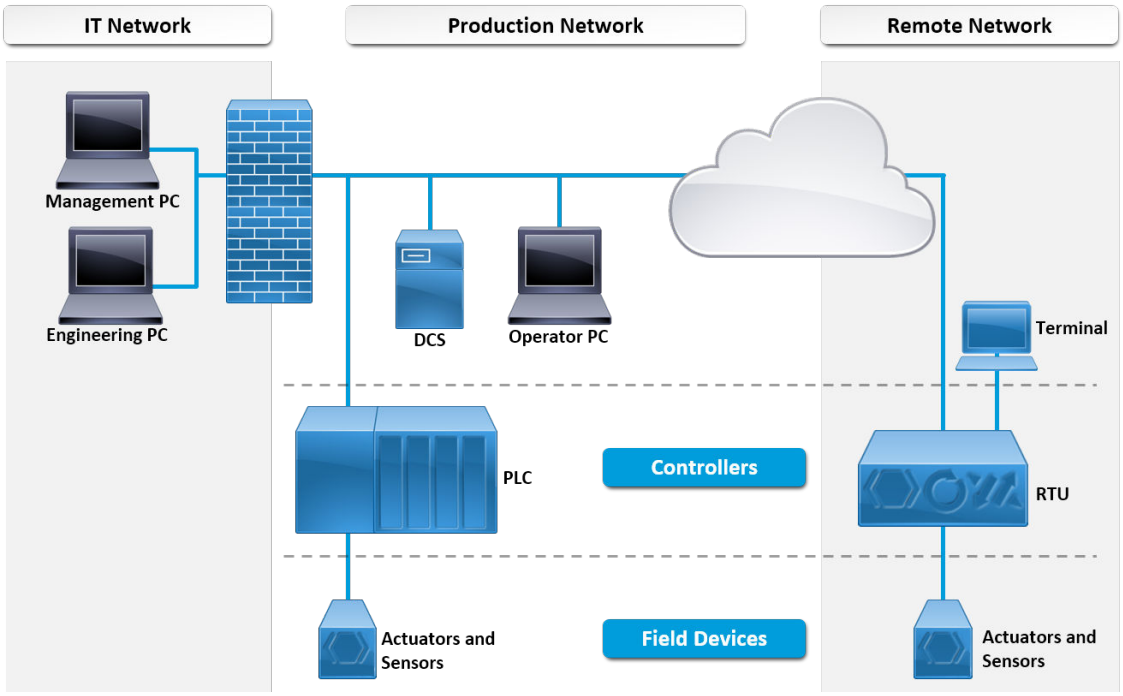


Figure 1-7: Industrial controllers and related devices.

Item	*Description*
PLC	Industrial control traditionally has been based on programmable logic controllers (PLCs). A PLC is comparable in some ways to a desktop computer or rack-mounted server in that it is a box containing a CPU and supporting components (e.g., motherboard, memory, storage). However, PLCs are optimized for use in industrial control applications. PLCs may be configured for rack mounting in a control room or in other form factors for standalone installation out on the factory floor or industrial complex. They are designed for durability in demanding environments, including resistance to electrical noise, vibration, smoke, humidity, and impact. The software on PLCs has been optimized to model traditional hardwired electronic controls, such as relays, which control motors, valves, lights, and so forth, using relatively simple programming schemes, such as ladder diagrams.
RTU	***Remote terminal (or telemetry) units (RTUs)*** monitor and control a remote asset (such as an oil well or water pump). Because they are often located in remote locations, these devices often need capabilities such as network communication (communicating with a centralized control system through radio, cellular telephony, and so forth), low power consumption, data logging and short term storage, ability to operate in adverse conditions (such as weather extremes), and so forth.

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Item	Description
PC	Personal computers (PCs) provide a user interface for viewing data and controlling industrial systems. Initially, PCs had required special adapter cards to communicate with industrial control systems, but as industrial control systems have incorporated PC networking standards, a common network connection (such as cabled Ethernet or Wi-Fi) is sufficient to make the connection, and other end-user devices, such as smartphones and tablets, can often be used to access control applications.
DCS	PLCs were initially developed to control single manufacturing processes and continue to be used as a basic building block for automating a single line or process. But ***distributed control systems (DCS)*** have been developed to control manufacturing on a larger scale, integrating production across multiple processes and locations. Often, a DCS will incorporate multiple PLCs and PCs.
PAC	Both PLCs and DCSes have evolved substantially over time, so, in many cases, it has become difficult to pinpoint precisely where one category begins and ends. ***Programmable automation controllers (PACs)*** combine functions found on DCSes, RTUs, and PCs. This makes them suitable for operations requiring coordination among multiple processes (like a DCS), while providing fine control over specific processes (features traditionally performed by a PLC and RTUs).
IPC	Another component that blurs the distinctions among industrial control systems is the industrial PC (IPC). In the 1990s, software for personal computers was developed to emulate functions of a PLC. Some manufacturers have developed IPCs, essentially "ruggedized" PCs adapted for industrial I/O tasks, to build upon this capability. IPCs can cost-effectively perform many of the tasks as a PLC and PC combined, making them well suited for small automation projects and applications where space is limited.

While traditional programming tools continue to provide backward support, more advanced programming tools are now available, including instruction sets specifically geared toward specific industries, such as brewing, oil, gas, and nuclear power plants.

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Summary

In this lesson, you identified components of a successful IoT project, including a general architecture that enables those components to work together. You also identified benefits of IoT and challenges you might encounter in implementing an IoT system.

IoT has the ability to transform both old and new businesses. How well you both understand and apply IoT and your industry domain knowledge before you dive in will determine, to a large extent, how successful your path will be.



Note: Check your CHOICE Course screen for opportunities to interact with your classmates, peers, and the larger CHOICE online community about the topics covered in this course or other topics you are interested in. From the Course screen you can also access available resources for a more continuous learning experience.

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2 Undertaking an IoT Project

Lesson Time: 2 hours, 5 minutes

Lesson Introduction

Your organization is sold on the benefits of Internet of Things (IoT), has a healthy respect for the risks, and is prepared to undertake its first IoT project. To pull the project off successfully will require careful planning.

Lesson Objectives

In this lesson, you will:

- Identify problems in your organization, and determine how you can use IoT technologies to help resolve them.
- Identify issues you'll need to consider throughout the IoT development lifecycle.

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TOPIC A

Real-World Applications for IoT

Implementing IoT within your organization may have numerous costs in terms of hardware, software, setup, configuration, integrating it into other systems, maintenance, and acquiring the skills needed to work with it. Before you undertake a project, you should identify where it has the greatest potential to benefit your organization.

IoT Market Sectors

According to IoT Analytics (**https://iot-analytics.com**), in 2018, the global share of IoT projects was distributed among various market sectors as follows:

- Smart city: 23%
- Connected industry: 17%
- Connected buildings: 12%
- Connected cars: 11%
- Smart energy: 10%
- Connected health: 6%
- Smart supply chain: 5%
- Smart agriculture: 4%
- Smart retail: 4%

Other sectors accounted for an additional 8% of the global share of IoT projects.

Smart City

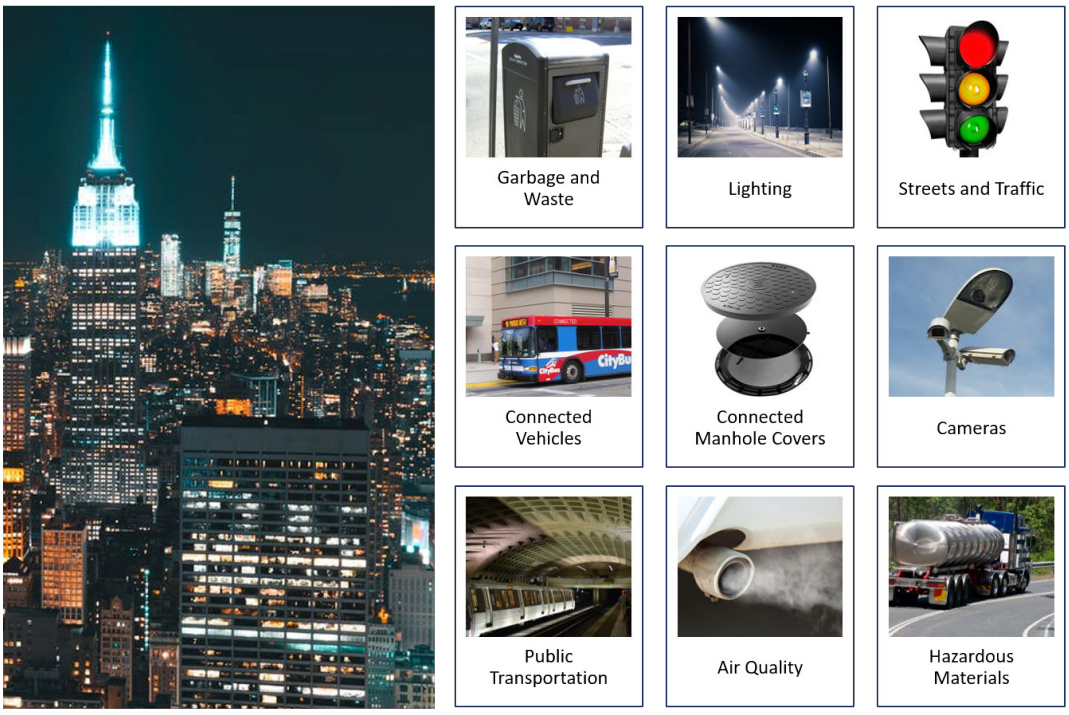


Figure 2-1: IoT automation in a smart city.

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With increasingly large concentrated populations, there is significant pressure on the limited resources and space within cities and other municipalities. IoT-based innovations can help them make better use of resources and provide a high quality of life to their occupants. A ***smart city*** uses sensors distributed throughout the environment and connected to a network in conjunction with applications that support data collection, analysis, and remote control to solve problems in real time or near real time, coordinate complex urban systems, provide efficiency, reduce costs and adverse environmental impact, and improve public services and safety.

The following are examples of ways that IoT technology may be applied in smart cities.

Garbage and Waste

Waste bins can be monitored so pickup routes can be optimized based on demand, saving operational costs, improving timeliness of collection, and reducing unnecessary traffic. Similar concepts can be applied to sewage systems, adjusting the flow of water pumps based on need and reducing water usage when demand for water increases elsewhere within the municipality.

Streets and Traffic

Sensors can measure traffic patterns of pedestrians, cyclists, automobiles, and public transportation through pressure sensors in walkways, turnstiles, and so forth, which can be used to improve long-term planning of city layouts. Smart parking meters can monitor parking, enabling apps and smart signs to direct drivers to open parking spaces through smartphone apps, GPS units, and smart cars. Inputs from traffic sensors throughout a city can be analyzed in real time and coordinated with messaging sent to street signals, signs, and directly to citizens' IoT devices (smartphones, smart cars, etc.) to efficiently coordinate traffic flow out of the city during emergencies or other high-traffic situations.

Lighting

Based on reduced need as measured by sensors measuring automobile and foot traffic, ambient light, and other inputs (such as scheduled sports events, festivals, etc.), street lamps can reduce or increase their light levels appropriately.

Public Transportation

Digital signs in train stations, bus stops, and smartphone apps can provide real-time updates on estimated arrival times and suggest alternate routes when appropriate, to keep riders informed, improve the public transportation experience, and encourage wider use of public transportation. Based on real-time data such as air quality (smog levels from automobiles), available parking spaces, and so forth, temporary discounts on public transportation prices can be offered to provide citizens with incentive to avoid driving. Real-time data regarding traffic flow, road closings, working accidents, number of people waiting at bus stops, number of seats already filled, and so forth can be used to dynamically optimize routes and the number of vehicles in service, to make the best use of resources while meeting demand.

Safety and Security

Sensors strategically located throughout the city can detect specific types of risks, such as the presence of chemical, biological, explosive, or radioactive devices, and authorities can be alerted. Sensors can monitor for impending natural disasters such as earthquakes, tsunamis, and landslides. Early warnings to citizens, road closures, evacuations, and other measures can be taken. Personal IoT devices such as smartphones and wearables can be used to send timely alerts to users as a supplement to traditional warning and messaging systems. In situations that unfold quickly, such as floods, landslides, and active shooting situations, the ability to quickly inform the public can save lives. Security and safety services, such as police presence and active visual monitoring, can be better coordinated based on sensor data that identifies such things as traffic patterns. While targeted tracking of individuals raises privacy issues, general tracking of numbers of people, noise levels, and so forth is less controversial. Analysis of ambient noises within a city can identify gunshots, car crashes, and other noise signatures that can alert authorities to an active situation in real time.

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Environment

Air quality in cities varies widely based on weather conditions, traffic, industrial processes, the operation of heating and power plants, and other factors. Various microclimates based on terrain, buildings, and other factors result in uneven air quality throughout the city on a given day. Sensor data from municipal systems and citizen-owned devices can be collected and analyzed to enable appropriate countermeasures to be taken as needed, such as coordinating the use of alternate energy sources, rerouting traffic, and so forth.

Infrastructure

The structural integrity of bridges, roads, tunnels, dams, levees, and other infrastructure can be monitored through sensors that measure stress, vibration, erosion, position, and so forth to support predictive maintenance, reduce inspection costs, and prevent dangerous and costly failures. Connected manhole covers can detect and report abnormal, costly, and dangerous events like theft, vandalism, and dislocation due to storm surges, enabling timely replacement. Sensors within manhole covers can also be used to detect sewer levels and flow, chemical and biological conditions, temperature, and other factors that can be analyzed in real time to identify sewer management problems.

Case Studies

Case studies describing various smart city IoT projects may be found at the following URLs.

- Berkeley County case study for smart meters

  https://www.smartcitiescouncil.com/sites/default/files/main/public_resources/Berkeley%20County%20Case%20Study_Final_0_0.pdf
- Seoul waste management case study

  https://www.smartcitiescouncil.com/sites/default/files/main/public_resources/Ecube%20city%20of%20seoul_0.pdf

Industry



Figure 2-2: IoT automation in industry.

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Traditional industrial automation has been accomplished through industrial control systems, such as programmable logic controllers (PLCs), process automation controllers (PACs), motor drives, robots, and so forth. The following are examples of additional ways in which IoT technology may be applied in industry.

Transportation and Distribution

Logistics, the detailed coordination of people, facilities, and supplies involved in manufacturing, production, and fulfillment, is often the primary difference between businesses that succeed and those that fail. Tracking of goods from their source to their destination has usually required someone scanning a barcode at each junction. However, using devices like radio-frequency identification (RFID) tags, this process can be automated. Goods in transit can connect to the cloud and share data on their status and location, eliminating most human intervention. The status of the entire supply chain can be evaluated and communicated at any time. This includes raw materials, components, production machinery, finished products, material handling equipment, packaging, transportation vehicles, other means of transport, delivery, and distribution.

Wearables

Wearable IoT devices such as watches, vests, jackets, and shoes can monitor workers operating in dangerous environments. They can detect exposure to chemicals, radiation, or other agents. They can sense when the wearer has lost consciousness or mobility (due to a cave-in in a mine, medical conditions like apoxia, stroke, or heart attack). They can detect falls from heights, and attempts to lift excessively heavy loads. Actuators in wearable devices can notify workers when someone nearby is in distress, through an audible alarm or through vibrations when the environment is loud or hearing protection is in use.

Environment

IoT devices can be used to monitor the quality of air, water, and soil in areas where contamination is a risk to provide early detection of leakage or contamination. Natural disasters such as tornadoes, floods, earthquakes, tsunamis, avalanches, mudslides, and forest fires can be detected. Early detection enables problems to be quickly identified and remediated. Affected systems can be quickly shut down to minimize the impact, and personnel can be notified.

Inspection and Quality

Data regarding quality parameters can be analyzed in real time, before the manufacture of a part is even completed, and necessary adjustments can be made. Quality control data from individual machine tools can be collected and sent to the cloud, where it can be analyzed remotely.

Quality inspections commonly involve visual confirmation that components are correctly assembled and have no defects. With the massive number and variety of products being manufactured today, traditional computerized scanning techniques are not practical, as they require extensive programming, testing, and refinement. However, by presenting machines with examples of defects, IoT systems can be trained quickly using artificial intelligence to recognize even subtle defects.

Maintenance

Predictive maintenance is forecasted to be one of the fastest growing and most profitable applications of IoT over the next few years. IoT can improve the ability to predict when production equipment and machinery requires preventive maintenance, with its ability to measure overall equipment effectiveness in real time.

Case Studies

Case studies describing various industrial IoT projects may be found at the URLs provided.

- IoT in Transportation and Logistics

  https://www.iotcentral.io/blog/iot-in-transportation-and-logistics
- Maersk and Ericsson collaborate for IIoT success story

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https://internetofbusiness.com/maersk-ericsson-iot-success/

- GE's Big Bet on Data and Analytics

  http://marketing.mitsmr.com.s3.amazonaws.com/PDF/57380-MITSMR-EY-GE-Case.pdf
- Leading Tools Manufacturer Transforms Operations with IoT

  https://www.cisco.com/c/dam/en_us/solutions/industries/docs/manufacturing/c36-732293-00-stanley-cs.pdf
- Hirotec Launches IoT Initiative

  https://www.ptc.com/en/case-studies/hirotec

Buildings



Figure 2-3: IoT automation in a smart building.

Large buildings with extensive mechanical, HVAC, electrical, security, and fire alarm systems are often controlled through smart building software called a building management system (BMS). This software may also be called a building automation system (BAS). The BMS may control other systems as well, such as turnstiles, access doors, and elevators that control who is allowed passage, as well as monitoring for security through closed-circuit TV and motion detectors.

The software may be proprietary, or may use open standards. The systems connected to a BMS might control up to 70% of the energy used in a building, so the BMS is an important tool for managing energy usage in smart cities. It is believed that 8% of the world's energy usage could be eliminated through better control of building management systems, which IoT technologies could support.

Building management systems are useful in the event of emergencies. For example, in a fire, the system can be used to locate occupants and shut HVAC dampers to limit the spread of smoke and fire.

IoT technologies support and extend the capabilities of building management systems to improve the experience of occupants, make better use of resources and building space, and improve safety and security. The following are examples of ways in which IoT technology may be applied to smart buildings.

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Building Traffic

Traffic into, within, and out of the building can be monitored and controlled through doorways, gates, toll gates, elevators, and escalators. Various types of keys associated with an individual can be used to track and control traffic, such as key cards, fobs, cipher locks, facial recognition, and other biometric scanners.

Occupants

Smart buildings can monitor how many people are in the building and determine where they are located. There are numerous applications for this capability. For example, it can be used to locate people during an emergency such as a fire. It can be used to help manage consumption of resources such as heating, cooling, and lighting. It can be used for billing tenants for actual usage of the building and to identify underutilized spaces and facilities.

Safety and Security

In an emergency, data showing who is currently in the building and where they are located, the locations where hazard and intrusion alarms have been triggered, and other factors, can provide first responders with extremely useful information to help guide a targeted and appropriate response.

Building Infrastructure

Buildings can be "self-aware"—actively monitoring themselves and making adjustments as needed to improve their status, such as adjusting air balance and control of HVAC systems, rather than blindly following pre-programmed routines that are outdated when conditions within the building change over time. Some building management functions can be monitored and controlled remotely. These features can provide building owners with greater flexibility in staffing, and can reduce the need for on-site specialists, while providing better service to tenants.

Inside and outside environmental sensors monitor factors such as occupancy, lighting levels, temperature, humidity, air quality, and weather, and systems make adjustments as needed to ventilation controls such as vents, louvers, and window blinds, as well as heating and air conditioning. Resources such as power and water can be obtained and stored during times when demand and costs are lower, to be used when demand and costs are higher. Similarly, waste disposal can be more efficient—for example, retaining waste such as greywater until periods when it is most ecological to release into the sewage system. Power from the electrical grid can be supplemented with renewable power from on-building sources, such as wind power, solar, and geothermal. IoT capabilities within the building enable it to balance power generated on site with power from the grid, enabling it to be distributed and used, or sold back to the power company.

Data from machinery such as HVAC systems, elevators, water pumps, and other equipment can be analyzed to support predictive maintenance of equipment.

Case Studies

Case studies describing various smart building IoT projects may be found at the following URLs.

- Intel Creates Smart Building Using IoT

  https://www.intel.com/content/dam/www/public/us/en/documents/case-studies/smart-building-using-iot-case-study.pdf
- From Concept to Reality: Designing Network Access Control Technology Into a Smart Building Complex

  https://www.fortinet.com/content/dam/fortinet/assets/case-studies/cs-smart-building.pdf

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Connected Cars

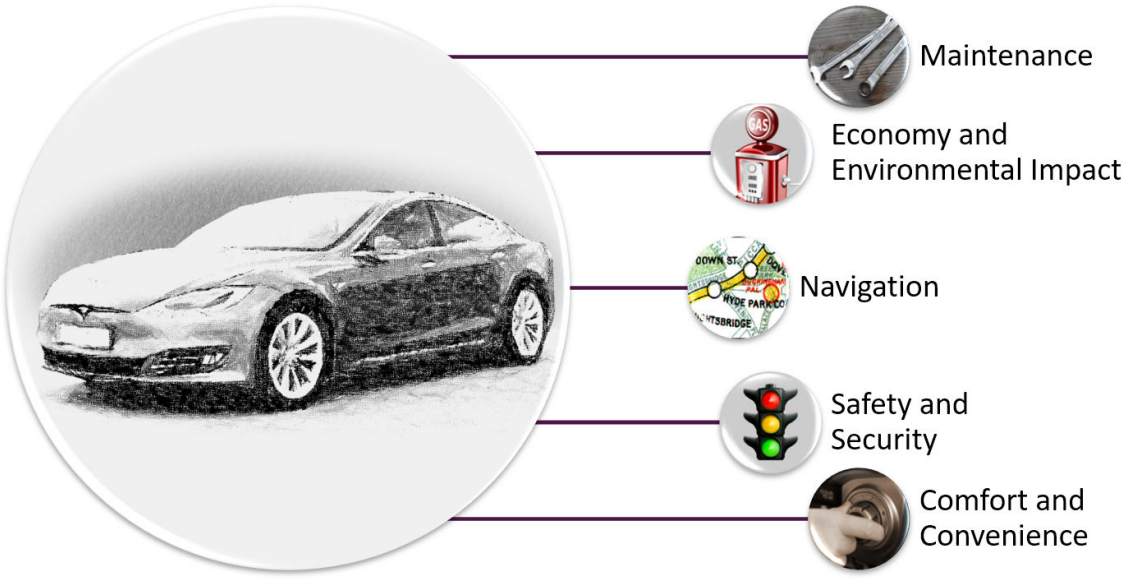


Figure 2-4: IoT automation in a connected car.

A single connected car may have the processing equivalent of 20 desktop computers onboard. Connected cars typically include a wireless Internet connection through cellular connections such as 3G, 4G LTE, and 5G. Onboard Wi-Fi enables passengers to route their smartphones or other personal devices through the car's cellular Internet connection.

The following are ways that IoT technology may be applied to connected cars.

Maintenance

Information such as location, fuel level, fuel consumption, maintenance data, load, and weight can be tracked for individual vehicles within a fleet. Analytics tools can be used to correlate that information for the entire fleet, facilitating scheduling, routing, and planned maintenance. Based on fleet data and local fuel/energy costs, and using connected cars, smartphones, or other onboard communication, drivers can be provided with information on where to find optimal refueling locations on their route. Connected cars can monitor driving patterns and assess the wear and tear on a vehicle. Insurance companies can monitor this information and calculate fairer premiums that are based on usage and maintenance of the vehicle. Service stations can use the diagnostic information to perform predictive analysis and contact vehicle owners to schedule a proactive service appointment.

Economy and Environmental Impact

Connected cars enable the driver to travel quickly, safely, and cost effectively. When vehicles can communicate with traffic signals and other infrastructure, they manage the accelerator better, slowing down before reaching a signal, and the highway infrastructure can also adjust the timing of lights to reduce unnecessary stops. These features will reduce fuel consumption and save drivers time.

Navigation

Many modern cars are equipped with GPS navigation systems, but connected cars will take this further, combining smart navigation features with services based on location and the car's servicing requirements, such as prompting you to stop and refuel. The car could determine the distance to the nearest fueling station and automatically guide you there using the navigation system.

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A connected car could integrate with your personal calendar to help you take the best route to your next meeting, based on real-time traffic and weather conditions. Based on your travel history, your current bearing, and traffic conditions, a connected car might figure out where you are headed and offer suggestions on alternate routing if you're approaching an accident, construction work, or other delay.

Safety and Security

When connected cars and infrastructure such as smart roads are able to communicate, traffic flow can be coordinated better, not only reducing stops and saving fuel, but also potentially reducing traffic accidents. Connected cars can avert collisions by tracking the speed and the proximity with other vehicles. Vehicles can provide drivers with real-time alerts regarding road and weather conditions and nearby accidents. A connected car can track its location and coordinate with owner data to inform customers when their vehicle may have been stolen and support its safe retrieval. A vehicle could capture the state of the driver through cameras and sensors, and warn them when they suffer from fatigue and tiredness. The system can even optimize the temperature, music, and seat functions to ensure the driver remains alert.

The U.S. Department of Transportation (DOT) has proposed a legal requirement that cars be able to communicate with each other using vehicle-to-vehicle (V2V) technology. If two vehicles are on a collision course, the vehicles could prevent an accident by applying brakes, slowing down, or taking other appropriate actions. The cars would coordinate regarding their movement and speed, and could provide evasive actions before drivers are even aware of the problem.

Comfort and Convenience

"Keyless" door locks can be accessed using key fobs, smartphones, or other tokens. Preferences for cabin climate and seat positions can be set for each driver and passenger. The power management system enables the car to be "awakened" under remote control to take readings or operate the car. Self-driving vehicles are already on the road. A common feature is automatic parallel parking, which automatically performs a task that some drivers find difficult.

Mobile payment systems are already in use. Digital wallets operating on smartphones enable consumers to safely and securely conduct transactions electronically, such as "paying at the pump." Automobile companies will enable similar electronic payment schemes directly through smart vehicles, enabling the customer to pay at toll gates, fueling stations, parking meters, parks, and campgrounds using a digital wallet embedded in the car dashboard.

Case Studies

Case studies describing connected car IoT projects may be found at the URLs provided.

- The IoT-Connected Car of Today—Case Studies

  https://www.digitaldoughnut.com/articles/2017/march/the-iot-connected-car-of-today-cases
- Cisco Jasper: Connected cars the fastest-growing IoT sector

  https://www.zdnet.com/home-and-office/networking/cisco-jasper-connected-cars-the-fastest-growing-iot-sector/
- Connected Cars: How Jaguar Land Rover works with BearingPoint to benefit drivers

  https://www.bearingpoint.com/files/IDC_buyer_case_study-Connected_cars-Jaguar_Land_Rover_BearingPoint.pdf

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Energy and Utilities

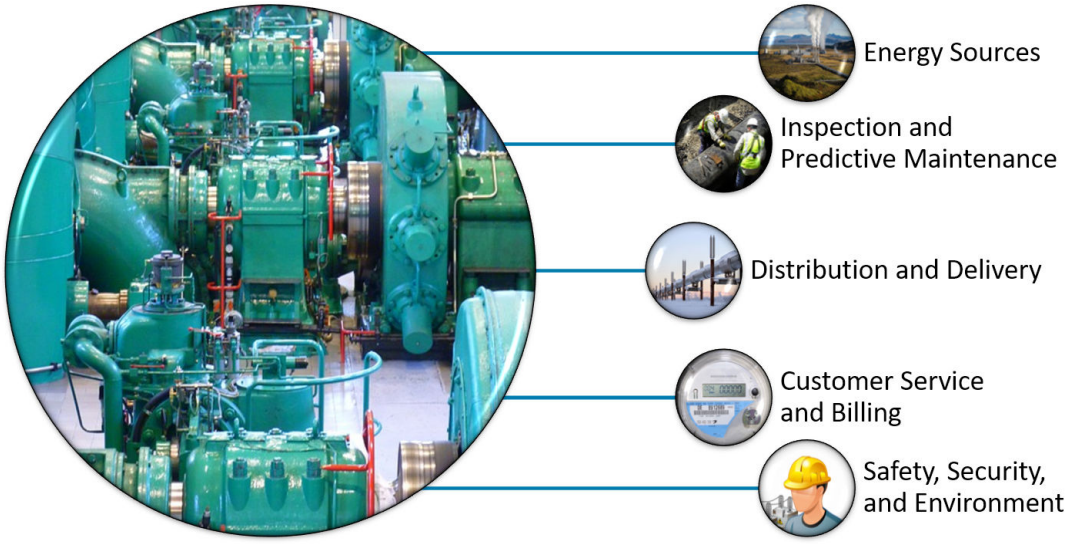


Figure 2-5: IoT automation in energy and utilities.

The rationale for implementing IoT solutions in public utilities primarily derives from three needs: to make the most efficient use of energy, to minimize and eliminate adverse environmental impacts, and to reduce the cost of producing and delivering it. IoT can provide a wide range of benefits to customers, utility company employees, investors, and the environment. By providing two-way communication between a power utility and its customers, combined with sensing along power transmission lines and smart electrical meters, it is possible to create a ***smart grid***, providing numerous benefits to customers, utility companies, and the environment. Many aspects of the smart grid will be managed through ***energy management*** applications that enable pipeline operators and energy managers to precisely monitor and control their systems, and make well-informed decisions that optimize energy efficiency, increase profits, and reduce greenhouse gas emissions. As with other industry, there are also potential benefits in the form of improved maintenance, safety, and security.

The following are ways that IoT technology may have applications in energy and utility industries.

Energy Sources

Oil-well flow rate, pressure, and temperature data collection can be done every minute, enabling production to be monitored and optimized in near real time. Customer-owned power generation systems and smaller commercial power plants, including solar, wind, bio-gas, and other renewable energy systems, can be controlled and monitored remotely, enabling better integration with the grid, improving power buyback mechanisms, distribution, and use of customer-provided energy sources. Oversupplies during peak production hours can be diverted to short-term storage, such as pump-storage hydropower plants, as needed. With careful monitoring and rules for automatic trading, ***energy trading*** can be performed in real time.

Inspection and Predictive Maintenance

Sensors in oil and gas pipelines can monitor pressure, flow, compressor condition, temperature, density, and other variables. Acoustic sensors can detect a breach by a variation in the acoustic signature. Fiber-optic sensors can detect deformations in pipe walls. Pinhole leaks can be detected before they can become large ruptures. Sensors can be sent inside pipes for inspection using a ***smart pig***, a drone that can detect cracks, defects, and corrosion using magnetic and ultrasound detection. Aerial drones can perform similar inspections outside a pipe. Cameras in remote power stations can use various wavelengths of light (e.g., infrared, visual light) to provide inspection and monitoring.

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Distribution and Delivery

Power outages can be detected and corrected faster, and small outages can be quickly isolated to contain them before they become large-scale blackouts. Gas, oil, and water pipelines can be monitored to identify leaks, enabling prompt detection and shutdown, reducing adverse environmental impact, loss of resources, and cleanup costs. Monitoring of oil pipelines and tank trucks can help to reduce theft, detecting ruptures and instantly alerting personnel when distribution and delivery systems have been compromised.

Customer Service and Billing

Automatic meter reading capabilities provide up-to-the-minute, accurate readings of power usage. Analytics can help customers visualize how they are using energy over time, to help them plan their use of power more strategically, during less expensive times of day, for example.

Safety, Security, and Environment

During outages, power restoration can be conducted strategically—such as routing electricity to emergency services first, and taking greater advantage of customer-owned power generation. Water consumption and water quality can be monitored. Rain water and grey water can be harvested and recycled intelligently—particularly in arid climates—to identify water suitable for cleaning, irrigation, and so forth.

Case Studies

Case studies describing energy and utility company IoT projects may be found at the following URLs.

- AWS Power & Utilities Case Studies

  https://aws.amazon.com/power-and-utilities/case-studies/
- Silver Spring Networks solution delivers smart grid-based outage restoration

  https://www.power-grid.com/td/silver-spring-networks-solution-delivers-smart-grid-based-outage-restoration/
- Three oil and gas IoT case studies

  https://enterpriseiotinsights.com/20170515/channels/fundamentals/20170515channelsfundamentalsthree-iot-case-study-oil-gas-industry-tag23-tag99

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Health, Medical, and Life Science



Figure 2-6: IoT automation in health, medical, and life sciences.

The healthcare industry has already begun using connected devices in a big way. As of 2018, the majority of healthcare organizations are using IoT for maintenance and monitoring of medical equipment and are leading business in adoptions of IoT technology. Patient-monitoring equipment is the leading use case, but a variety of other applications are emerging. Consumer-oriented devices such as fitness watches have become commonplace. The following are various ways in which IoT technology is used in health, medical, and life science organizations and by consumers.

Personal Health and Welfare

Wearable devices such as smartwatches can measure a user's physical activity, heart rate, and body temperature. Better techniques for unobtrusively monitoring blood pressure, irregular heart rhythms, blood glucose levels, body chemistry, and so forth continue to be developed. Patients' use of prescription medicines can be monitored through devices such as connected pill boxes, injection devices, and inhalers. At-risk patients can wear devices that monitor them for medical issues or falls, and automatically alert care providers when there is a problem and provide germane data, such as where the patient is currently located.

Medical Devices

Home-use medical devices such as ***continuous positive airway pressure (CPAP) devices*** (which facilitate a patient's healthy breathing while sleeping), defibrillators, cardiac monitoring devices, and insulin pumps are now commonly designed to be connected devices that can be set up and monitored by clinicians in a remote location using a Wi-Fi or cellular data connection.

Hospital Logistics

Safety is a primary concern for healthcare organizations. A critical component of safety is the ability to track assets such as medical staff, patients, and equipment throughout the medical campus, and to know the status of all devices and instruments. IoT and real-time location systems facilitate asset tracking, making it possible to know the location and status of the people and things involved in any healthcare scenario.

IoT's ability to track logistics—the location and status of goods—in real time has been demonstrated through retail organizations like Amazon. Logistics capabilities can help solve the

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problem of verifying the chain of custody in the pharmaceutical supply chain. New regulations require organizations to ensure that drugs have been protected from tampering by tracking the global path of a pharmaceutical product from raw material to end consumer.

Medical Services

Cost reduction is a major benefit of IoT in healthcare. Telemedicine, sleep studies, cardiac monitoring, and other tasks that used to be performed in clinics are now often performed outside the clinic, as patients wear small devices that perform the monitoring tasks and go about their normal lives. This reduces the workload of healthcare providers, and frees up space in clinics, reducing overall costs.

Telemedicine can be very beneficial in countries and locations where health facilities are unavailable or inaccessible. The portability of these devices makes them very useful in areas with limited resources such as floods, earthquakes, and war zones.

Robotic surgery provides a number of potential benefits. Robots may be able to perform certain operations with precision exceeding that of a human surgeon. And because robots can be autonomous or controlled from a distance, robotic surgery may have applications in telemedicine as well, enabling procedures to be performed in remote locations that can't easily be reached by a surgeon, and enabling surgeons to extend their coverage. Another type of surgery involves using miniature robots that can perform surgery inside the body, minimizing incisions and exposure to pathogens, and making possible various types of surgery that would previously have been impossible or dangerous to perform.

Case Studies

Case studies describing IoT projects in health, medical, and life science organizations may be found at the following URLs.

- Internet of Things case study: Boston Children's Hospital and smarter healthcare

  https://www.iottechnews.com/news/2017/feb/28/internet-things-case-study-boston-childrens-hospital-and-smarter-healthcare/
- 365mc Unveils World's First AI-assisted Liposuction System 'M.A.I.L System' Developed in Collaboration with Microsoft

  https://www.businesswire.com/news/home/20171018005647/en/365mc-Unveils-Worlds-First-AI-assisted-Liposuction-System-M.A.I.L-System-Developed-in-Collaboration-with-Microsoft
- Reimagining healthcare with Azure IoT

  https://azure.microsoft.com/en-us/blog/reimagining-healthcare-with-azure-iot/

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Supply Chain

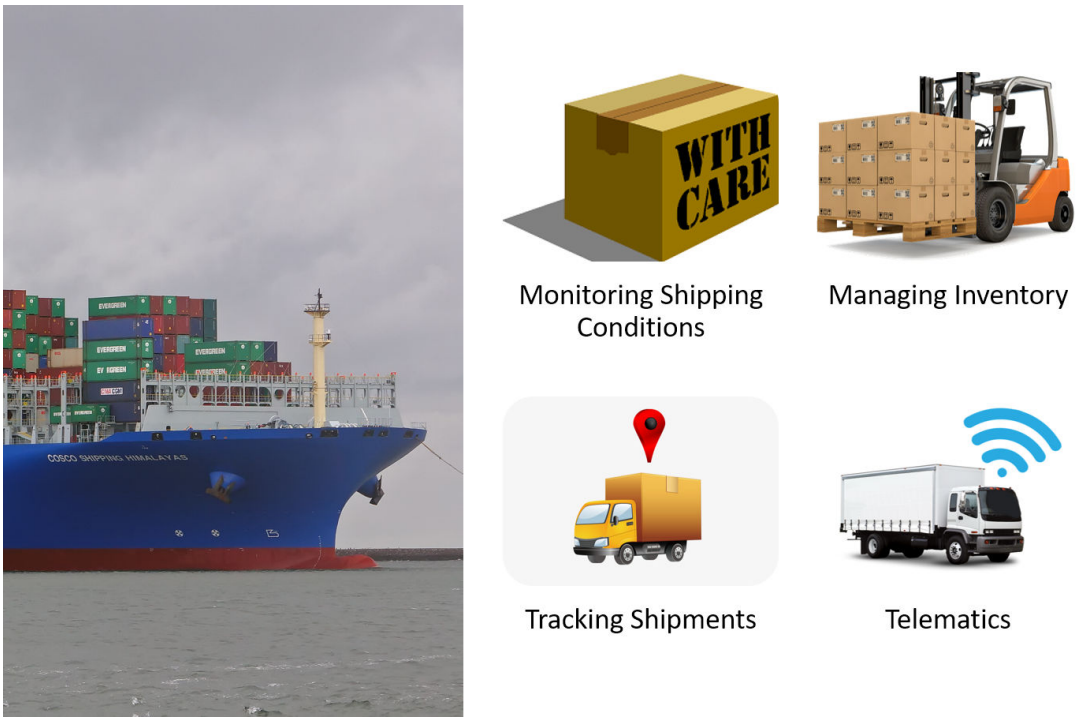


Figure 2-7: IoT automation in supply chain management.

IoT sensors can be made small, rugged, and self-supporting, so they can follow goods along for the ride as they make their way through the supply chain. IoT devices can relentlessly track and report details about the world around them while requiring minimal human intervention, making them ideal for tracking shipments and the status of inventory.

The following are various ways in which IoT technology is used in supply-chain management.

Monitoring Shipping Conditions

IoT devices at the source can provide measurements of conditions in which inventory is shipped, from the source to the destination. Products like pharmaceuticals, paint and chemicals, food, and so forth require special handling to ensure quality is maintained during shipping. Sensors can measure conditions such as ambient temperature, humidity, and air pressure, and either prevent operations or re-route if the conditions fall outside required parameters.

Managing Inventory

In the past, managing inventory involved numerous counts, recounts, and/or paperwork as products were used or sold. With IoT, the process is automated through RFID tags on each item, which communicate with sensors to provide real-time data on the location of inventory. As inventory is depleted, sensors can monitor the count, weight, or volume of the remaining inventory, automatically ordering or replenishing as needed. Connected devices, tags, and sensors help to eliminate guesswork, thereby reducing risk.

Tracking Shipments

Tracking shipments have been handled similar to inventory. As a package moves from one hub to another, a barcode scan is performed to keep shipper and receiver informed about the package's location. The barcode scan was an improvement over manual accounting, but in the IoT model, tracking information can be acquired from tags that automatically check in when they reach a hub.

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Other sensors can report on shipping conditions, such as pressure, jolts, vibrations, noise, temperature, and so forth. If an item is damaged in transit, the cause and location can be pinpointed through this data. IoT can ensure that perishable products such as food have been kept in a controlled temperature during the entire shipping process, verifying what is called a "cold chain." Lab samples en route for testing can be tracked in a similar way. These are the sorts of tasks for which IoT is ideal.

To provide customers with transparent reporting of cold chain and other tracking data, companies are using blockchain to record the various data points as products are shipped from their source.

Telematics

Telematics uses telecommunications technology to monitor the location, movements, status, and behavior of a vehicle or fleet of vehicles. A company's fleet might include vehicles such as ships, vans, and trucks.

Case Studies

Case studies describing IoT projects involving supply-chain management may be found at the following URLs.

- Orbcomm launches new cold chain monitoring solution

  https://enterpriseiotinsights.com/20170816/internet-of-things/orbcomm-launches-new-cold-chain-monitoring-solution-tag23
- Bringing Blockchain, IoT, and Analytics to Supply Chains

  https://hbr.org/2021/12/bringing-blockchain-iot-and-analytics-to-supply-chains
- Virtualization of food supply chains with the Internet of Things

  https://www.sciencedirect.com/science/article/pii/S026087741530056X

Agriculture



Figure 2-8: IoT automation in agriculture.

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Large farms have already seen significant productivity gains through IoT. Improvements in the measurement and analysis of weather, the condition of soil, plants, and livestock, and market data combined with effective analytics enable farmers to improve their yields while making more efficient use of land, water, fertilizer, and other resources, and reducing environmental impact. These improvements help farmers obtain a higher return on investment.

Better information can also help farmers provide consumers with greater transparency, enabling them to more accurately monitor how much water and chemicals were used, and when and how the food was harvested.

Large investments in sensors and equipment may cost more than small farmers can afford. But some small farmers have figured out how to take advantage of IoT on a smaller scale, and are experiencing productivity gains. IoT can improve the efficiency of small farms, making them more viable and resilient to adverse conditions.

The following are examples of various ways in which IoT technology may be applied in agriculture.

Crops and Livestock

IoT sensors can monitor plants for signs of stress, and make adjustments or recommendations to the farmer as needed. IoT analytics can evaluate crop data, weather conditions, and market data to determine the best timing for harvest. Biosensors on and around livestock can be used to acquire data that will help make better decisions regarding feeding, breeding, and the use of antibiotics.

Soil and Environment

Soil chemistry can be monitored in very small areas, enabling precise adjustments of fertilizer and amendments to conserve on their use while improving soil quality. Local weather can be monitored and forecasted, adjustments to irrigation and drainage can be made as needed, and spraying and application of fertilizer can be scheduled. Aerial drones provide a closer, more accurate, affordable, and continuous view of farms, enabling them to be divided into manageable zones. Soil and crops can be managed on an extremely local basis, accounting for different microclimates within a single field, for example.

Equipment

Tractors, harvesters, planters, sprayers, and other equipment can be tracked and monitored to support predictive maintenance. Fuel use can be managed by monitoring equipment use, and making adjustments or recommendations. GPS-connected tractors can plow furrows more accurately and use land more efficiently. Water, fertilizer, and amendments can be applied more precisely under machine control, and refer to soil and location data.

Irrigation Systems

Data such as soil humidity, temperature, groundwater levels, weather forecast, and recently applied fertilizer or chemical interventions can guide where irrigation is needed to conserve water while improving results. Actuators can use GPS to determine the current position of fertilization and chemical apparatus, and can precisely adjust the amount being applied as determined by previously acquired data and analysis.

Drainage and Waste Systems

Chemical sensors can determine the composition of animal waste and prioritize its use for fertilization or energy production through bio-gas processing.

Case Studies

Case studies describing IoT projects involving agriculture may be found at the following URLs.

- Can a Cow be an IoT Platform?

  https://www.iotcentral.io/blog/can-a-cow-be-an-iot-platform
- Drones in Precision Agriculture

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https://medium.com/@Unfoldlabs/drones-in-precision-agriculture-331c23cefc0b

Retail



Figure 2-9: IoT automation in retail businesses.

IoT technologies provide new ways to manage retail operations and provide store security, market products to customers, and manage sales transactions.

The following are ways in which IoT technology may have applications in retail businesses.

Inventory Management and Distribution

Retail involves significant logistics operations that can be assisted through IoT, including the distribution of products to warehouses, from warehouses to stores, the transfer of inventory between stores, and shipping products to customers.

Customer Traffic and Presence

IoT can be used to monitor customer traffic and presence in stores. This includes the use of floor sensors or light beams to track foot traffic, detecting the presence of mobile phones (through beacon or Wi-Fi triangulation functionality), and other types of sensors. Monitoring in-store traffic patterns can help to inform where products should be placed along popular routes. Combined with demographic data, traffic patterns can be more deeply analyzed to reveal how different types of customers move through the store, further informing product placement.

Advertising

Analysis of in-store traffic patterns can be correlated to advertising in digital signs and store loudspeakers to determine their impact in real time. Online stores target their marketing based on customers' previous purchases or patterns of products they have looked at. Similar approaches can be used in brick-and-mortar stores by detecting the customers' presence and location, identity (whenever possible), and using digital signs, self-serve kiosks, smartphone apps, wearable devices, and other methods to deliver advertising messages. Based on data provided by the customers' smart-home appliances (such as smart refrigerators and thermostats), customers can be reminded to purchase needed consumables (such as milk or furnace filters).

Security

Using sensors such as RFID and smart cameras, IoT systems can track unpurchased items being removed from the store. Using infrared cameras or chemical and metal detectors, people entering or

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leaving the shopping facility can be scanned for weapons and dangerous materials. When security events occur, the system can trigger appropriate interventions.

Vending and Payment

A wide variety of mobile payment systems and digital wallets (using smartphones or other devices) can be used to enable customers to safely and securely conduct electronic transactions. Wireless technologies such as Bluetooth® Low Energy (BLE) and near-field communication (NFC) beacons make it unnecessary for customers to retrieve their devices or wallets to conduct transactions. Self-serve kiosks can be provided to help customers locate products in the store, request human assistance, bag items, and check out.

Case Studies

Case studies describing IoT projects involving retail businesses may be found at the following URLs.

- Amazon Robotics: IoT in the Warehouse

  https://www.informationweek.com/strategic-cio/amazon-robotics-iot-in-the-warehouse/d/d-id/1322366
- The New Retail Revolution: Connected Store

  https://www.slideshare.net/IBMIoT/watson-iot-for-retail
- Leading from the front: Digital Reinvention in retail

  https://www-01.ibm.com/common/ssi/cgi-bin/ssialias?htmlfid=GBE03873USEN&

Defense



Figure 2-10: IoT automation in defense.

Military organizations, in many ways, have many of the same information management needs as business and industry. The armed forces support their own medical facilities, have buildings and vehicles to maintain, an enormous and complicated supply chain to manage, and even operate their own publishing, broadcasting, and retail facilities. As it does for the private sector, IoT has much to offer these sorts of operations.

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But military organizations also have special needs. And IoT presents special risks to the military as well, as evidenced by recent reports that a hidden military base was revealed by a fitness app innocently used by a soldier who frequently jogged on and around the base.

To be fit for military use, IoT must be implemented with some differences from infrastructure supporting IoT in the private sector. Hacking is always a threat. Military IoT should not depend on centralized, publicly accessible clouds. Secure networks are essential. The U.S. has already created a classified data network that spans 48,000 miles. Using consumer-oriented devices will be unacceptable for many military scenarios. Such devices are not made to support protocols used in the military, and are not up to its stringent standards.

Nonetheless, the special needs of military organizations can be served by IoT technologies, even if they're not always provided in the same form as they are for consumer products.

Matériel

Matériel such as tanks, aircraft, drones, ships, submarines, and satellites may be computer controlled and network connected. Vehicles can form a peer-to-peer network within seconds of forming a convoy. Data can be communicated between vehicles on the ground, in the air, and at sea, and with bases of operation throughout the world.

Logistics

IoT has already improved the military's logistics and supply-chain management. In war situations, IoT's logistics capabilities enable tracking of supplies and equipment from their source to where they are needed on the battlefield.

Connected Warfighter

Soldiers in combat can wear devices to enable them to communicate and coordinate with each other. The small size and low power requirements of IoT sensing and communications devices makes them ideal for use in forward situations. Soldiers are provided with enhanced situational awareness. Combined with aerial drones, satellite images, and night vision, soldiers literally have a bird's-eye view across hostile terrain. Internetworking capabilities enable information to be relayed from soldiers' sensors and communication channels back to command and support facilities.

The armed forces collect data from a range of different platforms, including aircraft, weapon systems, ground vehicles, and troops in the field. Through IoT analytics, the military can increase the effectiveness of their intelligence, surveillance, and reconnaissance systems, enabling armed forces to identify key threats faster and with more accuracy.

Case Studies

Case studies describing IoT projects involving security and public safety may be found at the following URLs.

- Rugged and Reliable IoT Device Integration

  https://objectcomputing.com/case-studies/rugged-reliable-iot-device-integration
- A systemic and cognitive vision for IoT security: A case study of military live simulation and security challenges

  https://hal.archives-ouvertes.fr/hal-01478323/document

Connected Services

Suppose you buy a watch. Your relationship with the store that sold you the watch and the company that manufactured the watch may end the moment you leave the store. Of course, you might contact them if you have a problem with the watch. And some day, you might want to buy a new watch. If you had a good experience with the previous watch, you might buy your new watch from the same store or company. It's hard to get new customers, so many organizations look for ways to retain customers and build loyalty by establishing a relationship that goes beyond the sales transaction.

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Connected devices can "phone home" to services provided by their manufacturer, so the device literally maintains a connection between the manufacturer and the customer. Of course, there must be a benefit—some sort of services provided—for customers to allow the connection. This is what is known as ***connected services***.

In the case of a connected fitness watch, connected services might include such things as:

- A dashboard for viewing data collected by the watch over several months, such as your heart-rate statistics, steps walked, calories burned, and so forth.
- Social networking with other watch users, such as competitions to see who gets the most exercise.
- Access to an app store, from which you can download apps and custom clock faces to the watch.
- Recommendations on exercise.
- The ability to examine watch data on other devices, such as computers, tablets, and smartphones.

Products that include connected services help to establish a relationship with customers, and provide more value to customers than a physical product alone. They also enable companies to learn more about their customers, which can help them make improvements to products and services over time.

Guidelines for Using IoT to Solve Problems

Follow these guidelines to use IoT to solve problems.



Note: All of the Guidelines for this lesson are available as checklists from the **Checklist** tile on the CHOICE Course screen.

Identify Goals for an IoT Implementation

When identifying your goals for your IoT project, look for opportunities to accomplish the following:

- Improve decision making.
- Understand customers better.
- Deliver new value to customers.
- Make operations more efficient and effective.
- Reduce adverse impact on the natural environment.
- Improve the bottom line and increase business value.

Identify Problems in Your Business That IoT Can Help to Solve

When planning an IoT solution, start by identifying problems your business needs to solve. Then consider which of the following questions apply to each problem. If the question applies to the problem, record your answer, which may become part of the requirements for your IoT solution.

- Can we obtain real-time data that would help us respond to this problem faster or better?
- Would it help if we could accomplish work in remote locations with less human intervention? (If "yes," then ask the following questions to determine what sorts of work should be accomplished.)
  - Is there data we should collect?
  - Is there data that should be analyzed immediately at the collection point to enable corrective actions to be taken using process automation?
  - Is there data that should be forwarded to the cloud for deeper analysis?
  - Are there automated processes that should be guided by a person in a different location (remote control)?
  - Are there automated processes that should be performed by a robot or process automation (autonomous)?
- Would it help to collect and analyze data about any of the following? (If "yes," then note which of the following information areas would be useful to have.)

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- Location and status of raw materials throughout the entire supply chain.
- Operations performed in your organization.
- Your equipment, assets, and inventory.
- How your operations impact the natural environment.
- How your customers interact with products.
- Customer shopping behavior.

Use Models to Design a Solution

Use a good reference architecture as a general model for your own projects, such as:

- IEEE P2413.2 Standard for a Reference Architecture for Power Distribution IoT (PDIoT): **https://standards.ieee.org/ieee/2413.2/7719/**
- Industrial Internet Reference Architecture (IIRA): **https://www.iiconsortium.org/IIRA.htm**

Use Vendor References to Guide Your Design

When basing an IoT solution on a particular vendor's platform, take time to learn the practices vendors recommend for their platform. Examples of vendor-specific guides include:

- AWS Pragma architecture: **https://aws.amazon.com/blogs/startups/iot-a-small-things-primer/**
- IBM IoT reference architecture: **https://www.ibm.com/cloud/architecture/architectures/iotArchitecture/reference-architecture/**
- Intel® IoT Platform reference architecture: **https://www.intel.com.au/content/www/au/en/internet-of-things/white-papers/iot-platform-reference-architecture-paper.html**
- Microsoft Azure IoT reference architecture: **https://azure.microsoft.com/en-au/blog/azure-iot-reference-architecture-update/**

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ACTIVITY 2-1
Using IoT to Solve Problems

Data File

C:\CNX0480Data\Undertaking an IoT Project\Using IoT to Solve Problems.docx

Scenario

Best Elevators specializes in elevators with a reputation for high quality and dependable customer service. Competition is on the rise and the company is constantly at risk of losing market share to competitors.

The company has identified several areas that have been a problem for them in the past. They are hoping to address these problem areas through IoT. They have expressed the problems as goals for improvement.

- Increase customer loyalty.
- Improve service maintenance and predictive maintenance.
- Manage spare-parts inventory more efficiently.
- Improve product design and technician training.
- Improve uptime and field service efficiency.
- Allocate scarce service technicians more efficiently.
- Communicate more effectively with suppliers.

In the spaces provided, write your ideas regarding ways that an IoT solution might address these problems. (If you'd prefer to type your ideas in a Word document, a worksheet is provided in the aforementioned data file location.)

---

1. How might IoT help Best Elevators increase customer loyalty?

2. How might IoT help Best Elevators improve service maintenance and predictive maintenance?

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3. How might IoT help Best Elevators manage spare-parts inventory more efficiently?

4. How might IoT help Best Elevators improve product design and technician training?

5. How might IoT help Best Elevators improve uptime and field service efficiency?

6. How might IoT help Best Elevators allocate scarce service technicians more efficiently?

7. How might IoT help Best Elevators communicate more effectively with suppliers?

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TOPIC B

The IoT Development Lifecycle

Your organization has decided to undertake an IoT project. To avoid the risks inherent in such a project, the organization should be sure to implement effective project management and development methodologies throughout the process.

Complexity of IoT Projects

According to a Cisco survey (**https://newsroom.cisco.com/press-release-content?type=webcontent&articleId=1847422**), a majority of IoT projects get bogged down at the proof-of-concept stage, with only 26% of completed IoT projects considered a success by those who implement them. IoT projects are often complex. IoT projects converge different technologies from different vendors involving different platforms. Some of the products involved in a solution may target consumers, while others target business or industry, and even custom electronic components may be involved.

Furthermore, those implementing IoT solutions often need to learn new approaches to network design, security, and business strategies. They may need to work with new business or technology partners. And they may be exposed to regulations and standards they never had to deal with before.

There may be significant payback on IoT projects, but there are also significant risks. Because of this, organizations should be careful to employ systematic development processes when working on IoT projects.

The IoT Development Lifecycle

Various models have been defined to describe essential phases in the ***system development lifecycle (SDLC)*** of an IT system or project. While different models exist, they all tend to promote the idea of following a methodology to ensure no important steps are left out at any phase, with the goal of ensuring quality (including cybersecurity, privacy, and safety), operating efficiently, and reducing costs.

The models tend to encompass some form of the following phases.

- **Initiate**: A need is identified, initial goals are set, and the project is initially staffed with those responsible for planning it.
- **Plan**: Planning may include preliminary analysis and development of a system concept. Alternative solutions are examined along with a prediction of costs and benefits. The preliminary analysis may include clarification of the problem and objectives. A decision may be made to leave the system as is, improve it, or develop a new system.
- **Analyze requirements**: Facts are gathered through documentation, interviews, observation, questionnaires, examination of existing systems, and other sources. These facts are interpreted to translate project goals into functional and quality ("non-functional") requirements for the project, with a focus on describing the desired outcomes, rather than the design of the solution.
- **Design**: Based on requirements, a solution is identified and recorded in documentation such as screen layouts, business rules, process diagrams, data-flow diagrams, and pseudocode to describe the desired features and operations in detail to those who will develop them.
- **Develop**: Based on requirements and design documents, code is written and other resources (data, images, etc.) are developed for use in the solution.
- **Integrate and test**: All of the pieces are brought together in an environment where they can be tested and checked for errors, bugs, security, privacy, safety, and interoperability with other systems. This phase ends when requirements have all been met, the project is accepted by

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stakeholders, and development is considered complete. The system is authorized for implementation.
- **Implement**: The system is installed and deployed. It is put into production, and actual business begins operation on the system.
- **Operate and maintain**: The system is monitored and periodically assessed to ensure it maintains proper functioning and performance, continues to meet requirements, and does not become obsolete. Periodic updates must be performed.
- **Dispose**: In this phase, plans are developed for discarding system information, hardware, and software in making the transition to a new system. The purpose here is to properly move, archive, discard, or destroy information, as well as hardware and software that is being replaced, in a manner that prevents any possibility of unauthorized disclosure of sensitive data. The disposal activities ensure proper migration to a new system. Particular emphasis is given to proper preservation and archiving of data processed by the previous system. All of this should be done in accordance with the organization's security requirements.

As part of the initial planning process, all of these phases should be planned. For example, the planning process might identify potential problems that could occur when the system is eventually retired or replaced. By identifying these problems upfront, the system can be designed so that these problems don't occur in the first place, which is much better than having to come up with a solution later on.

Return on Investment

One of the greatest challenges reported by enterprises planning and deploying IoT projects is demonstrating a return on investment. IoT presents a wide array of interesting use cases for business, but it can be difficult to estimate the economic value of these benefits. Likewise, it can be hard to estimate the cost of numerous challenges that need to be addressed, such as network design, cybersecurity, aggregating data from many sources and data formats, and bringing them all together to allow complex analytics and decisions to be made.

Focus on Business Goals

IoT is transformative. Numerous organizations have already implemented IoT solutions and are reporting their successes with it. It is easy to get tempted by the hype, focusing on the interesting possibilities of IoT and looking for ways to shoehorn them into your business process, when instead you should focus on specific problems your organization needs to address, and then consider how IoT and other key technologies might help you address the issues you have identified.

Cybersecurity Throughout the IoT Lifecycle

Unfortunately, in many IT projects, cybersecurity may be dealt with as an afterthought. But when security problems are late in development or after the system is deployed, security problems may have expensive consequences and at the very least are more expensive to correct than if they are found and corrected early in the project.

As with any aspect of software quality, to ensure successful implementation, security should be dealt with in every phase of the development lifecycle. From the very start, as you plan and identify requirements, security requirements should be identified. Those requirements should be designed and developed into the solution, and testing should verify that the requirements have been met. As you deploy and maintain the system over time, security should be monitored and necessary updates applied as needed to maintain security.

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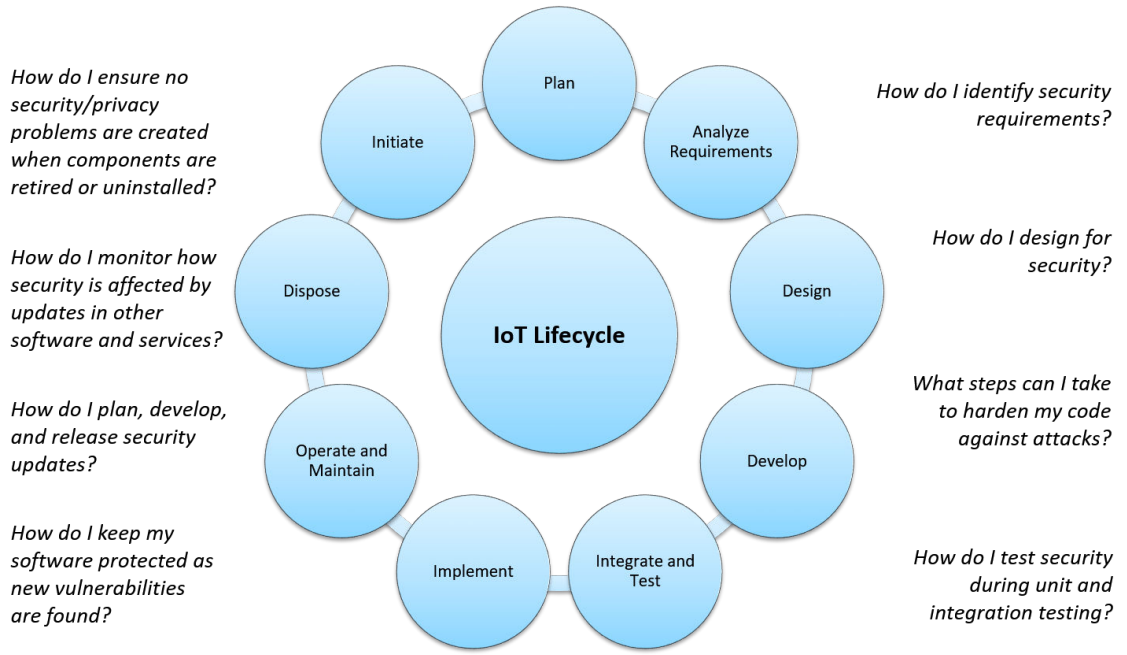


Figure 2-11: Cybersecurity throughout the IoT lifecycle.

Guidelines to Prepare for an IoT Initiative

Follow these guidelines to prepare for an IoT initiative.

Prepare for an IoT Initiative

Even before your organization commits to undertaking a major IoT project, there are steps you can take to prepare the organization, and by starting small, you can minimize risks and help to ensure the successful adoption of IoT.

- **Understand the organization's current state**: An early part of the planning process is to identify the results you're looking for. This requires that you understand the current state of the organization's business processes, know where there is need for improvement, and what type of information and process deficits are involved. This will help to reveal where new information flows and process automation provided by IoT can help.
- **Perform pilot projects and experimentation**: Starting small can help the organization to avoid risks involved in climbing a learning curve on a major project. Consider developing a proof of concept to test ideas first on a small scale. You can use small pilot projects to minimize risk, develop and improve your internal processes, evaluate vendors and products, and refine integration with your other systems. As you gain successes on smaller projects, you can scale them up over time. Pilot projects give you better data regarding costs and benefits, which you can use to estimate your return on investment on larger projects.
- **Manage change and corporate culture**: As you plan your IoT project strategy and pilot small projects, you can identify and eliminate gaps in your infrastructure, core skills, tools, and resources. By avoiding large failures, you can gain acceptance of IoT throughout the organization. As the use of IoT increases, involve more groups and individuals, sharing successes and lessons learned.

Follow Best Practices for IoT Development

To help ensure your success in your IoT projects:

- Work with project stakeholders to define business goals and identify requirements for IoT projects that will help the organization reach those goals.

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- Build security and privacy into every phase of development, rather than treating them as an afterthought. For example, include threat modeling early in the design process to plan countermeasures and defenses in advance.
- Employ a dedicated IoT architect to provide central coordination of IoT projects and provide guidance on design and best practices.
- Update skillsets and job descriptions as needed throughout the organization to distribute responsibility for supporting IoT systems.
- Include the IoT architect, information technology (IT), operational technology (OT), and business groups within the organization when identifying, planning, and documenting the organization's target architecture for IoT.
- Honestly appraise the organization's legacy infrastructure to determine whether it can meet the fundamentally different requirements of IoT, and if not, where it needs to be upgraded or replaced.
- Honestly appraise the organization's capabilities, and do not try to implement IoT projects relying solely on existing internal resources and IT generalists.
- Provide staff with training opportunities and add staff as needed. Work with trusted partners or vendors who can fill in your organization's gaps in IoT expertise.
- Utilize staff and resources from the organization's IT, OT, and business groups when evaluating potential IoT partners, vendors, and products.
- Design the entire solution as a system. Ensure the team understands the capabilities of all components within the system, and configure them as required to keep the overall solution as simple as possible, and configured correctly to meet performance, security, and other requirements.

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ACTIVITY 2-2
Preparing for an IoT Initiative

Scenario
Working with your colleagues at Best Elevators, you have identified how you will address various problem areas within the company's operations. The following is a summary of the solution.

As for the first problem, *increasing customer loyalty*, the company wants to establish a closer relationship with customers, provide services to help customers meet their own needs, and provide them with peace of mind that they associate with using Best Elevators' products and services.

They plan to achieve this goal by quickly moving to a *connected-services* strategy for their elevator product lines. When a company chooses to use Best Elevators, they will not only buy a high-quality product that provides good value for the money spent, but they will also invest in services that make Best Elevators a better choice than its competitors.

Each elevator will come with services that include:

- **Continuous monitoring of elevator safety and performance factors**, with data viewable through applications provided by Best Elevators.
- **Comprehensive data dashboards** to show the status of all elevator systems in real time, including usage reports and other information facility managers will find useful.
- **Faster maintenance and repairs**, made possible through better information, and the ability to anticipate problems before they occur and before they lead to other problems.
- **Reduced elevator downtime**, made possible through continuous monitoring and real-time notifications when problems are anticipated.

As IoT systems are installed and used, the other problem areas will be addressed in phases, as follows.

Problem Areas	*How Addressed Through IoT*
• Improve service maintenance and predictive maintenance	Initially, Best Elevators wants to provide its customers with data collection from sensors, and analytics to improve service maintenance and predictive maintenance. The IoT solution must provide early-warning data and remote diagnostics to enable the organization to dispatch technicians promptly, ensuring they are equipped with the necessary information, tools, and parts to restore the equipment to safe operating conditions quickly.
• Manage spare parts inventory more efficiently	IoT can be very helpful for keeping spare part inventories in warehouses and service vehicles stocked appropriately, which is normally a notoriously difficult task to manage cost-effectively. On the supply side, inventory can be tracked using typical IoT devices such as RFID tags. On the demand side, elevator equipment can be monitored and analyzed using data analytics and Best Elevators' own domain expertise to predict future needs. Combining historical data of part failure rates with analytic algorithms, Best Elevators can optimize their part stocks in the warehouse and technicians' service vehicles, reducing inventory costs and improving repair time.

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Problem Areas	*How Addressed Through IoT*
• Improve product design and technician training • Improve uptime and field service efficiency • Allocate scarce service technicians more efficiently	Once IoT systems are in place, the system will be able to collect and analyze usage patterns over time. This information can be used to improve product design and technician training, which will further improve uptime and field service efficiency, and allow more efficient allocation of scarce service technicians.
• Communicate more effectively with suppliers	Parts distributors and value-added resellers will also see additional opportunities to leverage the insight garnered from the IoT connected-service solution. Best Elevators can share its experience and knowledge with equipment manufacturers, creating a value-based competitive advantage in a market that is highly cost-driven with pricing volatility.

At this point, Best Elevators has identified the problems they need to address and have set goals for the project. Staff have been identified to carry out the project. A preliminary analysis has been performed, and a general system concept has been described (summarized previously). After much deliberation, costs and benefits have been identified, and the team has decided to proceed with the IoT implementation.

1. What would need to be done next in this project?

2. What sorts of change-management issues might arise as the company shifts from primarily selling physical goods to selling services?

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Summary

In this lesson, you examined how IoT technologies can help to resolve real-world problems, and you identified issues you'll need to consider throughout the IoT development lifecycle.

Where does IoT have the greatest potential to benefit your organization?

What steps should your organization take to ensure its success in IoT projects?



Note: Check your CHOICE Course screen for opportunities to interact with your classmates, peers, and the larger CHOICE online community about the topics covered in this course or other topics you are interested in. From the Course screen you can also access available resources for a more continuous learning experience.

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3 Data Science Fundamentals

Lesson Time: 1 hour, 15 minutes

Lesson Introduction

In today's business environment, everyone is talking about data science and how it can transform success from mild to wild. But what exactly do the pundits mean by data science? Before an organization can take full advantage of the benefits of the data it accumulates, individuals at every level of the organization need to have some basic knowledge of what data science is, what types of data are readily (and not so readily) available, and the phases of the data science lifecycle.

Lesson Objectives

In this lesson, you will:

- Explain concepts and terminology related to data science.
- Describe the types of data normally associated with data science.
- Describe the data science lifecycle.

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TOPIC A

What Is Data Science?

Emerging trends in business and technology almost always generate an abundance of new buzzwords. To grasp the full potential of new ideas and how they might benefit your organization, you must cut through the fog and discover the information you need to discuss these ideas clearly and intelligently. In this topic, you will explain concepts and terminology related to data science.

Data Science

Data science is the scientific study of data to gain knowledge. It combines the fields of mathematics, statistics, machine learning, and computer science to collect, analyze, and interpret large amounts of potentially diverse data, with the end goal to provide insights and other information that can help improve an organization's decision making, strategic planning, and overall operations.

Data science involves:

- **Accumulating data**. This includes concepts like big data, data types, and data preparation.
- **Analyzing data**. This includes data analytics and machine learning.
- **Extracting value from data**. This also includes data analytics and machine learning.
- **Presenting the value in a meaningful way**. This includes creating visualizations and reports to share with stakeholders.

The Elements of Data Science

You can think of data science as an umbrella term, which encompasses several functions, including:

- Data analysis, or the detailed examination of data.
- Data analytics, the systematic computational analysis of data or statistics.
- Data cleaning, or preparation of data for analysis.
- Statistical analysis, or the interpretation of data to discover patterns and trends.
- Data engineering, or the creation of systems to collect and store data.
- Machine learning (ML) engineering, or the implementation of statistical or ML models into a live production system.

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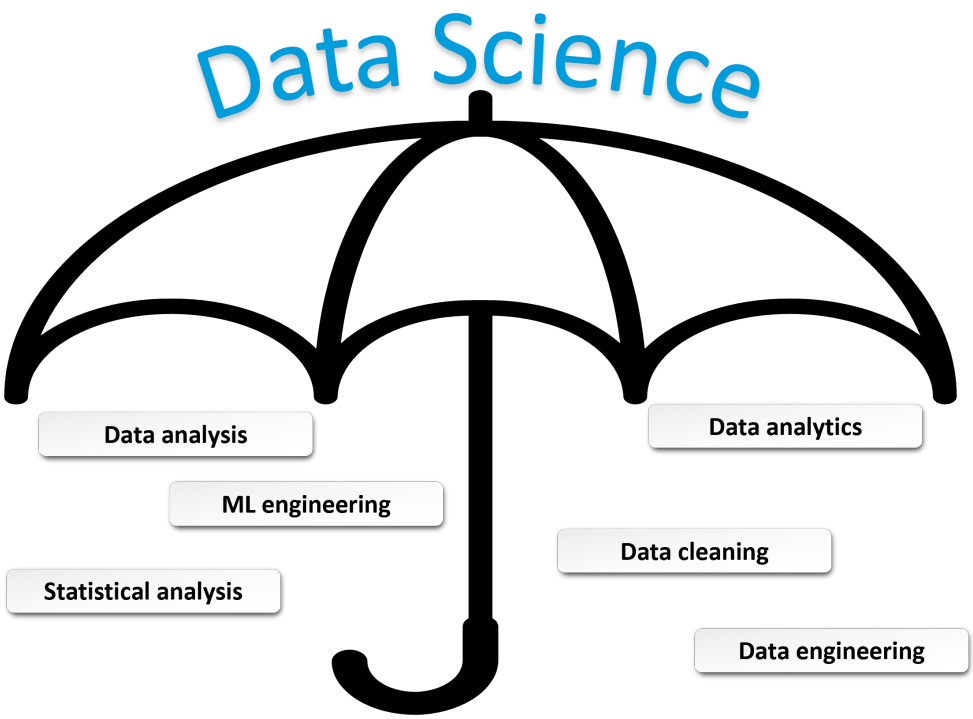


Figure 3-1: Data science encompasses several functions.

Data Teams

You might ask yourself, "How can anyone have all the requisite skills to accomplish all of the analysis and other work involved in data science?" It would be impossible for one individual to excel in each of the disciplines under the data science umbrella. A ***data team*** is a group of individuals involved in the strategic development, implementation, and management of business analytics technologies, processes, and the overall success of business analytics at the organization. Some key data team members and their roles are described in the following table.

Role	*Responsibilities*	*Knowledge, Skills, and Experience*
Data scientist	Cleans, massages, and organizes data.	• Knowledge of languages like R, Python®, MATLAB®, Structured Query Language (SQL), Hive™, Pig™, and Spark™ is a must. • Skills in mathematics, statistics, and machine learning are also required. • Most data scientists have experience with distributed computing, predictive modeling, and applying visualization techniques.

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Role	Responsibilities	Knowledge, Skills, and Experience
Data architect	Designs data management systems that integrate, centralize, protect, and maintain data resources.	• Knowledge of languages like SQL, Extensible Markup Language (XML), Hive, Pig, and Spark is a must. • Skills in data modeling and systems development are also required. • Most data architects have experience with data warehousing solutions; spreadsheet tools; extract, transform, and load (ETL) tools; and BI tools.
Statistician	Collects, analyzes, and interprets qualitative and quantitative data by using statistical theories and methods.	• Knowledge of statistical theory and methodology, as well as languages like R, SQL, SAS®, SPSS®, MATLAB, Python, Perl, Hive, Pig, and Spark is a must. • Skills in data mining and distributed computing are also required. • Most statisticians have experience with database systems, machine learning, and cloud tools.
Database administrator	Manages and maintains databases.	• Knowledge of languages like SQL, Java®, XML, Ruby, and Python is a must, along with enterprise resource planning (ERP) and business knowledge. • Skills in data modeling and design are also required. • Most database administrators have experience with database systems, distributed computing, data security solutions, and backup and recovery solutions.
Data engineer	Develops, tests, and maintains architectures for databases and large-scale processing systems.	• Knowledge of languages like SQL, Hive, Pig, R, Python, MATLAB, SAS, and SPSS is a must. • Skills in data modeling, data API development, and implementation are also required. • Most data engineers have experience with database systems, ETL tools, and data warehousing solutions.

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Role	Responsibilities	Knowledge, Skills, and Experience
Data analyst	Collects and processes data and performs statistical data analysis.	• Knowledge of languages like R, Python, HTML, SQL, JavaScript®, and C/C++® is a must. • Skills in mathematics, statistics, and machine learning are also required. • Most data analysts have experience with spreadsheet tools, database systems, and applying visualization techniques.
ML engineer	Feeds data into the models developed by the data scientist, and scales the theoretical models into production where they can handle massive quantities of real-time data.	• Knowledge of languages like R, Python, C/C++, and Java is a must. Knowledge of applied mathematics and machine learning approaches and in-depth technical knowledge of data science algorithms is also essential. • Skills in computer programming are also required. • Most data engineers have experience with distributed computing.
Data-team manager	Provides overall leadership for the data team, communicates with stakeholders, and manages data science projects.	• Knowledge of business goals and domain-specific information is a must. • Skills in problem solving, analysis, and communication are also required. • Most data-team managers have experience in management and project management.

Required Skills and Knowledge

Another way to look at data science is to examine the skills and knowledge required to perform data science, including:

- **Mathematical skills such as statistical analysis**. The following roles commonly contribute mathematical skills:
  - Data scientist
  - Statistician
  - Data analyst
- **Computing skills such as coding or programming**. The following roles commonly contribute computing skills:
  - Data scientist
  - Data architect
  - Statistician
  - Database administrator
  - Data engineer
  - Data analyst
  - ML engineer

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- **Industry or domain knowledge**. The following roles commonly contribute domain knowledge:
  - Database administrator
  - Data-team manager

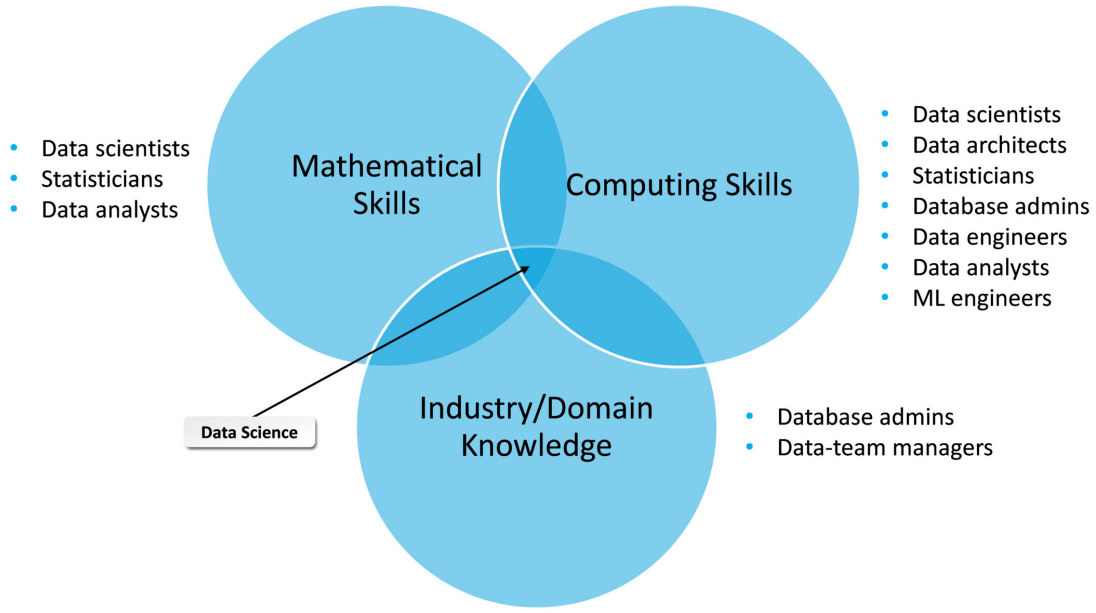


Figure 3-2: Data science requires mathematical skills, computing skills, and domain knowledge.

Data Analytics

Data analytics is the science of examining raw data to derive insights and conclusions that help businesses make better decisions or to verify or disprove existing models or theories. Analytics is one of the mathematical parts of the data science equation. There are four generally accepted types of analytics that, when used together, can help you forecast more accurately:

- Descriptive analytics
- Diagnostic analytics
- Predictive analytics
- Prescriptive analytics

Descriptive Analytics

Descriptive analytics is the form of data analytics that uses data to summarize past behaviors. Past behaviors can be anything that occurred before the moment in question, whether it is one minute ago or one or more years ago. You can think of descriptive analytics as providing the answer to the question "What has happened?"

The majority of analytics fall into this category, such as basic mathematical operations like summations and averages. Often, the underlying data is filtered and manipulated before the basic math is applied. Examples include things like total stock in inventory, average sales per department, and month-over-month sales growth. These are commonly included in reports that provide historical insights regarding the company's production, financials, operations, sales, finance, inventory, and customers.

Use descriptive analytics when you need to understand what is going on in your company at a high level, and when you want to summarize and describe different aspects of your business.

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Types of Descriptive Analytics

Commonly used types of descriptive analytics include:

- ***Business intelligence (BI)***: A recognized industry term for organizational analytics, including past, present, and future views of business operations. For descriptive analytics, the analysis of historical data is key.
- ***Data mining***: The process of pulling actionable insight out of a set of data and putting it to good use. Descriptive data mining identifies patterns and provides information on past or recent events.
- ***Data visualization***: The practice of communicating meaningful data in a visual format, such as graphics, charts, and dashboards.



Figure 3-3: Charts used in data visualization.

Diagnostic Analytics

Diagnostic analytics is a form of data analytics that focuses on examining past data to identify the causes of specific events. It goes a step beyond descriptive analysis to provide insight into the reasons behind past behaviors. You can think of diagnostic analytics as providing the answer to the question "Why did this happen?" Another term used for diagnostic analytics is ***root-cause analysis***.

Common examples of diagnostic analytics include:

- Analyzing market demand. It's common for some products to be more popular than others. Determining the reasons behind extreme popularity (or unpopularity) makes it easier to set up future product offerings to be more successful.
- Identifying reasons behind customer churn. In most businesses, it is more cost effective to retain customers than to find new ones. By examining the reasons why a customer decides not to purchase from your company, you may be able to identify actions the company can take to increase customer satisfaction and retention.

Types of Diagnostic Analytics

Common types of diagnostic analytics include:

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- **Identifying reasons behind changes in market demand**. When descriptive analytics shows a change in market demand, you can use diagnostic analytics to determine factors that correlate to the change, and then investigate those factors to learn more about the increase or decrease in demand.
- **Explaining customer behavior**. If a company collects customer data, it can analyze that data to provide a better understanding of customer behavior. For instance, companies that provide subscription services may solicit information from customers who want to cancel their subscriptions. If numerous customers cite the same reason for canceling, the company can focus on changing or improving that factor to help retain customers.
- **Identifying technology issues**. This is probably the most familiar use of diagnostic analytics. Virtually anyone who has used a computer has run some sort of software that tests the device's configuration and performance, identifies potential problems, and suggests actions to resolve the identified issues.

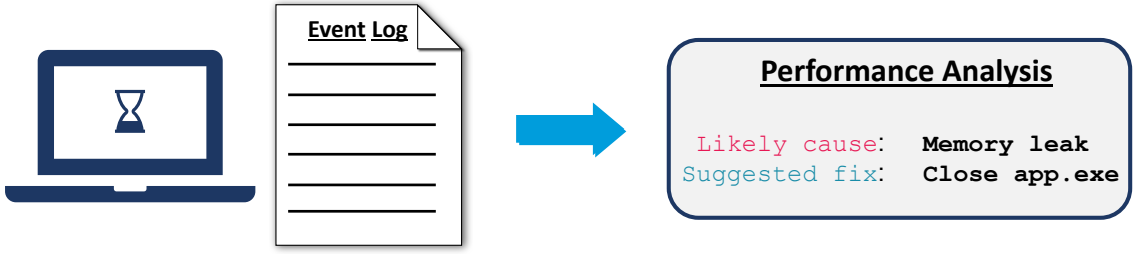


Figure 3-4: The data in a device's event log is used to diagnose performance issues.

Predictive Analytics

Predictive analytics is a form of data analytics that uses data to forecast future behaviors. It provides estimates of the likelihood of a future outcome based on existing data, statistical models and algorithms, and probability theory to interpolate missing information or to extrapolate what might happen in the future. You can think of predictive analytics as providing the answer to the question "What could happen in the future?"

Predictive analytics is often used for forecasting in various parts of an organization. For instance, it can be applied to:

- Anticipating customer behavior.
- Identifying purchasing patterns.
- Tracking sales trends.
- Predicting inventory levels.
- Predicting staffing needs to support operations.
- Anticipating the need for hardware and software purchases.

Types of Predictive Analytics

Commonly used types of predictive analytics include:

- **Predictive modeling**: The practice of analyzing past performance to assess how likely a customer is to exhibit a specific behavior in the future, or how well a product will be accepted by the marketplace.
- **Root-cause analysis**: A problem-solving method that is used for identifying the underlying causes of faults or problems.
- **Data mining**: The process of sorting through large datasets to identify patterns and establish relationships to solve problems. Predictive data mining suggests future results based on past events.
- **Pattern identification and alerts**: The automatic discovery of similarities in data by using algorithms. Alerts can be raised when anomalies in the pattern arise.

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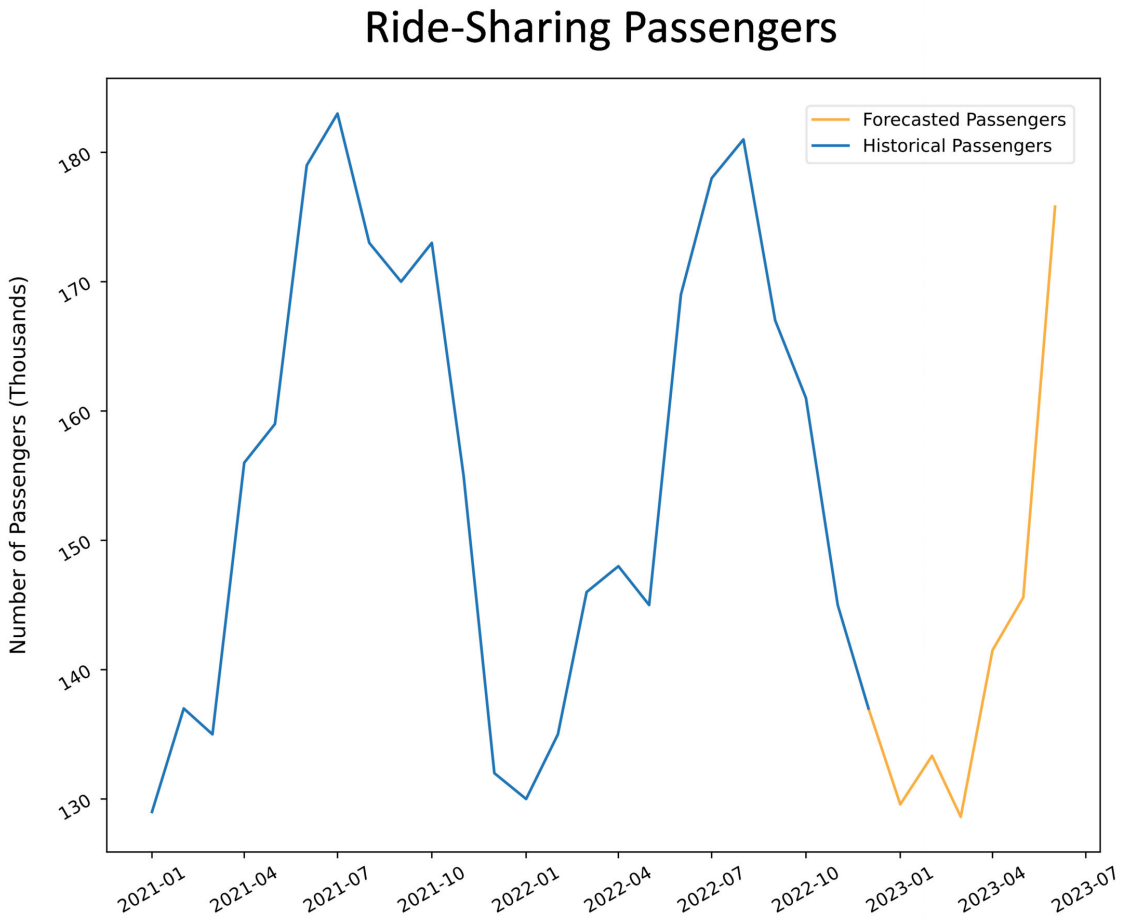


Figure 3-5: Predicting the number of passengers for a ride-sharing service based on trends in past data.

Prescriptive Analytics

Prescriptive analytics is a form of data analytics that uses data to predict multiple possible outcomes and then offers recommendations for actions based on those predictions. Going beyond the bounds of predictive analytics, it provides the answer to the question "What should we do?"

Business rules, algorithms, machine learning, and computational modeling are all used in prescriptive analytics. Data inputs include historical data, transactional data, real-time data, and big data.

Types of Prescriptive Analytics

Commonly used types of prescriptive analytics include:

- **Simulation**. In the context of prescriptive analytics, creating a simulation involves building a digital replica, or model, to represent a business process or other scenario on which you want to perform prescriptive analytics. The model should include details that remain constant while different configurations are applied. The different results are recorded and compared to one another by using machine learning algorithms. The results are then evaluated against an established baseline to determine a scoring scheme that data scientists can use to determine the best approach to solving the original business problem. You might use a simulation to determine the best layout for a warehouse or retail operation.
- **Optimization**. In the context of prescriptive analytics, optimization is the creation of a mathematical model that includes variables and equations that, when solved, provide the best

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solution to a business problem. In some instances, the time it takes for the model to be solved may be excessive, given the variables involved. And some issues, such as equipment failure or lack of raw materials, are virtually impossible to factor into a mathematical model. You might use an optimization model to determine the best delivery route for a truck driver.

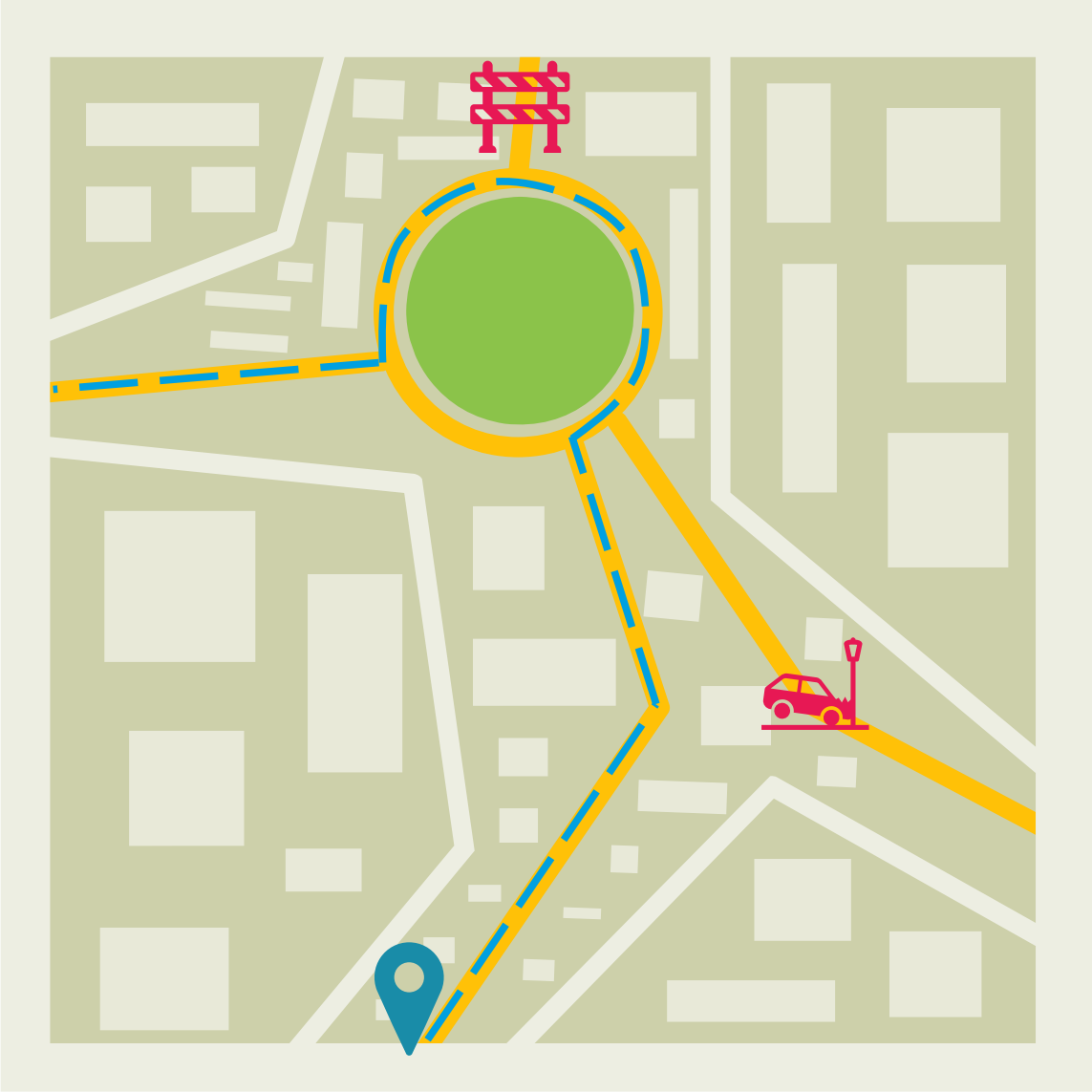


Figure 3-6: Using prescriptive analytics to suggest an optimal route through a city given various factors.

Statistical Analysis Concepts

To better understand how various forms of analytics work, you should review a few related concepts. They are described in this table.

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Concept	Description
Statistical modeling	The combination of statistical assumptions and mathematical models that describes the generation of sample data. Two common types of models are linear models and Bayesian models. • Linear models describe a continuous response variable as a function of one or more predictor variables. • Bayesian models build on the existing knowledge about a population to predict or estimate unknown variables.
Statistical inference	The process of using data analysis to predict the properties of a set of data. This is sometimes referred to as ***computational statistical inference***.
Hypothesis testing	A statistical inference process that enables you to prove (or disprove) an assumption. For diagnostic analytics, construct hypotheses that act as a reminder of what you are aiming to discover.
Regression analysis	A statistical inference method that estimates the relationship between a dependent variable and one or more independent variables, often in a graphical format. Primarily used for prediction and forecasting, it can also be used for identifying causal relationships.
Linear regression	Describes the relationship between variables by deriving a linear equation based on existing data. Two forms of linear regression are commonly used: single linear, which estimates the relationship between one dependent and one independent variable, and multiple linear, which estimates the relationship between one dependent and more than one independent variables.
Correlation and causation	A ***correlation*** is a statistical association between two or more variables, while ***causation*** means that a change in one variable has a direct effect on another variable. There are many instances where a correlation exists, but changes in either variable do not affect the other (no causation).
Sampling	The process of studying a population through the gathering and analysis of information about the population.
Distribution of data	The spread of individual data points across all possible values for a specified variable. Distributions can be categorized as displaying frequency distribution, central tendency, and variability. • Frequency distribution displays how often data occurs. When data values are plotted on a graph, duplicate values indicate more commonly occurring values. The best-known frequency distribution is probably the bell curve, which is often referred to as the normal distribution. • Central tendency displays the data's center point of distribution. • The variability of a dataset displays its degree of dispersion, or how far the data is spread out.



Note: A more in-depth exploration of statistical analysis is beyond the scope of this course.

Sampling Techniques

The following tables list commonly used sampling techniques according to their methods.

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Probability Sampling Technique	Description
Simple random sampling	Any item has an equal chance of being selected as a sample. Also known as representative sampling.
Systematic sampling	An initial item is selected, and subsequent items are selected by using a sample interval. For example, if you select the third item of a population of 500, and the sample interval is 50, then items 3, 53, 103, and so forth will be included in the sample.
Stratified sampling	The total population is divided into groups based on selected characteristics, and items are randomly selected from each group.
Clustered sampling	One or more clusters, or subsets, of items are selected from the total population, and items are randomly selected from the cluster(s).

Non-Probability Sampling Technique	Description
Convenience sampling	Items are selected directly from the population expressly because of their availability to the researcher.
Consecutive sampling	Items are selected and results are analyzed; then, additional items are selected and analyzed in sequence and as needed.
Quota sampling	Items are selected to represent the population based on specific characteristics, and groups are added to ensure the sample generally represents the population.
Judgmental sampling	Items are selected solely on the researcher's knowledge. Also known as purposive or authoritative sampling.
Snowball sampling	Items are selected based on unique and difficult-to-find characteristics, and selected items are asked to find other, similar items. Also known as chain-referral sampling.

Related Concepts and Technologies

Other concepts and technologies related to data science include business intelligence, artificial intelligence, deep learning, and computer programming tools and techniques. The following table summarizes these terms and their meanings.

Concept or Technology	Description
Artificial intelligence (AI)	The ability of machines to exhibit humanlike intelligence, as well as the scientific discipline concerned with this idea.
Algorithm	A process or set of rules for calculating results or solving problems.
Generative AI	A type of AI that can produce text, images, and other content based on information provided by the user.
Machine learning (ML)	An AI discipline where a machine can gradually improve its predictive capabilities without having received explicit instructions. ML is often used to conduct predictive analytics.

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Concept or Technology	Description
Deep learning (DL)	A type of ML that makes complex decisions based on multiple layers of information. DL is also used in predictive analytics.
Programming tools	Common programming languages used in data science include Python, R, and SQL. For Python, the NumPy and pandas libraries are the primary tools for numerical analysis.
Application programming interface (API)	A means for two or more computer programs to communicate. In data science, APIs support the extraction and sharing of data.
Web scraping	The process of using automation to extract content and data from a website.
Business intelligence (BI)	Technologies, applications, and practices that assist in the collection, integration, analysis, and reporting of information, with the goal of supporting decisions regarding business practices.

Putting It All Together

These concepts are all interrelated:

- Data science is one way to gain BI, with data analytics and data mining being tools to gather relevant information from data sources such as big data.
- Data analytics—particularly predictive and prescriptive analytics—can be automated by the use of AI.
- AI encompasses both machine learning and deep learning, but they are not the only disciplines that fall under the AI scope.

At the beginning of the topic, you saw that data science is a melding of mathematical skills, computer skills, and domain knowledge. Factoring in related concepts and technologies, you can get a better sense of how it all comes together.

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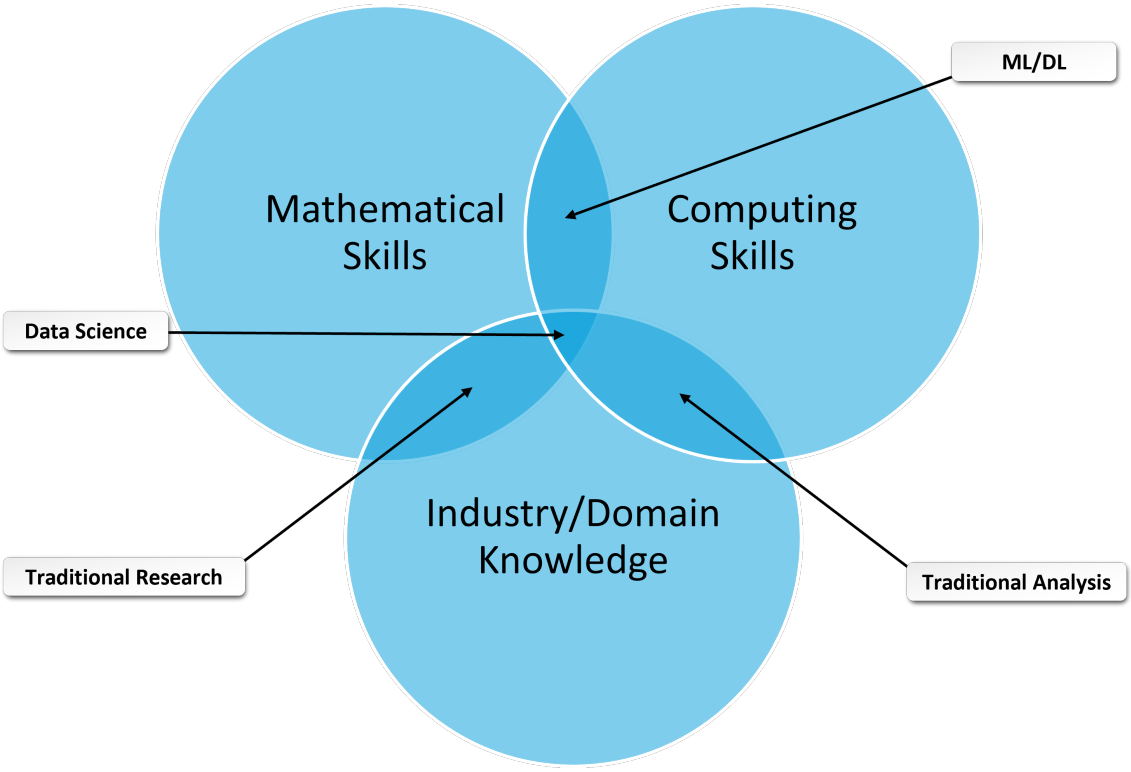


Figure 3-7: The data science Venn diagram.

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ACTIVITY 3-1

Discussing Data Science and Related Technologies

Scenario

Greene City Emporium (GCE) is a small chain of retail stores that operates in the fictitious city of Greene City, Richland, and surrounding municipalities. GCE sells a variety of products from many different manufacturers—everything from food, to housewares, to sporting goods, and more. The business is currently expanding into new territories within Richland, and it's also expanding its product catalog to meet the needs of its customers.

You are an assistant manager at the Esker Road store in nearby Olinger. You received an invitation to a cross-functional meeting to investigate the possibilities of implementing data science practices at GCE. You've heard a lot of buzz about data science, analytics, big data, and machine learning, so you are eager to attend the meeting to hear what your colleagues think about the proposal. You're also interested to learn if there might be new career opportunities for seasoned employees.

You are friends with Susan Sims, one of the IT technicians who works at the central office, so you arrange to have coffee with her two days before the meeting is scheduled. You are hoping to get a preview of what might be discussed during the meeting.

---

1. Susan begins by asking you how you would define data science.

2. You then spend a little time discussing the parts of data science. How would you describe the skills and knowledge required to perform data science?

3. You mention to Susan that part of your duties at the Esker Road store includes reviewing shrinkage reports to determine the percentage of loss due to theft each month. What type of analytics is this activity?

4. Susan asks if you are involved in setting up employee schedules for your store, and you agree that this is part of your responsibilities. What type of analytics is this most closely related to?

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5. As you wrap up your session with Susan, she suggests that if GCE is truly going to embrace a data-driven culture, other technologies will probably be needed. What types of technology is she referring to?

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ACTIVITY 3-2

Identifying Data Science Opportunities from a Dataset (Optional)

Data File

C:\CNX0018Data\Data Science Fundamentals\store_trial_data.csv

Scenario

GCE has finished running a trial in which several hundred customers volunteered to have their store purchases tracked, as well as some personal information, over a three-month period. This trial was designed to help reveal useful information about customers' shopping habits. More importantly, your team should be able to extract insights from the data that will guide the business into making decisions that better support GCE's customers, as well as increase sales.

Although you are not directly responsible for performing analysis on the data, you'll get a better sense of how this data can be valuable by taking a brief look at it. You'll get acquainted with the data and come up with some ideas for how various analytical techniques might make good use of it.

---

1. From the course data files, open the trial dataset.
   a) Open the **store_trial_data.csv** file using the spreadsheet program or text editor of your choice.

   Note: This is a text file in a comma-separated values (CSV) format. As the name implies, each value in the dataset is separated by a comma, ultimately forming rows and columns. The data will be easier to read in a spreadsheet program than a text editor.

   b) Verify that you can see the rows and columns of the dataset.
   Each row represents a transaction at the store. Each column is an attribute of that transaction.

2. Briefly review the data, focusing on the columns and what they reveal about each transaction.

   What kind of information about each transaction is recorded in the dataset?

3. **How might you use descriptive analytics with this dataset?**

4. **How might you use diagnostic analytics with this dataset?**

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5. How might you use predictive analytics with this dataset?

6. How might you use prescriptive analytics with this dataset?

7. When you're done, close the data file.

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TOPIC B

Types of Data

Now that you have a better idea of what data science is, it is time to look at the various types of data that can be involved in a data science project. Data can be characterized in different ways, depending on its source, its format, and the unique needs of the business.

Big Data

Big data are collections of data that are so large and complex that they require advanced tools to process and analyze them. Often, big data resides in enormous databases and is described in terms of three, four, five, or even six "Vs":

- **Volume**: Big data implies enormous volumes of data.
- **Velocity**: Working with big data requires you to deal with the massive and continuous flow of data from its various sources.
- **Variety**: Data comes from many sources and can be of many different types.
- **Veracity**: One of the pitfalls of working with big data is dealing with the bias, noise, and abnormality inherent in large amounts of data culled from different sources.
- **Validity**: It is imperative that the data being used is correct and accurate, or any conclusions that are made are subject to error.
- **Volatility**: You also need to determine how long a set of data is valid and how long it should be retained.

Structured, Unstructured, and Semi-Structured Data

Data comes in many forms and formats. One way to describe data is to identify the level of structure that bounds the various data points. Three common designations are structured, unstructured, and semi-structured data.

Structured data is data that contains elements that are easily accessible for effective analysis. The most common example of structured data is data that resides in a relational database, which is known for using tables with defined rows and columns, and keys that define relationships among the tables.

Unstructured data is data that is not organized in any established manner and does not conform to any data model. Examples include text-based data or data that contains both text and graphics.

Semi-structured data is data that does not reside in a relational database but includes organization characteristics that support some sort of analysis. Often, a semi-structured data format is ideal when space is a consideration. Common examples include email and documents based on a markup language such as XML or Hypertext Markup Language (HTML).



Figure 3-8: Structured, unstructured, and semi-structured data.

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What Makes Email Semi-Structured?

Email is considered semi-structured because information about the sender, receivers, date, and subject is structured, but the message contents are not structured.

Open and Proprietary Data

Data can also be categorized as open data or proprietary data.

Open data is data that is freely available for use. It can be shared or built on by anyone for any reason. Open data should be open in two ways:

- Legally open, meaning that the data is licensed to permit commercial and noncommercial use and reuse without any restrictions.
- Technically open, meaning that the data is available in a machine-readable format and can be processed by software applications.

Common forms of open data include census and economic information collected by governments, industry-specific information, and public data provided by companies like Kaggle™ Inc., Amazon Web Services (AWS™), and Google™.

Proprietary data is data that is owned and controlled by an individual, organization, or group. It can be protected by copyright, patent, or trade-secret legislation. Proprietary data might be:

- Internal private information such as your company's quarterly sales data, or
- Data that you can purchase such as from Gallup®, Acxiom®, and other data brokers.

Data Sources

Data can come from numerous sources—from structured relational databases to free-form social media posts and almost everywhere in between. Because data science can pull from so many different sources, it can be difficult to ascertain the veracity of each dataset. This can have a significant impact on the acquisition and preparation of the data.

Examples of data sources include:

- ***Transactional data***, or data that results from buying and selling products and services. This includes both online commerce and brick-and-mortar commerce and runs the gamut from invoices and payment orders to information stored in customer resource management (CRM), human resources management (HRM), and enterprise resource planning (ERP) systems.
- ***Machine data***, or data that is generated by industrial equipment, sensors in Internet of Things (IoT) devices, and even logs and counters that track visits to a web page.
- ***Social data***, or data that is garnered from social media (such as Twitter®, Facebook®, LinkedIn®, and YouTube™) or other websites.

Some data sources use a combination of these. For example, machine-sourced data is often derived from social data.

Data Repositories

Several terms are used to refer to the various types of repositories where data is collected and stored. Although these terms are meant to represent different concepts, you might find them used interchangeably.

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Data Repository	Description
Data lake	• **Purpose**: Machine learning, big data analytics, predictive analytics, and data discovery. Data could be used at any time or never at all. A specific purpose for keeping the data might not yet exist, but it is kept for possible future needs. • **Source**: Structured and unstructured data from a variety of sources, including sensors, websites, business apps, mobile apps, server logs, and so forth. • **Structure**: Varies widely. Data is typically retained in its original format until it needs to be used. Consuming and storing such data can be expensive and difficult.
Operational data store	• **Purpose**: Collects, aggregates, and prepares data for use in operational systems. May feed into a data warehouse. • **Source**: Transactional data captured from multiple enterprise applications and other sources. • **Structure**: Data is structured for fast and easy access, but additional preparation might be required prior to transferring the data to a warehouse.
Data warehouse	• **Purpose**: Business intelligence, batch reporting, and data visualization. • **Source**: Relational data captured from multiple sources such as applications, transactional systems, and operations databases. • **Structure**: Data is structured for fast and easy access.
Data mart	• **Purpose**: Data used by a particular department or business function to support analytics. • **Source**: A subset of the data warehouse contents that houses data specifically intended to support a particular department or business function. • **Structure**: Data is structured for fast and easy access.

Note that most of these repositories incorporate relational databases. Relational databases are not so much a separate repository as they are a technique of storing data, typically in an organized, consistent, and normalized format. Such databases usually support applications directly, like applications that process transactions.

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ACTIVITY 3-3

Discussing Types of Data Used in Data Science Projects

Scenario

As the assistant manager at GCE's Esker Road store, you already work with different types of data as part of your regular activities.

1. Store inventory is kept in a database consisting of a series of linked tables. What type of data is this?
   - ○ Structured
   - ○ Unstructured
   - ○ Semi-structured
   - ○ Big data
2. GCE has a web presence, with each store location having a page to highlight local employees and events. It also contains a link for customers to provide feedback via a web form that includes clickable fields for identifying age ranges and departments and free-form text fields for the actual feedback. What type of data is the customer feedback?
   - ○ Structured
   - ○ Unstructured
   - ○ Semi-structured
   - ○ Transactional
3. What type of repository is most likely used to store the customer feedback?
4. Your manager said that when they had the soft opening for the store, they used census data to select certain potential customers for incentive savings offers. What type of data is census data?
   - ○ Big data
   - ○ Open data
   - ○ Proprietary data
   - ○ Unstructured data
5. What type of repository is most likely used to store the census data?

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TOPIC C

The Data Science Lifecycle

With a better idea of what data science is and the types of data commonly used in data science, you can explore the stages and processes that make up the data science lifecycle. Using a systematic, process-based approach like a lifecycle ensures that all relevant activities are performed and that the efforts of the data science team have a positive effect on the organization.

Lifecycle Stages

Like software development and other complex business initiatives, it can be helpful to identify a framework for the activities involved in completing an initiative. In many cases, a lifecycle can show progression towards the end result as well as the cyclical nature of such endeavors. As data science evolves and more companies get involved in the practice, no single lifecycle is recognized as superior to any of the others. For the purposes of this course, the data science lifecycle consists of the following stages:

1. Problem identification, including formulating questions that define a business problem and the business goals that can be achieved by solving the problem. Identification of potential data sources is also part of this stage.
2. Data collection.
3. Preprocessing, including data wrangling and munging.
4. Exploratory data analysis.
5. Modeling, including feature selection, training, and evaluation.
6. Model deployment.
7. Communication of results.

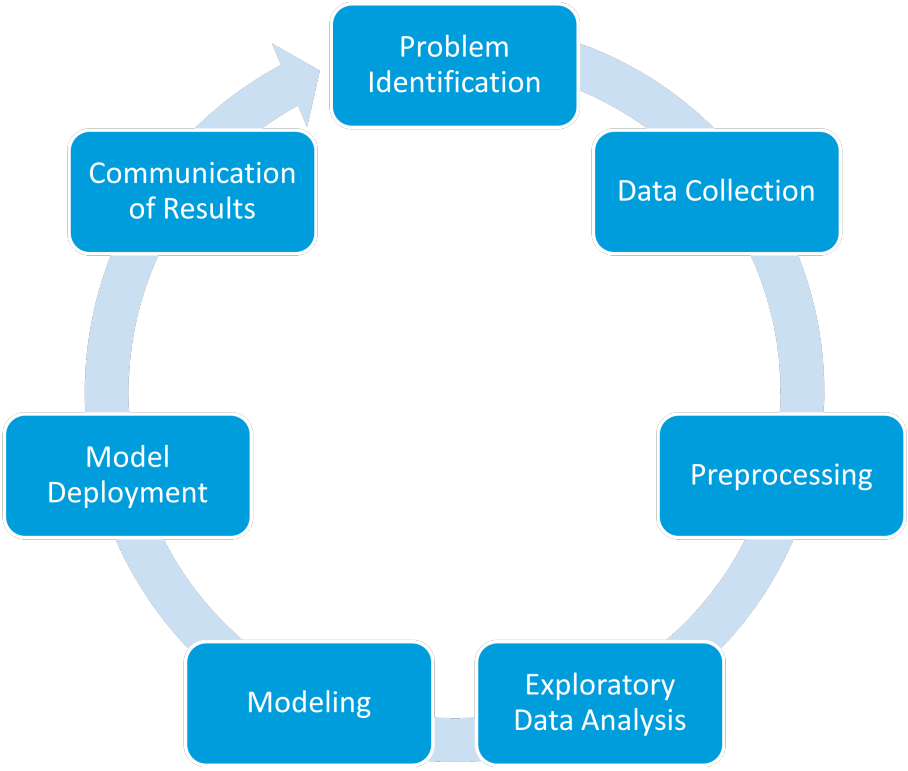


Figure 3-9: The data science lifecycle.

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As you review these stages, keep in mind that certain roles and activities will be involved in more than one phase. The stages that are relevant to most business professionals are likely to be those in which they have input and can provide feedback.

Examples of Data Science Lifecycles

As mentioned, no single data science lifecycle will be able to meet the needs of every organization. Here are a couple of examples of other lifecycles, most of which include similar stages:

- The Microsoft Team Data Science Process (TDSP) lifecycle:
  1. Business understanding
  2. Data acquisition and understanding
  3. Modeling
  4. Deployment
  5. Customer acceptance
- The Data Science Process Alliance's general data science lifecycle:
  1. Problem definition
  2. Data investigation and cleaning
  3. Minimal viable product/model
  4. Deployment and enhancement
  5. Data science operations
- The Cross-Industry Standard Process for Data Mining (CRISP-DM):
  1. Business understanding
  2. Data understanding
  3. Data preparation
  4. Modeling
  5. Evaluation
  6. Deployment

Problem Identification

The **Problem Identification** phase of the data science lifecycle encompasses the processes and activities that lay the foundation of the entire system. This phase requires domain knowledge to have any chance of success. You can think of this as the "business" portion of data science. Key parts of the **Problem Identification** phase include:

- Developing questions that define a business problem and the business goals that can be achieved by solving the problem. Determine the ***model targets***, which are those business variables that the data analysis needs to predict. What kinds of information do you want to extract? Though some companies just amass a great volume of data and then see where it leads them, it can be more expedient to identify specific issues that you want to investigate.
- Identifying bounding or restrictive parameters such as priorities, specifications, and budgets.
- Establishing that you have the necessary personnel, technology, time, and data to attain the established goal. This also includes identifying plans to acquire resources, such as data acquisition, sourcing, and collection options.

A general process for problem identification might include the following steps:

1. Clearly state the problem to be solved.
2. Describe the potential value of solving the problem.
3. Identify stakeholders, required resources, and possible risks.
4. Record and communicate a project plan.

The amount of effort an organization expends in this phase cannot guarantee success, but it does establish a scope and provides the environment for the remaining phases.

Key roles in this phase include:

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- Stakeholders such as C-level executives and managers, and possibly key personnel in affected departments.
- Data scientist.
- Data-team manager.

Tools

Common tools used in problem identification include:

- Organizational policies and procedures.
- Periodic company and departmental reports.
- Budgets and time frames.

Data Collection

The **Data Collection** phase of the data science lifecycle focuses on the collection, exploration, and profiling of data. The data acquisition and sourcing options elicited in the **Problem Identification** phase are implemented here.

Key activities in this phase include:

- Gathering data from identified sources. In some instances, data scientists are asked to work with existing datasets that have been collected in the course of other investigations. How that data is gathered and stored can limit the questions it can answer, and relevant data is not always immediately available. Some of the considerations surrounding data collection include:
  - Sourcing, or selecting the right data. Internal sources include log data, transactional data, email messages, and departmental or enterprise databases. External sources include social media and multimedia files such as audio, photos, and videos.
  - Identifying what collection methods to use. You can create online surveys, develop collection software, leverage open data, subscribe to data feeds, and work with companies that use similar sorts of data.
  - Identifying how to store the data as it is collected. When massive amounts of data are in play, you will likely want to use a cloud-based storage solution that is highly scalable.
- Performing an initial investigation of the data to ensure it contains the information required to solve the business problem. Results of this activity include a description of the data and its structure, and confirmation of the relevance of the data. ***Data profiling*** is the process of examining, analyzing, and summarizing a dataset to determine the level of data quality of the dataset. When performed prior to preprocessing and cleaning, profiling can help to identify how much effort may be involved in those tasks.

Again, although expending effort in this phase cannot guarantee success, having high-quality, consistently formatted data provides the optimal input for your data science project.

Key roles in this phase include:

- Data scientist
- Data analyst
- Data-team manager
- Database administrator

Tools for Data Collection and Generation

Tools for collecting or generating data include:

- Business-to-consumer (B2C) applications, such as e-commerce and customer-loyalty apps.
- Sales and customer relationship management (CRM) systems such as Salesforce®, Microsoft® Dynamics®, and HubSpot®.
- Web page traffic analyzers such as Google Analytics™ and Sprout Social®. You can also garner this information from internal-server log files.

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- Web page scraping tools such as Beautiful Soup, Mozenda, and Octoparse.

Tools for Data Storage

Data storage tools can be local, cloud based, or a combination of both:

- **Local storage** is designed to hold structured data. Common examples include:
  - Traditional relational database management systems (RDBMSs), such as Microsoft SQL Server® databases, Oracle® databases, SQL databases, and MySQL® databases. Traditional storage is present in most enterprises and can serve as a starting point, but once you reach a certain volume of data, the performance is affected. At that point (if you have not already), you will need to investigate more scalable solutions.
  - Data warehouses. Unlike RDBMSs, which are designed to optimize the update of transactional data, data warehouses are optimized to support data aggregation.
  - Data lakes. While a data warehouse is likely to hold databases that contain bounded information (like sales data or inventory data), a data lake can hold many categories of data in a central repository.
- **Cloud-based storage** is inherently scalable and well suited to large, diverse datasets. Common examples include:
  - Amazon Web Services (AWS) technologies such as Amazon Redshift®, AWS Lake Formation™, and Simple Storage Service (S3®) buckets.
  - Google Cloud Platform™ technologies such as Cloud Storage™, Cloud SQL™, and Cloud Datastore™.
  - Microsoft Azure® technologies such as Blob Storage™ and Data Lake Storage™.
- **Hybrid storage**, which provides access to local and cloud storage to provide flexibility, security, and performance in a cost-effective manner. Common examples include:
  - Apache Hadoop®, a free, open-source framework for storing and managing massive datasets.
  - OpenStack® Swift, an open-source object-storage system designed to store unstructured data.
  - Apache Cassandra®, a distributed database designed to store structured data across multiple server locations.
  - Cloud-service providers like AWS, Google, and Azure also support hybrid storage.

Preprocessing

Getting data from multiple sources means that it is probably not going to be formatted consistently and that there may be missing or duplicate information. ***Preprocessing*** is a series of activities that improve the quality of data to facilitate in-depth analysis. Common preprocessing tasks include:

- ***Data wrangling*** (or ***data munging***) is the process of transforming raw data into a format that is better suited to analysis.
- ***Data cleaning*** refers to the manipulation of data to remove inconsistencies and address missing data points.

Related to preprocessing is loading data into a ***data sandbox*** or ***data workbench***, which is a scalable development platform used to enable collaboration and exploration of large datasets. This is sometimes referred to as data integration. Techniques for loading data include ***extract, transform, and load (ETL)***, which pulls data from one or more sources, cleaning and formatting it before loading it into the sandbox.

The results of ETL often provide insight into the data model. A ***data model*** is a description of the relationships among various types of information. It organizes data elements and establishes standards for how the elements interrelate. Data models can be conceptual, physical, or logical:

- A conceptual model describes a high-level view of data. This is typically equivalent to the end user's view. For example, you might identify sources such as Salesforce or Oracle databases.
- A physical model is a schema or framework for how data is stored in a database. Some questions relevant to physical models include:

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  - How will you access the data?
  - Will you need APIs developed?
  - If you will use ETL, which fields should be accessed?
  - What permissions are required for accessing various data sources?
- A logical data model sits between the physical and conceptual levels and enables the logical representation of data to be separate from its physical storage. To accomplish this, you need to find the relationships between sources and the data they hold.

Key roles in this phase include:

- Data scientist
- Data analyst
- Data-team manager
- Database administrator

Data Cleaning Activities

Some general steps for cleaning data include:

- Remove extra spaces.
- Find and address blank cells.
- Convert numbers stored as text into numeric format.
- Convert dates and times to a consistent format.
- Perform data matching (John Q. Public, John Public, and Jack Public might all refer to the same individual, even though there are separate records under each name).
- Remove duplicates.
- Remove outliers (those values that fall far from the rest of the data).
- Highlight and correct errors.
- Change text casing to be consistent.
- Check for and correct all spelling errors.
- Remove all formatting.

Other Data Loading Techniques

Although ETL is quite commonly used in data science projects, you might hear other techniques referred to, such as ELT or even ETLT.

- Extract, load, and transform (ELT) pulls data from one or more sources, cleaning and formatting it after it has been loaded into the sandbox.
- Extract, transform, load, and transform (ETLT) pulls data from one or more sources, cleaning and formatting it both before *and* after it has been loaded into the sandbox. The first transform cycle is often implemented to eliminate noise from the data.

Tools

Common data preprocessing tools include:

- Microsoft Excel. While Excel itself does not scale to enterprise-level data wrangling or cleaning, its vast presence on business computers makes it a common first choice for data cleaning tasks.
- R, which is a data-analysis programming language that supports statistical analysis, predictive modeling, and data visualization.
- Python, which is a high-level programming language that supports data manipulation and repetitive tasks.
- MATLAB, a proprietary programming language developed by The MathWorks®, Inc. that enables programmers to manipulate matrices, plot functions and data, implement algorithms, and other tasks.
- SAS, a statistical software-application suite developed by SAS Institute® for managing and analyzing data.

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- IBM® SPSS, a software platform that provides advanced statistical analysis, a library of ML algorithms, and other features. Two of the main applications in the suite are SPSS Statistics™ and SPSS Modeler™, which respectively provide top-down and bottom-up analysis.
- Apache Spark™, an open-source distributed cluster-computing framework that supports big data processing, as well as ML, visualization, and other data science–related computing tasks.
- Informatica® PowerCenter®, an ETL tool used for building data warehouses.
- RapidMiner®, an integrated environment for data science tasks such as data preparation, ML, and predictive analytics.

Exploratory Data Analysis

Exploratory data analysis (EDA) is a form of analysis that examines datasets, summarizes their main characteristics, and provides the summary information in a visual format. Characterized by using statistical functions such as averages, means, and medians, **EDA** provides initial insights into your data by:

- Calculating summary statistics.
- Identifying dependent and independent variables or features.
- Analyzing key features of data to work on.
- Defining the spread of data and identifying outliers.

Identifying relationships between variables, by performing some initial analytics to better understand the patterns inherent in the data, helps you establish a basis for the algorithms to be implemented. For instance, ***aggregation*** is the process of gathering and summarizing data in a summarized report.

For thorough data analysis, different plots can be used to visualize the data for better understanding for everyone. Data scientists explore the distribution of data inside distinct variables graphically by the usage of bar graphs. Relations between distinct aspects can be captured via graphical representations like scatter plots and heat maps.

EDA and preprocessing can inform each other. For example, information you discover during EDA may require additional preprocessing to address outliers or inconsistent data types. EDA can also uncover features that are either required or unnecessary for modeling. Performing another round of preprocessing can ensure the most important features are selected. Many data teams find that including one or more iterative cycles of preprocessing and EDA result in a better model and more accurate predictions.

Key roles in this phase include:

- Data scientist
- Data analyst

Tools

Common tools used for EDA include:

- Applications like Tableau® and Power BI®
- Python
- R
- MATLAB
- SAS
- IBM SPSS
- Apache Spark™
- RapidMiner

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Modeling

The **Modeling** phase of the data science lifecycle is where you develop a machine learning model that will perform the analytics required to solve the business problem. ML models consist of one or more ***algorithms*** and the model data. Algorithms are mathematical procedures or techniques that, when converted into code, enable a computer to identify patterns, make predictions, or perform tasks without explicit instructions. Activities in this phase include:

- ***Feature engineering***, which is the process of using domain knowledge, mathematics, and statistics to manipulate a dataset to facilitate the training of an ML model. It is imperative that the most relevant variables are used as input for your ML model so that it can successfully predict outcomes.
- ***Model training*** is the process of feeding a small set of data (training data) into an ML algorithm, evaluating the results of the algorithm, and optimizing the algorithm for performance. Model training is an iterative process and is often performed on several algorithms until one or more algorithms attain sufficient performance levels.
- ***Model validation*** occurs after training and confirms that a model actually achieves the intended purpose. During validation, a different set of data (testing data) is fed into an algorithm, and input, calculations, and output are all tested for accuracy.



Note: With several types of models involved, things can get confusing. Remember, data models represent the structure of data, ML models learn from data and make predictions, and statistical models infer information about a population based on sample data.

Key roles in this phase include:

- Data scientist
- Data architect
- Database administrator
- Data engineer
- ML engineer
- Data-team manager

Tools

Common modeling tools include:

- Microsoft® Analysis Services, which is available for SQL Server®, Power BI®, and Azure®.
- R
- Python®
- Jupyter® Notebooks, which is an open-source web application that enables you to create and share documents containing live code, equations, visualizations, and narrative text. It is used for data cleaning and transformation, numerical simulation, statistical modeling, data visualization, and machine learning.
- SAS/ACCESS®, which is software designed to integrate SAS and third-party relational databases.

Model Deployment

Once the model is built, trained, and validated, you can deploy it. The **Model Deployment** phase includes the following activities:

- ***Operationalizing***, or moving the data and the model out of the sandbox and into a live environment. The goal is to gather repeatable, scalable, and flawless results from data. Sandbox testing can help with the first and third part of the goal, but historically the greatest challenge has been scalability. Identifying tools that can automate the production of APIs or executable packages can significantly reduce the time and effort it takes to operationalize.

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- Monitoring and analyzing the data periodically to ensure that the model is generating the expected results. When acceptable bounds are exceeded, the team can revisit earlier stages in the lifecycle to address any required adjustments.

Key roles in this phase include:

- Data engineer
- Database administrator
- Software engineer
- Implementation engineer
- Operations engineer
- Operations manager

Tools

Common tools used in model deployment include:

- R and Python
- Amazon Web Services (AWS) cloud infrastructure services
- Microsoft Azure cloud infrastructure services
- Informatica
- Databricks® Unified Data Analytics Platform™
- RapidMiner
- DataRobot®

Communication of Results

To ensure that you have achieved the goals set in the **Problem Identification** phase, you should gather the results of the initiative and communicate them to the stakeholders. That is the primary goal of the **Communication of Results** phase. Activities in this phase include:

- Identifying the key findings that were documented in each stage of the process.
- Creating visual representations of the insights you have gained.
- Packaging the findings and visualizations into reports and other assets that all stakeholders can understand and use.

Key roles in this phase include:

- Stakeholders such as C-level executives and managers, and possibly key personnel in affected departments.
- Data scientist.
- Data-team manager.

Tools

Common tools for communicating results include:

- Microsoft Excel
- Tableau
- Python
- Jupyter® Notebooks
- Microsoft Power BI

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ACTIVITY 3-4
Discussing the Data Science Lifecycle

Scenario
During the cross-functional meeting, you and your colleagues agree that a systematic approach to data science projects is vital to the success of these initiatives. You volunteer to do some research about the lifecycles commonly used for data science projects, and report back to the team.

1. In which stage of the lifecycle is data wrangling performed?
   - ○ Problem Identification
   - ○ Data Collection
   - ○ Preprocessing
   - ○ EDA
   - ○ Modeling

2. In which stage of the lifecycle are ML algorithms tested?
   - ○ EDA
   - ○ Modeling
   - ○ Model Deployment
   - ○ Communication of Results

3. Which stages of the data science lifecycle would you (as a manager/business professional) expect to have the most influence over? (Choose two.)
   - ☐ Problem Identification
   - ☐ Preprocessing
   - ☐ Model Deployment
   - ☐ Communication of Results

4. Why did you select the stages you did in the last step?

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Summary

In this lesson, you identified several fundamental concepts related to data science, including basic definitions of many terms, the types of data that data science can leverage, and the phases of the data science lifecycle. With this information at your disposal, you will be able to communicate about data science more effectively.

Of the various practices that exist under the data science umbrella, which practice are you most intrigued with and why?

Which data science lifecycle stage will be most challenging for your organization to implement? Briefly explain your response.



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4 Functions of Data Science in Business

Lesson Time: 1 hour

Lesson Introduction

Business professionals bring vital domain knowledge to every data science project. It is imperative that companies expend money and effort towards those projects that will offer the most beneficial results. Even with this consideration, there are a few use cases that almost any company might try to enhance by using data science. In this lesson, you will identify functions of data science in business.

Lesson Objectives

In this lesson, you will:

- Improve customer experience (CX).
- Improve marketing efforts.
- Optimize organizational and transactional security.
- Enhance operational practices.

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TOPIC A

Improve Customer Service

Establishing and maintaining customer satisfaction is essential in most businesses, especially those that rely on repeat business to maximize their profitability. In this topic, you will review ways that data science can improve customer experience.

Customer Experience (CX)

Customer experience (CX) is the overall perception of a company based on customers' interactions with that company. Based on the internal, subjective responses of individuals, CX encompasses consumer reactions from each stage of the customer journey—enquiry, comparison, purchase, installation and use, and loyalty. CX is highly individualized in the details, but it is generally accepted that improving CX is a benefit to any business. When you add data science to the mix, several strategies for improving CX emerge as possible use cases:

- Personalization
- Sentiment analysis
- Recommender systems
- Self-service support
- Chatbots and virtual assistants

Personalized CX

The effort of personalizing CX and providing a specific set of interactions and services that is tailored to each individual consumer often helps customers feel valued, which in turn can lead to increased loyalty and the potential for repeat business. Predicting which services and interactions each customer will react positively to depends not only on the collection of customer data, but also on the correct interpretation of that data. This is where data science comes into play.

Often, different types of data are leveraged to provide personalization. Each stage of the customer journey enables a company to gather information from their customers, so personalization can begin almost immediately.

- **Initial enquiry**. At the start of the customer journey, consider providing more than one "front door" to potential customers. Some people appreciate a self-service approach, while others prefer more human interaction right from the start. Whether a customer's initial experience with your company is through direct contact with a sales representative or via your website or other online marketing materials, be sure to collect basic contact information (name, email, phone number, etc.) and at least a general idea of what the customer is interested in. Whenever possible, try to collect the same types of information no matter the source of the initial enquiry. In addition, consider following up the initial contact with a short survey to enable customers to provide more specific information about whatever it is they are seeking, as well as feedback about their impressions of the company so far.
- **Comparison**: When customers look for alternatives to past purchases, you want to position your company and its products and services as the most viable alternative. This stage can be considered a "conversation" between your company and the customer, with your goal being to encourage the customer to make a purchase. Gathering information from the customer about past experiences and current expectations enables you to tailor the information you share to provide enough information to facilitate decision making, but not so much information that potential customers are overwhelmed with unnecessary information. You can also leverage demographic data, browsing history, and purchase history to create different messages for different population segments. Another possibility is to create personalized offers and content based on the information gathered so far.

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- **Purchase**: Your main goal at this stage is to make the purchasing process as effortless as possible for each customer. For online and offline purchases, collect basic invoice data such as the specifications (such as quantity, size, and so forth) for each product, billing details, and shipping information. Offer the purchaser membership in existing rewards programs. Consider conducting another survey so that customers can provide feedback about the purchasing process.
- **Implementation and use**: After a sale, you will want to follow up with the customer to ensure that they are satisfied with the products and services that they have purchased. Elicit any areas of confusion or other trouble spots, and refer specifically to those when you respond to the customer. Ask customers to rate the product or service according to how closely it meets their needs, and ask for suggestions on how to improve the process.
- **Loyalty**: Your ultimate goal is to create and foster long-term relationships with your customers. Ensure that customer service personnel are trained to rapidly respond to questions and concerns, and to verify that customer concerns have been adequately addressed. Consider implementing tiered loyalty programs that offer different incentives for future purchases.

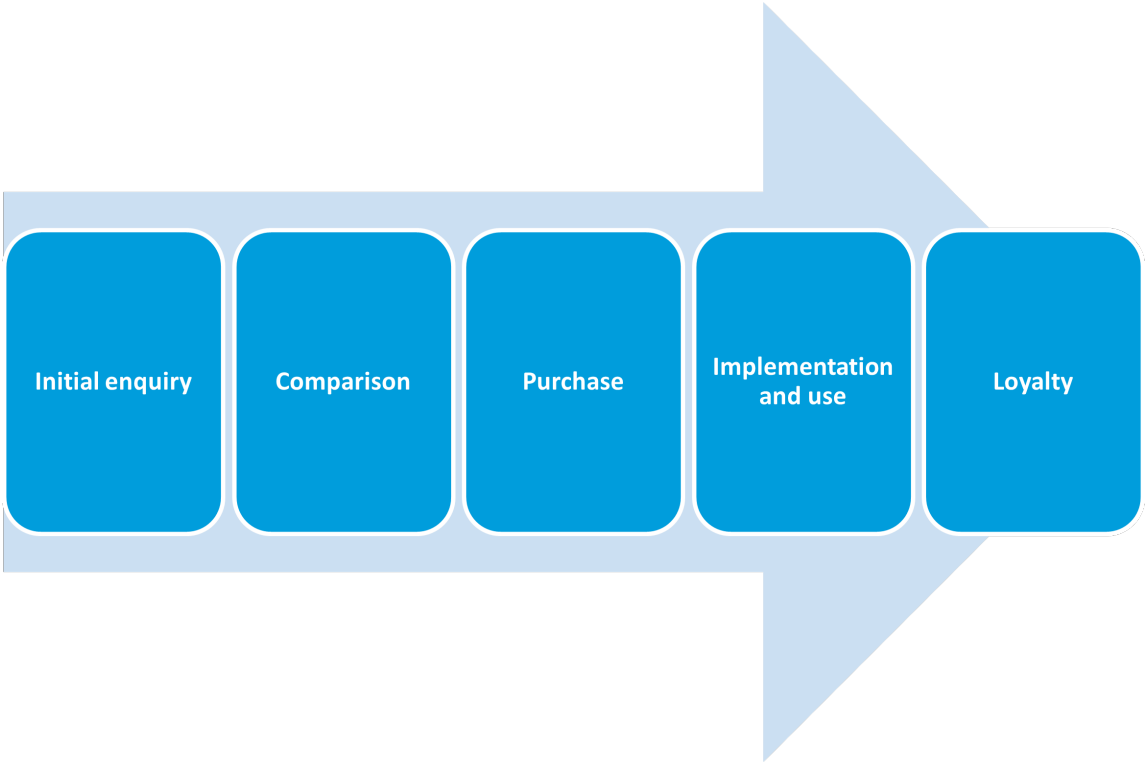


Figure 4-1: Try to collect data at every stage of the customer journey.

Sentiment Analysis

Sentiment analysis is a process that determines the emotional tone of textual data, often grouping results in positive, negative, and neutral categories. Sources for the textual data commonly encountered in sentiment analysis include emails, customer-support chat transcripts, online reviews, and other comments made on social medial platforms. The technologies used for sentiment analysis include machine learning (ML), statistics, and ***natural language processing (NLP)***, which is a form of artificial intelligence (AI) that enables computers to comprehend, generate, and respond to human language.

Sentiment analysis can provide insight into your customers' mindsets and uncover possible ways to increase customer satisfaction.

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Recommender Systems

A ***recommender system*** is a form of machine learning that suggests potential items of interest based on a customer's profile and search history. By comparing this data with metadata about available inventory, a recommender system can suggest similar products and services that the customer may find relevant.

For example, if you search an online bookstore for a certain book title, you might receive recommendations such as additional books by the same author or in the same genre. Other online stores can suggest companion products to facilitate the use of items in your shopping cart, like the correct batteries for a battery-operated item you want to buy. Finally, if you are shopping for a certain type of clothing and you have viewed several options before making a selection, you might be shown other items in similar styles or colors. Each of these examples supports the company goal of making sales while driving customer satisfaction levels.

Self-Service Support

For customers, the availability of self-service support enables them to receive assistance when they need it—days, nights, or weekends—without the need for a company to hire customer service representatives to work around the clock. Self-service support is particularly beneficial for resolving basic issues, such as lost passwords, quickly and efficiently. Options for implementing self-service support include:

- Chatbots
- Knowledge bases and other documentation
- Online forums and communities

As convenient as self-service support can be, it cannot meet the needs of all customers. When more complex issues arise, or when the customer prefers human interaction to solve a problem, customer service and support personnel can provide extra guidance to help customers feel confident about the products and services they use.

So, how does the use of data science relate to self-service support efforts? You can collect data about support calls and analyze it to determine the most commonly experienced issues or questions, and to classify the issues or questions as involving basic or advanced support. You could then create frequently asked questions (FAQs) content to address the top 10 basic issues or questions.

Chatbots and Virtual Assistants

Perhaps the ultimate tool to personalize CX is the use of ***generative AI***, which is a deep learning application of AI that is used to generate content such as text, audio, images, and video, in a way that appears convincingly humanlike. Popular implementations include chatbots and virtual assistants. These are similar, and may be used interchangeably in some instances, but they do have different roles and purposes.

- A ***chatbot*** is a program or feature that is designed to simulate human conversation through text or voice messages. Early chatbots used forms of pattern matching to interpret user messages, while more modern chatbots leverage AI (in particular, machine learning). A few of the functions performed by chatbots include providing instructions to complete a task, gathering and sharing product or contact details, setting calendar appointments, and integrating with a knowledge base or self-service portal. Often deployed on websites, within apps, or on messaging platforms, chatbots use a conversational interface similar to standard text messaging to communicate with users. Popular chatbots that you may have experience with include the CNN® chatbot on Facebook® Messenger, Spotify®, and OpenAI® ChatGPT®. The primary goal of a chatbot is to provide customer support.



Note: The first chatbot was developed in 1966 by Joseph Weizenbaum, professor emeritus at the Massachusetts Institute of Technology (MIT).

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- A ***virtual assistant*** is a software-based agent designed to assist users with performing simple tasks. Examples of tasks include checking calendar appointments, setting alarms, initiating phone or video calls, typing text messages, automating workflows, providing reminders, and offering directions. Virtual assistants do not need to use an interface to communicate with users, and they are commonly built into devices or appliances. Popular virtual assistants include Apple® Siri®, Amazon® Alexa®, Microsoft® Cortana®, and Google® Assistant™. The primary goal of a virtual assistant is to help users conduct business or personal activities.

If your company is looking to implement AI-based technology to improve CX, consider using chatbots as opposed to virtual assistants.

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ACTIVITY 4-1

Improving Customer Experience

Scenario

Consider the following discussion questions to help review your understanding of methods for improving CX.

1. Customers who use the GCE online store and other online resources have been experiencing substantial wait times to speak with customer service agents. This has led to some dissatisfaction and less-than-optimal online reviews.

   How could the company address this issue?

2. To increase sales of some less-popular, but high-profit items, a suggestion was made to create some in-store displays that show the items in a typical scene. For example, a seasonal display including a beach umbrella, chairs, towels, and a drinks cooler.

   How can you tell how effective these displays are?

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TOPIC B

Improve Marketing Efforts

The primary goal of any marketing strategy is to maximize revenue while minimizing customer acquisition costs. An accurate understanding of the target audience is crucial, and it should be no surprise that data science can help. In this topic, you will identify how data science can improve marketing efforts.

Audience Segmentation

It's unlikely that a single marketing campaign can appeal to everyone. Things that appeal to parents in their 20s and 30s are probably different from the things that a teenager finds interesting. Understanding these differences enables you to isolate the market into categories and offer specific selling points to each category. This makes your campaigns more precise because you can focus on serving smaller, more refined segments. ***Audience segmentation***, sometimes referred to as *customer segmentation* or *market segmentation*, is the process of dividing individuals into groups based on common characteristics. Target markets can be divided in numerous ways, but a common approach is to use four main types of audience segmentation:

- Demographic segmentation divides an audience by variables such as age, gender, education level, occupation, and income.
- Geographic segmentation divides an audience by previously determined geographical boundaries.
- Psychographic segmentation divides an audience by intrinsic traits such as values, personalities, interests, attitudes, motivators, and opinions.
- Behavioral segmentation divides an audience by how they make decisions and work through the purchasing process.

Targeted Advertising

One of the goals of audience segmentation is to provide targeted advertising to consumers. Focusing on specific traits, preferences, and interests enables you to differentiate the messages sent to potential and existing customers. For this type of advertising, the following data sources are commonly used:

- Web tracking, most often in the form of third-party cookies.
- Location tracking, through the use of technology like GPS signals, cellular triangulation, and Bluetooth® connectivity.
- Cross-device tracking, which uses data from multiple devices to collect user data and preferences.

While studies show that well over half of consumers find targeted advertising appealing, others may be concerned with privacy and other ethical issues.

Campaign Optimization

As the name implies, campaign optimization is the process of increasing the effectiveness of marketing and advertising campaigns. The ultimate goal of any campaign is to maximize the return on investment (ROI) for the company, and data science can help.

Campaign optimization software such as Google Analytics™, WordStream®, and others can identify and track metrics that signify the success of a campaign. They may even suggest ways to improve the metrics and therefore the effectiveness of the campaign. These metrics, often referred to as key performance indicators (KPIs), include:

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- Return on ad spend (ROAS): Comparing the conversion rate to the amount spent on the campaign.
- Cost per click: Identifying the unit cost of the investment.
- Conversion rate: Identifying how many visitors were converted into customers.

Marketing Measurement Analysis

Similar to campaign optimization is the concept of ***marketing measurement analysis***, which is the process of collecting, analyzing, and interpreting data to measure the effectiveness of marketing activities. While campaign optimization focuses on a particular campaign, marketing measurement analysis can also compare the effectiveness of different campaigns. Using information about the relative success of previous marketing activities can help drive the success of future campaigns.

Many organizations use a variety of metrics for this analysis. Some popular metrics include those described in the following table.

Metric	*Description*
Reach	The number of people who are exposed to a marketing message.
Frequency	The number of instances a person is exposed to a marketing message.
Engagement	The level of interaction with a marketing message.
Conversion	The number of people who react to a marketing message as was intended by the organization.
ROI	The amount of money generated by a marketing campaign divided by the amount of money spent on the campaign.

Common tools for collecting and analyzing marketing data include web, social media, and email analytics tools.

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ACTIVITY 4-2
Improving Marketing Efforts

Scenario
Consider the following discussion questions to help review your understanding of methods for improving marketing efforts.

1. The executive team is concerned that GCE's marketing campaign related to new products such as beach gear and home leisure is not getting to enough of the right people, because initial sales numbers are below expectations.

   How can the use of data science help alleviate their concerns?

2. After GCE's marketing department implements the suggested changes for the new product campaign, they want to see how this campaign stacks up against campaigns from last year.

   How can data science help address this issue?

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TOPIC C

Optimize Organizational and Transactional Security

In addition to the customer-facing issues discussed so far, most businesses have immense interest in ensuring the security of their systems and data. Not surprisingly, data science can help with security issues, too. In this topic, you will identify ways that data science can help optimize organizational and transactional security.

Fraud Detection

Fraud detection is an area where data science can benefit a business. Banking organizations analyze transactional data and can flag anomalous activity that is not overtly fraudulent, but AI has determined is possibly related to fraud, such as several consecutive small transactions followed by a larger transaction. Once the flag has been raised, the bank can contact the account holder to verify the transactions and take corrective action.

Minimization of Loan Defaults

Financial institutions can use data science techniques to more accurately predict the likelihood that a potential borrower will default on a loan. For instance, analyzing historical data in the form of past loan applications and whether or not the loans were paid in full can provide some insight as to the types of loans that are most likely to go into default. The organization may decide to assign weights to various loan types and then, when an individual applies for a particular type of loan, that weighting can be factored into the overall determination of risk. Other factors that could be considered include:

- The applicant's identifying information (name, taxpayer identification number, etc.).
- The applicant's financial history.
- The applicant's current work status and financial standing.
- The estimated value of the property to be acquired.
- The loan amount.

Reduction of Intellectual Property Theft

Intellectual property (IP) is a category of property that includes intangible creations of human intellect, such as literary and artistic works, inventions, designs, and so forth. Probably the best-known examples of IP are patents, copyrights, trademarks, and trade secrets. As you can imagine, IP theft results in adversity for the victim, including losing a competitive edge, damage to organizational reputation, slow or stagnant growth, and losing customer trust.

There are numerous best practices to prevent or reduce the impact of IP theft, many of which relate to data security. For example, having systems in place to continuously monitor user activity with regard to IP data provides companies with a warning mechanism that supports investigation in real time. So, how can data science help to reduce IP theft? Extending the concept of user monitoring, user and entity behavior analysis (UEBA) uses AI to establish a baseline of user behavior. If the user performs anomalous activities, security personnel can be alerted in real time. Another example is seeding a honey trap—using AI and NLP to create and publish false information intended to lure and mislead adversaries.

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Risk Identification and Mitigation

The previous sections identify several specific types of organizational risk and how data science can help address these risks. On a more general note, data science can assist in identifying and mitigating many forms of operational and strategic risks. Operational risk includes disruption of day-to-day business processes, while strategic risk arises when a company cannot react to market conditions quickly enough. Both of these types of risk can be lessened somewhat by the implementation of data science.

For example, one form of operational risk is failure of mission-critical equipment. Predictive modeling could be used to anticipate component failure and allow for the components to be replaced on a preventive maintenance schedule. The schedule itself might be derived from data about number and size of orders, any work that is scheduled for other equipment, and disaster recovery data regarding mean time to failure (MTTF) and mean time to repair (MTTR).

Data science is just as effective at mitigating strategic risk. Prescriptive modeling can not only identify changing customer requirements or industry trends, but it can also offer suggestions for how to adapt to those changes.

Cybersecurity Considerations

The use of data science to strengthen cybersecurity measures is a key component of most of the popular cybersecurity strategies currently in use. Some examples of this phenomenon include:

- The first stage of any cyber attack is likely to be reconnaissance, where the attacker gathers information about potential targets. ***Network traffic analysis (NTA)*** is the process of capturing and analyzing network data packets to gain insights into the behavior, performance, and security of a computer network. It involves examining the patterns, protocols, and volume of data flowing through a network to understand how devices and systems communicate with each other. Companies can use NTA devices to identify unusual behavior, alert cybersecurity personnel to the situation, and possibly even block the continuation of the anomalous activity.
- Monitoring user behavior to identify patterns and anomalies can also uncover possible vulnerabilities. ***User and entity behavior analytics*** is a cybersecurity tool that uses algorithms and ML to detect anomalies in the behavior of the users in a corporate network, as well as the routers, servers, and endpoints in that network. For instance, if user Gio normally downloads a few small files over the space of a week, but starts downloading multiple large files every day, the UEBA system can flag the activity and alert security personnel of a potential threat.
- Analyzing historical data to identify and prioritize cybersecurity risks can help organizations focus their resources on protecting against the most critical threats.
- Implementing machine learning and NLP to analyze online content for potential cybersecurity risks can help organizations respond to cyber attacks in real time.

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ACTIVITY 4-3

Optimizing Organizational and Transactional Security

Scenario

Consider the following discussion questions to help review your understanding of methods for optimizing organizational and transactional security.

1. Management is concerned about the security of the online store, especially with regard to minimizing the effects of identity theft and other fraudulent activities.

   How would you explain possible uses of data science to prevent fraudulent activity in the online store?

2. Store managers received an email warning of people trying to return counterfeit goods for store credit.

   What security issues might this be an example of, and how would you suggest that the issues be addressed?

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TOPIC D

Enhance Operational Practices

The final category of business uses discussed in this lesson involves using data science practices for the optimization of common day-to-day tasks. In this topic, you will identify ways that data science can help enhance operational practices.

System or Component Failure

A common hallmark of reliability for a business is system availability, which is expressed as a percentage, such as 99.999%. Many organizations maintain a high level of availability by employing ***predictive maintenance***, which is the practice of monitoring the overall performance and condition of specific equipment during regular operations to avoid unplanned downtime.

Many companies use predictive analytics to estimate the probability of system or component failure based on historical information about past failures. This information enables these companies to schedule planned downtime to replace components or otherwise ensure that systems do not fail.

For examples of predictive maintenance, visit: **https://blog.brewerscience.com/six-examples-of-predictive-maintenance**.

Sales Forecasting

Another common use for data science is creating and refining sales forecasts. By analyzing historical data, identifying trends and patterns, and applying machine learning models, businesses can make better decisions about staffing, inventory, marketing, and pricing. Forecasting can be short term or long term, identifying monthly, quarterly, and annual benchmarks. Common methods for sales forecasting include:

- Time-series forecasting uses data from previous periods (like monthly or quarterly sales) and estimates future outcomes.
- Regression forecasting uses a combination of historical data and linear regression to analyze how numerous factors change (such as market condition and economic trends) and affect sales efforts over time.
- Historical forecasting analyzes past sales performance to identify trends and other patterns and uses that information to predict future sales.
- Opportunity-stage forecasting use data from current transactions (those within a production or fulfilment pipeline) and other factors to anticipate customers' future needs.

Dynamic Pricing

Dynamic pricing is the process of adjusting the price of goods or services according to customer behavior or changes in the market. A well-known example of dynamic pricing is the fluctuation of airline ticket prices, where factors such as seat type, number of available seats, and the number of days until the flight takes place are considered when a ticket-price quote is requested.

Customer Churn

Customer churn is the tendency of customers to cancel subscriptions to a service, or otherwise cease their relationship with a business. Basically, you can think of customer churn as the opposite of customer retention. Although some churn occurs naturally, it is important to recognize increases in customer churn, the causes of the change, and actions that the company can take to minimize

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churn. Data science, and predictive analytics specifically, can help identify the factors that lead to customer churn, which may include:

- Negative customer experiences, such as the perceived low quality of products and services, lack of effective customer support, and pricing increases without added value.
- Changes in customer priorities, including changing economic situations or no longer needing a product or service.
- Switching to a competitor with better pricing or product selection.

Once the main causes are identified, the company can address them, possibly by improving customer service, offering discounts or loyalty rewards, and similar actions.

The type of data required to calculate and predict customer churn includes demographic and behavioral data as well as information about interactions with the company, such as beginning and end dates for a subscription, pricing data, and monthly recurring revenue (MRR). This data is analyzed to identify characteristics that are common to former customers, and the resulting information is compared to existing customer data to determine if there are any customers who fit the identified pattern. Those customers could then be contacted and offered an incentive to stay with the company.

Talent Acquisition

Recruiters, HR personnel, and other hiring managers have historically relied on perceptions and intuition to determine which candidate is the best fit for an open position, with understandably mixed results. Due to the virtual explosion of data science over the past several years, companies can now implement ***talent acquisition (TA) analytics*** to analyze recruitment data and candidate data to optimize the recruiting and hiring processes for the organization. Many HR/recruiting software packages include features such as automated candidate sourcing and tracking, résumé parsing, and email communication, all of which save humans from performing tedious and repetitive tasks related to the recruitment process.

Metrics that are useful in TA analytics include:

- Time to fill, or the number of calendar days between approval for posting the position and the hire date of the selected candidate.
- Time to hire, or the number of calendar days between a candidate applying for a position (or being contacted by a recruiter) and accepting an offer of employment.
- Offer acceptance rate, or the percentage of candidates who accepted offers out of the total number of candidates who received offers.
- Recruitment channel effectiveness, or the number of candidates obtained from various sources such as company websites, career search engines, social media platforms, and external recruitment agencies.
- Quality of hire, or the value a candidate brings to the company.

Transportation and Logistics

Some of the ways data science can help optimize the functions of transportation and logistics include:

- Dynamic route planning takes into consideration real-time data such as weather, traffic conditions, and active construction zones to provide the best possible route for deliveries.
- Demand prediction and dynamic pricing can enable companies to offer reduced rates to common delivery destinations.
- Warehouse management and optimization enables companies to monitor inventories and optimize the placement of items to facilitate handling and shipping.
- Operational capacity planning, or evaluating production resources to ensure that a manufacturer can meet product demand.

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For more transportation and logistics use cases, visit **https://rtslabs.com/the-evolution-of-predictive-analysis-in-transportation-and-logistics/** and **https://acropolium.com/blog/big-data-in-logistics-key-benefits-3-real-use-cases/**.

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ACTIVITY 4-4

Enhancing Operational Practices

Scenario

Consider the following discussion questions to help review your understanding of methods for enhancing operational practices.

1. Last month, a colleague's store did not meet its forecasted sales goals, and the shift managers mentioned that they have noticed a sharp decrease in the number of customers shopping before and after the peak hours of 11:00 A.M. to 6:00 P.M. The store manager has asked for suggestions on how to improve the numbers.

   What do you think the issue is? What advice would you offer?

2. As part of its expansion efforts, GCE is building two new stores and needs them fully staffed within two weeks of the completion of construction. HR is concerned about finding and hiring qualified candidates within the allotted time frame.

   How can TA analytics help streamline the recruitment and hiring process?

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Summary

From increasing customer satisfaction to optimizing standard operating practices, almost all of a company's strategic goals can be addressed, at least in part, by the implementation of data science practices. In this lesson, you reviewed several business uses for data science, including the improvement of CX and marketing efforts, the optimization of several forms of security, and the enhancement of operational practices.

Which CX enhancements are presently used at your workplace? Which would you suggest be considered?

Which of the organizational and transactional security issues are most relevant to your company? Were any issues discussed that your workplace does not address?



Note: Check your CHOICE Course screen for opportunities to interact with your classmates, peers, and the larger CHOICE online community about the topics covered in this course or other topics you are interested in. From the Course screen you can also access available resources for a more continuous learning experience.

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5 Implementing Business Requirements for Data Science

Lesson Time: 1 hour

Lesson Introduction

So far, you have explored data science fundamentals and identified numerous compelling business uses for data science. To put this information to work and transform your business, there are certain requirements you should meet to ensure that the transformation is smooth and successful. In this lesson, you will implement business requirements for data science.

Lesson Objectives

In this lesson, you will:

- Develop a data-centric organization.
- Develop an implementation strategy.
- Identify the impact of data science on business.
- Identify governance measures.

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TOPIC A

Develop a Data-Centric Organization

The successful and sustainable transformation of a business into one that embraces the philosophies and benefits of using data science requires some preparation at the organizational level. In this topic, you will examine the requirements for developing a data-centric organization.

What Is a Data-Centric Organization?

You may have heard the terms "data driven" and "data centric" used interchangeably, but there are some notable differences between the two. You can think of data driven as the mindset that supports making strategic decisions based on data, while data centric goes further to encompass the concept and the practice of positioning data as a core asset of the organization.

A ***data-centric organization*** is one where management and employees not only believe that data analytics are an integral component of the overall business strategy, but also implements a framework in which data is stored independently of the applications and tools that rely on it. The intent of a data-centric organization is to generate relevant information and use it in a collaborative manner to achieve successful business outcomes.

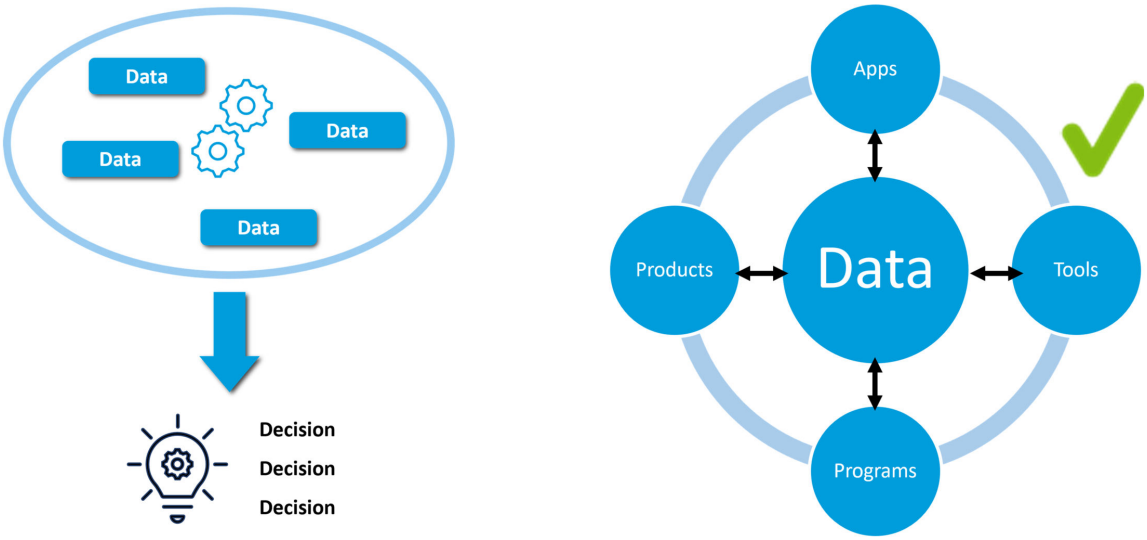


Figure 5-1: Data-centric organizations place data at the core of the organization.

Challenges Associated with Building a Data-Centric Organization

As you might expect, building a data-centric organization is a complex undertaking that can provide significant rewards when proper planning is incorporated into the process. A few of the common challenges include:

- **Data silos**. Data silos occur when data is stored in separate systems that cannot communicate with each other. This makes it difficult to access and analyze data from different parts of the organization. Migrating data from silos to central data storage provides the core of the data-centric environment.
- **Data quality**. Data quality is essential for any data-driven organization; however, it can be difficult to ensure that data is accurate, complete, and up to date. This is especially true when

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data is collected from multiple sources or when it is manually entered. Include sufficient time and resources to ensure that the consolidated data is clean and free of errors.

- **Data governance**. ***Data governance*** is the set of policies and procedures that govern how data is collected, stored, and used. It is essential for ensuring the security and privacy of data, as well as for ensuring that data is used ethically, but data governance initiatives can be complex and time consuming to implement. Organizations need to ensure that they are compliant with all applicable data privacy and security regulations. This can be a complex and evolving area of law, and organizations need to stay up to date on the latest requirements.
- **Trust**. Several levels of trust are crucial to ensuring the success of the data-centric organization.
  - Organizational management and IT must establish and maintain trust in each other and in the benefits of the changes being implemented.
  - Employees and customers need to trust that their data is being used responsibly and ethically. If people do not trust the organization with their data, they will be less likely to share it. This can make it difficult to collect the data that is needed to make informed decisions.
- **Changing roles and responsibilities**. As part of the consolidation of data that is distributed throughout an organization, the roles and responsibilities of individuals who work with the data must also change.

Some organizations that have worked through the process to become data centric include:

- The Boeing® Company, a multinational manufacturer of aerospace products.
- JPMorgan Chase™ & Co., a multinational financial services company.
- Domino's®, a multinational restaurant chain.

Organizational Preparation

Preparing a company to be a successful data-centric organization involves the following:

- **Obtaining management buy-in**. As with any major endeavor that results in fundamental change to an organization, the backing and support of upper management is required. Having one or more data evangelists in a leadership position can help ensure that data quality and management issues will be addressed. This can also ensure that lines of communication between all levels of the organization remain open and fruitful.
- **Achieving alignment with organizational objectives and values**.
  - Make sure you understand the overall business objectives of the organization, then examine the existing data infrastructure and its relationship to those objectives. Bringing these into alignment provides a data strategy that supports using and interpreting data as a central asset of the organization.
  - Identify organizational resources (people, equipment, and processes) that may be required to achieve these goals. Compare this inventory with what the organization presently has in place to identify gaps or the need for updating or replacing resources.
  - Consider the culture of the organization. What parts of the culture lend themselves well to a data-centric approach? Characteristics like design thinking, objectivity, clear channels of communication, and a focus on learning and professional development all help contribute to a data-centric mindset. Determine what enhancements are required to position the company for success.
- **Identifying potential contributors**. Although the overall initiative will affect everyone at some point, there are several individuals or roles that can greatly assist the transformation process. Some organizations create specific data teams, while others prefer to use an ad hoc approach. The common thread is that potential contributors have a clear understanding of the vision behind creating a data-centric organization, including the anticipated benefits to the organization. Commonly recognized stakeholders include leaders and other decision makers, data owners, data science practitioners, and data consumers. Each organization will have varying criteria for selecting the appropriate personnel to drive the necessary changes.
- **Preparing teams to succeed**. Ensuring a successful business transformation is impossible without a concerted effort to provide teams with the tools they need to be successful.

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Team Preparation

As part of organizational preparation, the company should have performed a resource assessment to identify equipment, processes, and personnel needed to achieve its goals. To fulfill the requirements for personnel resources, most companies will use a combination of training existing employees and adding new personnel, either as employees or through the use of contractors, with the goal of having a talent ecosystem that has the required level of ***data literacy***, which is the ability to read, analyze, communicate, and reason with data.

- **Training**. Transforming to a data-centric organization requires some form of training for all existing employees. After all, processes, tools, and tasks are likely to change, and each employee needs to be aware of how their roles and responsibilities are affected by the transformation. Some job roles may be eliminated, others augmented, and still others created from scratch, so the people in those roles may need ***upskilling***—training that gives employees the skills and knowledge to help them perform their jobs more effectively.
- **Talent recruiting and acquisition**. It's quite likely that there will be new or existing job roles that need to be filled by new personnel. Deciding whether to add full-time employees, use contractor or other third-party talent, or implement some combination of the two will obviously depend on each organization's unique needs.
- **Combining roles**. Many organizations find that combining certain job roles provides an optimal solution. Obviously, a large enterprise is in a better position to hire more experienced professionals than a small or medium business is, but the need remains the same. In many cases, companies will combine job roles and ask team members to extend themselves to perform tasks that might otherwise be performed by another.

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ACTIVITY 5-1

Discussing the Development of Data-Centric Organizations

Scenario

Consider the following discussion questions to help review your understanding of considerations related to developing a data-centric organization.

1. The executive team has embraced the idea of incorporating data science practices throughout the organization.

   What recommendations would you make to ensure the success of the endeavor?

2. After performing a resource assessment, several new job roles were identified as needing to be filled.

   Which staffing option or options would you recommend the company use to optimize costs?

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TOPIC B

Develop an Implementation Strategy

Recall the data science lifecycle discussed earlier in the course. As you might expect, there are several considerations to be taken into account when this lifecycle is operationalized in the workplace. In this topic, you will develop an implementation strategy.

Selection of Business Cases for Implementation

Earlier in the course, you discovered numerous business cases for using data science, including audience segmentation, recommender systems, and fraud detection. It could be very disruptive to attempt to implement these use cases all at once. For a successful implementation, each organization should examine the possible use cases to determine things like:

- Which use cases will provide the most benefit to the overall organization.
- Which use cases can be implemented at a relatively low cost.
- Which use cases can be implemented in a relatively short time period.

Once this information has been obtained, the company can select the order of implementation that is best suited to its needs. You may also find that some of the use cases will not provide enough benefit to outweigh the costs associated with implementing them. For example, if a particular project requires that the company spend a large amount of capital for new equipment but the anticipated benefits of implementation are not mission critical, it may be advantageous to delay the project, or even drop it altogether.

Implementation Investments

As you plan for the implementation of data science projects, it's crucial to consider the types of investments each project will need, such as:

- **People**. Your data team needs the skills and experience to collect, clean, analyze, interpret, and communicate data and insights. In most cases, investment in staffing and training will be substantial.
- **Technology**. Your organization needs certain tools and physical resources to ensure the success of its data science initiatives. These tools and resources include software tools for collecting, cleaning, analyzing, and visualizing data, and hardware resources such as servers, storage, and networking.
- **Data**. Access to large amounts of high-quality data is necessary for every data science project. Data can come from a combination of internal and external sources, with varying levels of quality, so investments in data can range from purchasing access to data to building a data pipeline that collects, cleans, stores, and moves data.
- **Governance**. Investing in the development of a solid data-governance program—one that ensures that data is secure, private, accurate, available, and usable—supports better decision making, improves cost controls, and contributes greatly to regulatory compliance.



Note: Data governance and regulatory compliance are discussed in depth later in this lesson.

- **Time**. Organizations must be aware of the time factor for data science projects. Rushing to completion will likely provide less than optimal results, and can even result in project failure.

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Data Collection Considerations

Different issues affect the various stages of the data science lifecycle. Consider the following as you prepare for data collection activities:

- **Make sure that relevant data is collected**. This actually begins during the **Problem Identification** stage. A clear definition of what the organization wants to achieve can help identify what data needs to be collected.
- **Identify options for collecting relevant data**. Options include actions such as collecting data directly—by using surveys, interviews, and other instruments—or collecting and acquiring data from a third-party source such as public databases, government websites, and social media platforms.
- **Identify and procure tools for generating synthetic data**. In some cases, it is more expeditious to generate ***synthetic data*** by using tools like synthetic data generators and ***data augmentation*** tools. Using tools to generate synthetic data for use in data science initiatives eliminates the need to provide access to confidential or sensitive information and can help enhance datasets so that they better represent the population.
- **Determine the best method for storing data**. If you are using structured data, a relational database might be the optimal storage method. For a mix of data types, a data lake might be the best method. Make sure you also consider the benefits and drawbacks of using on-premises storage or cloud storage, or a combination of both.
- **Identify how data from different sources will be combined**. Initially, you might combine, or integrate, multiple datasets into a single dataset to get a comprehensive view of the data. If you find that an existing dataset should include more or different information, you can update or enrich the dataset by appending data to it. The appended data can be entire new records or new fields incorporated into the existing records.

Data Preparation Considerations

Consider the following as you prepare for data preparation activities such as preprocessing and exploratory data analysis (EDA):

- Preprocessing activities such as data wrangling and cleaning improve the quality of data to facilitate in-depth analysis. Identify and use tools that streamline the preprocessing of data.
- EDA activities such as examining datasets, summarizing main characteristics, and visually displaying summary information provide initial insight into the data and can facilitate the selection of algorithms and model types. Select and use data analysis tools that support each of these activities.

Modeling and Deployment Considerations

Consider the following as you prepare for modeling and deployment activities such as feature engineering, model training and validation, operationalization, and monitoring:

- Select algorithms and machine learning (ML) models that identify patterns, make predictions, or perform tasks that support solving the identified business problem.
- After a model is selected, tune the model's hyperparameters (those parameters that control the learning process and that cannot be derived from the data) to optimize performance.
- During model testing, identify and use multiple algorithms to ensure the model provides acceptable results.
- Evaluate the model's performance and its results. Getting the wrong answer quickly is of little benefit.
- Avoid deploying a model before the performance and results of the model are acceptable.
- Verify that the deployed model provides repeatable, scalable, and flawless results from data.
- Monitor output data over time to ensure that the model continues to provide the expected results.

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Results Communication Considerations

Consider the following as you prepare for activities related to communicating the results of a data science project:

- Identify the audience that will receive the results, and adjust your communications accordingly. Remember that jargon and high levels of technical detail may not be appropriate for everyone.
- Provide a clear, concise summary of the project and any limitations associated with it to provide context to the audience.
- Highlight and explain the key findings to ensure that the audience understands the results.
- Consider using visualizations such as tables and charts to illustrate your findings to promote audience understanding. Select a data visualization tool that enables you to create compelling visualizations.
- Combine the summary, key findings, and visualizations into reports and other assets for stakeholders.
- Encourage follow-up questions and feedback to ensure that your message was received as you intended, and to identify possible improvements to your communication skills.

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ACTIVITY 5-2

Developing an Implementation Strategy

Scenario

Consider the following discussion questions to help review your understanding of the issues involved in developing an implementation strategy.

1. You are part of a cross-functional team that needs to brainstorm possible data science projects for implementation over the next two quarters. At the past two meetings, the team has recorded numerous ideas ranging from recommender systems for the online store to reducing customer churn. The next step is to identify the order in which the ideas should be addressed and present your recommendations to the executive team.

   What are some determining factors you would use to prioritize the project ideas?

2. The executive team has decided to move forward with implementing the project related to reducing customer churn.

   What is one issue that you would suggest the implementation team consider before beginning data collection?

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TOPIC C

Identify the Impact of Data Science on Business

Now that you have a clearer picture of the basics of data science and several examples of its implementation, you can examine factors that have a positive or negative impact on your business. This information is crucial to making informed and intelligent decisions on how and where to incorporate data science into your business practices. In this topic, you will identify the impact that data science can have on the business.

Effects on Overall Business Operations

In addition to the changes that are implemented in various pockets of the business, data science can have both positive and negative effects on overall business operations.

- **Reputational impact**.

  Here are some common positive effects that data science can have on an organization's reputation:

  - Improved decision making, which can lead to increased profits, improved customer service, and more effective risk management.
  - Enhanced customer experience, which can result in increases in customer satisfaction and loyalty.
  - Improved product development, which can lead to more sales and higher profits through the development of products and services that are projected to be more successful than their predecessors or the competition.

  Here are some common negative effects that data science can have on an organization's reputation:

  - Data breaches, which can result in financial losses, reputational damage, and legal problems.
  - Perception of unethical practices such as discriminatory decision making or using datasets with known biases, which can cause unfair treatment of employees or customers, damage to the company's reputation, and legal problems.
- **Impact on talent pool**.

  Here are some common positive effects that data science can have on an organization's talent pool:

  - Attracting top talent due to the organization's adoption of data science practices and/or a data-centric operating philosophy. Companies that are willing to invest in the resources required for success often appear more attractive to data science professionals.
  - Improvement of employee retention. Effective use of data science can provide staff with more challenging and satisfying job tasks, as well as plenty of learning and growth opportunities.
  - Increased employee productivity. When data science is used to automate repetitive tasks, the employees who formerly performed those tasks are free to focus on more strategic tasks.

  Here are some common negative effects that data science can have on an organization's talent pool:

  - Skills gaps. The explosion of data science as a business tool has caused there to be a shortage of experienced practitioners. This could lead to companies competing to recruit those individuals with impressive skill sets.
  - Prohibitive costs. Implementing data science projects can be costly, especially due to the efforts to acquire the best possible talent to the organization. When large capital expenditures are required, this can strain the implementation budget to make it more difficult to attract and retain talented employees.

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- **Legal and regulatory compliance**.

  Here are some common positive effects that data science can have on an organization's compliance efforts:

  - Improvements in compliance. Organizations that use data science as part of their risk identification and mitigation strategies often realize an increase in their compliance status as a direct result. Companies can also use data science practices to drive compliance management.
  - Reduced costs. Some compliance tasks are relatively easy to automate through the use of data science, leading to lower staffing costs attributed to the company's compliance efforts.

  Here are some common negative effects that data science can have on an organization's compliance efforts:

  - As mentioned previously, data breaches can lead to legal repercussions.
  - Unethical practices such as bias and discrimination are the target of many forms of legislation around the world. Each organization needs to be aware of which regulations apply so that it can implement practices that keep the organization in compliance with all applicable requirements.

	**Note:** Legal and regulatory issues are covered in depth in the next topic.

Effects on Business Processes and Practices

Data science can also have an effect on various business processes and practices, including:

- **Cost management efforts**. Implementation of new or enhanced processes that support the use of data science can bring with it a large initial investment, especially if the organization is undergoing the change to data-centric principles. As with every other organizational initiative, optimizing the return on investment (ROI) for data science projects can be a concern. Even if a project succeeds in providing insights, could those insights have been achieved with more traditional analytic methods that required fewer resources?
- **Management of infrastructure requirements**. Managing complex infrastructure requirements can be daunting for the most seasoned professionals. However, with the help of data science, several types of efficiencies can be realized. For example, using data science to predict component failure or the need for more computing power and storage helps organization plan enhancements to meet future needs. Automation of management and optimization tasks helps ensure that the infrastructure operates at peak efficiency.
- **Employee impact**. Organizations that use data science should see varying levels of impact to their employees depending on job roles and other factors. In addition to the impacts on the overall talent pool, companies are advised to manage these changes in a way that provides employees with opportunities for upskilling and ways to increase their productivity.
- **Organizational changes**. For large-scale organizational changes like transforming to a data-centric organization, the effects of using data science can be widespread. However, increases in the quality of decision making, efficiency, innovation, and communication provide numerous benefits to help offset any potential negative effects.

Data Issues

It should be clearly apparent that implementing data science practices in a business will affect organizational data; however, a few notable data issues that are common in most organizations are described in the following table.

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Data Issue	Description
Data security	Data science projects should be designed and implemented with security in mind. Organizations need to protect not only the data, but also the analytics themselves, from attack, theft, or other malicious actions. The nature of data science makes it potentially vulnerable to compromise whether the project is on-premises, cloud based, or a hybrid project with components in both areas.
Data controls	Data controls are measures and practices that help mitigate data risks. Common data controls include security measures to prevent unauthorized access; backup and recovery procedures to ensure the timely restoration of data that is lost for any reason; governance practices that describe how data is collected, used, and managed; and employee education to support the protection of data, including how to identify and report suspicious activity.
Data quality	The quality of data has an enormous impact on the success or failure of any data science initiative. If you think about it, the entire **Preprocessing** phase of the data science lifecycle deals with data quality. Also consider the effect of pooling data from different sources into a dataset. If your proprietary data is high quality, but you augment it with data of lesser quality, the quality of the entire dataset is degraded.
Data privacy	Data science can have profound effects on the protection of individuals' privacy. Datasets contain a vast amount of information (often from different sources), and any person or organization that controls the datasets also has control over the individual chunks of information. Because this information is likely to contain personal information, privacy becomes a significant concern. The processing of data might also threaten privacy. Combining data from different sources may reveal personal characteristics or other information that was not meant to be shared.

As mentioned previously, developing and implementing a solid data-governance plan can help address each of these data issues.

AI-Related Risks

Risks related to the implementation of artificial intelligence (AI) systems include:

- **Lack of formal governance for AI**. When organizations don't adopt governance measures for AI systems, problems can arise with regard to biased models, privacy issues, security issues, and accountability issues. It is highly recommended that any organization looking to leverage AI in their business practices develop formal governance in the form of:
  - Policies and procedures for the development, deployment, and use of AI systems.
  - Employee training designed to teach the responsible use of AI systems.
  - Monitoring and auditing requirements to ensure that AI systems are used safely and responsibly.
  - Methods for ensuring transparency and explainability surrounding the collection of data, the use of data, and the possible risks inherent in using AI systems.



Note: Governance measures are covered in depth in the next topic.

- **Acquisition of third-party data and AI products**. Using third-party data and AI systems means that you have very little control over important items such as data quality and provenance. This can lead to problems such as data bias, privacy issues, security vulnerabilities, and accountability issues. To counter these risks, perform a thorough review of the data and the

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terms of the agreement between your organization and the vendor, and ensure that all applicable compliance requirements are met.

- **Lack of domain expertise during creation and testing of AI systems**. Domain expertise is an integral component in the creation and use of AI systems to provide data science capabilities. Without domain expertise, it is difficult to discern if training data is truly representative of real-world data or is even accurate. Be sure to include domain experts in the creation and testing of AI systems to avoid bias and inaccuracies, as well as security vulnerabilities.
- **Use of black-box models and technologies**. In the world of AI, a black-box model is one that is not transparent with regard to its decision-making processes, which leads to difficulty in understanding the reasoning behind the decisions the system makes. Without that understanding, organizations are at risk of incorporating bias into the model; experiencing unintended consequences that can have adverse effects on people, society, or the environment; and leaving the model vulnerable to security attacks. Whenever possible, use interpretable models that use a transparent decision-making process. If you must use a black-box model, consider implementing safeguards such as using multiple models or requiring human review and oversight of all decisions made by the model. Finally, make sure you communicate the fact that a black-box model is being used and explain the risks associated with its use.

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ACTIVITY 5-3

Discussing the Impact of Data Science on Business

Scenario

Consider the following discussion questions to help review your understanding of the impact data science can have on business.

1. You continue your work with the cross-functional team for implementing data science across the organization. After the executive team selected the customer churn project as the first to implement, they ask if the remaining projects can be categorized according to the level of impact they will have on the overall operation of the business.

   Of the issues discussed (impact on reputation, impact on talent pool, and legal/regulatory compliance), which do you think carries the most weight? Why?

2. A suggestion is made to control costs and streamline implementation by using third-party data and AI systems, and a lively discussion ensues, with approximately half of the team agreeing with the suggestion and the other half disagreeing.

   What side of the discussion would you take?

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TOPIC D

Identify Governance Measures

So far, you have identified several strategies for implementing business requirements to support data science, including an introductory look at some ethical considerations and some legal and regulatory considerations. This topic expands on the introductory information to provide a more focused examination of these types of issues. In this topic, you will identify governance measures.

Ethical Considerations

Data ethics is a collection of moral principles that govern how organizations gather, protect, and use data. A fundamental goal of data ethics is to consider and protect the rights and interests of people and institutions outside of the organization. As a business decision maker, you won't necessarily be asked to take on the deep philosophical issues involved with data science. However, you are still responsible for ensuring that your data science projects follow general ethical principles. In some organizations, ethics have actually become a standard part of corporate board reporting. This course highlights five major ethical principles in emerging technologies like data science:

- **Privacy**. As mentioned earlier, protecting the privacy of individuals, groups, and other entities who are represented by the data found in datasets is of particular concern.
- **Accountability**. Accountability refers to the practice of ensuring that people are held responsible for their actions, but when machine learning is used to implement data science, who is responsible for the insights produced by the model?
- **Transparency and explainability**. Transparency refers to the idea that people should be able to see into the inner workings of the technology, while explainability refers to the ability to understand how and why the technology works to arrive at a result. Probably the best example of transparency and explainability in data ethics is the black-box problem described earlier.
- **Fairness and non-discrimination**. The principles of fairness and non-discrimination protect people from being treated improperly because of who they are. They are primarily undermined by bias, which is a major issue in AI and ML. If the source data used to train a model contains bias, it's highly likely that any results generated by the model will also contain bias. Organizations need to take steps to ensure that training data contains minimal bias by addressing any disparities in representing specific groups, or minimizing the reinforcement of stereotypes within the dataset.
- **Safety and security**. While security is generally acknowledged to refer to protecting organizational assets, the idea of safety refers primarily to protecting human assets by minimizing the chances for physical harm or compromise of property. Building security controls into data science projects can help mitigate safety and security risks, but organizations need to understand that such techniques are not guaranteed to work in every situation.

Legal and Regulatory Considerations, Frameworks, and Guidelines

Numerous forms of laws and regulations exist to try to govern emerging technologies like data science practices. Some of the most well known include:

- The General Data Protection Regulation (GDPR), a comprehensive regulation adopted by the EU to safeguard against data privacy violations and to provide regulations to ensure compliance by affected organizations.
- The Payment Card Industry Data Security Standard (PCI DSS) is a set of security standards designed to protect credit card data and prevent fraud.

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- The OECD Privacy Guidelines are a set of international guidelines for the protection of privacy. They were developed by the Organization for Economic Co-operation and Development (OECD) in 1980, and they have been revised several times since then. Although they are not legally binding, they are widely considered to be the international standard for privacy protection. They have been adopted by many countries, and they are often used as a reference point for national privacy laws and regulations.
- The Asia-Pacific Economic Cooperation Cross-Border Privacy Rules (APEC CBPR) is a set of voluntary privacy principles that aim to facilitate the safe and secure transfer of personal data across borders within the Asia-Pacific region.

With the rapid advancement of AI implementations, this is another arena where governments and other regulatory agencies are working to regulate the development and use of AI technologies. At the time of publication:

- China has adopted provisional regulations governing the use of generative AI.
- AI legislation is proposed in numerous countries, such as the EU's Artificial Intelligence Act and Canada's Artificial Intelligence and Data Act (AIDA).

Organizations must be aware of which laws and regulations apply to them. For instance:

- In the U.S., there is no comprehensive data privacy legislation, but several states have adopted laws to protect personal information.
- EU legislation, frameworks, and regulations like the GDPR can have a wider range than the geographical boundaries of the EU.
- According to the United Nations Conference on Trade and Development (UNCTAD), 79% of all countries have some sort of privacy or data protection legislation in place, and another 9% have draft legislation in the process of being ratified.

The landscape of legal and regulatory requirements is constantly changing. Some regulations are not put in place yet, but they *are* forthcoming. What does not affect your company today may do so in the future. It is crucial that your organization review new developments and respond accordingly to regulatory changes.

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ACTIVITY 5-4

Discussing Governance Measures

Scenario

Consider the following discussion questions to help review your understanding of the governance measures discussed in this topic.

1. During data collection and preprocessing for the customer churn project, Kevin, who is performing data wrangling and cleaning, points out that a large percentage of the data appears to represent only a few small geographical areas.

   Would you consider this a cause for concern? Why or why not?

2. Of the laws, regulations, and guidelines discussed in this topic, which do you think has the largest reach?

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Summary

In this lesson, you implemented business requirements for data science. You examined the process of developing a data-centric organization, developed a strategy for implementing data science projects, identified the effects of data science on business, and identified governance measures that are applicable to data science projects.

Try to envision your company transforming to a data-centric organization. In your opinion, what do you think the greatest roadblocks will be?

Of the various effects on business discussed in this lesson, which do you think will have the largest impact at your workplace?



Note: Check your CHOICE Course screen for opportunities to interact with your classmates, peers, and the larger CHOICE online community about the topics covered in this course or other topics you are interested in. From the Course screen you can also access available resources for a more continuous learning experience.

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6 AI Fundamentals

Lesson Time: 1 hour, 15 minutes

Lesson Introduction

Before you explore the many ways in which artificial intelligence (AI) can transform the business, you must first grasp the fundamentals. In this lesson, you'll identify what exactly AI is, as well as gain a high-level understanding of the major concepts that make up AI. This will provide you with a framework for understanding the practical implications of AI, as well as enable you to speak intelligently about this revolutionary phenomenon.

Lesson Objectives

In this lesson, you will:

- Describe the history of AI.
- Describe fundamental AI concepts.

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TOPIC A

A Brief History of AI

To begin, you'll focus on what AI is, how it's evolved over the years, and where it's at now. That way, you can place AI in the proper context as you start thinking about ways in which it can help your business.

What Is Artificial Intelligence (AI)?

Artificial intelligence (AI) is, at its core, the ability of machines to exhibit human-like intelligence. What exactly is meant by "human-like intelligence" is a source of much debate—some would even argue that true AI is not yet achievable, given strict requirements for "intelligence." However, there are some factors that make up a common understanding of "intelligence," including:

- The ability to perceive one's own environment.
- The ability to take actions for the purpose of achieving goals.
- The ability to obtain knowledge and apply it to a problem in order to solve that problem.

In a more practical sense, AI is a scientific discipline that governs the study, creation, and operation of machine intelligence. While this discipline has existed for decades, it has seen a recent resurgence in many different facets of society, including business. AI has the potential to revolutionize every sector of industry; in many cases, that revolution has already begun. Although AI is a highly technical field comprising many different complex elements, you don't need a great deal of technical knowledge to understand its impact and how you can leverage it in your organization.

Timeline of AI

It's helpful to contextualize AI as it has progressed over the years. Not only does this reveal where AI has been, it can also reveal where AI is headed.

The history of AI has its roots in antiquity. In a fictional sense, myths like those of Ancient Greece told of artificial beings—a precursor to the modern idea of robots. Realistically, humans have been creating self-operating machines for thousands of years. Later, the rise of formal reasoning and, eventually, the advent of computer science, paved the way for AI. However, it wasn't until after World War II that research into the "thinking machine" really started in earnest.

The following is a timeline of major events, contributors, and technologies, directly leading up to, and in the wake of, the beginning of AI as a scientific discipline.

Time Period	*Events, Contributors, and Technologies*
1950	Alan Turing proposes the Turing test as a way of measuring the intelligence of a machine.
1956	The "birth of AI"—the true start of AI as a scientific discipline—occurs at a workshop at Dartmouth College. The term "artificial intelligence" is coined at this workshop.
1961	Unimate, the first industrial robot, begins work at a General Motors assembly line.
1964	Joseph Weizenbaum develops ELIZA, one of the first natural language processing programs.
1966	Development begins on Shakey, the first mobile robot to exhibit rudimentary reasoning abilities.

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Time Period	Events, Contributors, and Technologies
1973	The Lighthill report advises UK Parliament that AI research has failed to achieve its far-reaching objectives.
1974–early 1980s	Due to the conclusions of the Lighthill report and a general lack of progress in the field of AI, the UK and US governments pull funding from AI research projects. This period is known as the "AI winter."
Early 1980s–mid 1980s	AI research experiences a boom with the advent of expert systems, which are programmed to answer questions within a specific area of knowledge.
Mid 1980s–early 1990s	AI research experiences a second winter as personal computing hardware makes specialized expert system hardware a needlessly expensive investment.
Mid 1990s	The dramatic increase in computing power and the rise of the Internet begins pulling AI research out of its winter. AI still has little acceptance in the business world, however.
1997	The Deep Blue supercomputer defeats world champion Garry Kasparov in chess, suggesting a new level of intelligence and creativity in AI systems.
2000	Cynthia Breazeal publishes a dissertation describing the Kismet robot, a head-shaped machine that simulates human emotions.
2004–2005	The Defense Advanced Research Projects Agency (DARPA) holds its first Grand Challenge, a competition for self-driving cars. Although no cars complete the course, it helps generate interest in the industry. The second Grand Challenge one year later sees several cars complete the course.
2006–2008	This period marks the rise of three important technologies that enable AI to flourish in the modern era: cloud computing, big data, and general-purpose graphics processing units (GPUs).
2011	IBM's Watson supercomputer beats champions Ken Jennings and Brad Rutter at *Jeopardy!* by a wide margin. Like Deep Blue before it, Watson is a milestone in AI research and application. That same year, Apple® releases Siri®, the first modern virtual assistant. The Google Brain™ team, which researches applications of machine learning and deep learning, also forms this year.
2016	Google's DeepMind™ project, which develops the AlphaGo program, defeats champion Lee Sedol at Go. Unlike Deep Blue and Watson, which were designed for specific tasks, DeepMind's programs are more general purpose and use deep learning techniques.
2020–present	OpenAI releases GPT-3, a deep learning model that can generate text of such high quality that it is often indistinguishable from text written by a human. It also releases DALL·E and its successor, DALL·E 2, which can generate realistic images based on user text prompts. In 2022, OpenAI releases ChatGPT, a chatbot that can generate human-like responses to queries and requests.

AI-Enabling Technologies: Cloud Computing

As mentioned in the timeline, 2006 to 2008 saw the rise of three of the most important technologies for AI development and research: cloud computing, big data, and general-purpose GPUs. These deserve further explanation.

Cloud computing is an approach to providing computing services and other resources over a network of distributed servers and other infrastructure. Businesses rely on cloud services for many, or even all, of their IT-related needs. Many organizations offload the computing costs of developing AI solutions to the cloud for two reasons: one, because AI, particularly deep learning, requires a lot

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of resources; and two, because the cloud can efficiently match the ever-changing demand for resources. Also, most organizations don't have the massive infrastructure that cloud providers like Amazon, Google™, and Microsoft® have.

Think of how you use virtual meeting software like Microsoft® Teams®, Zoom, or something similar. Your organization probably isn't hosting its own virtual meeting servers. Instead, servers in the cloud are establishing and maintaining these virtual meetings on behalf of the participants. So, every time you call someone using these apps, your voice and video are being sent over the Internet to a cloud server, where they are processed by that server and then sent along to the other meeting participants. This is possible thanks to cloud computing.

In 2006, Amazon released Elastic Compute Cloud (EC2™), the flagship service of their Amazon Web Services™ (AWS™) platform. This was the first publicly available Infrastructure as a Service (IaaS) offering from a cloud vendor. Since then, there have been many more cloud services from different vendors, including services that are specifically designed to support AI development.

AI-Enabling Technologies: GPUs

A ***graphics processing unit (GPU)*** is a computer chip designed to process graphics, images, and other visual media. GPUs were originally created to support video game graphics, but most GPUs these days are considered general-purpose GPUs, meaning they can carry out tasks that were traditionally assigned to central processing units (CPUs).

Comparing CPUs and GPUs is a bit like comparing a sports car to a big delivery truck. The sports car (CPU) can carry only a small load, but it is agile and can stop, turn around, and pick up passengers quickly. A big delivery truck (GPU) is not nearly as agile, but can handle a much larger load.

GPUs excel at AI processing tasks whereas CPUs tend to struggle. These tasks require a lot of memory bandwidth, and a lot of linear algebra calculations—both things that GPUs do better than CPUs. Therefore, GPUs have become the de facto processing equipment for AI, particularly deep learning.

If you've played a modern, graphically intensive video game or created/edited images and video on a computer, you have likely used a GPU, even if you weren't aware of it. If not, you've still probably benefited from GPUs indirectly. For example, the recommendations you get from Amazon.com or other online stores are probably using GPUs to process your data and determine the best products to recommend.

Although GPUs have existed since the 1990s, the first general-purpose GPU architecture was released by Nvidia in 2007, named Compute Unified Device Architecture (CUDA). More recently, high-performance GPUs designed to accelerate AI tasks have become much more common.

AI-Enabling Technologies: Big Data

Big data refers to collections of data that are so large and complex that they require advanced tools to process and analyze them. Big data is typically described using several "Vs": volume (the huge amount of data), variety (the many different sources the data can be pulled from), velocity (the high speed at which data processing must occur), and more.

AI is a highly data-driven technology; in fact, data is the first and foremost requirement of creating an AI solution. In general, the more data you have, the more capable your solution will be at solving a given problem. So, before the advent of big data, businesses struggled to collect enough data to build truly useful AI solutions. Now that the volume of available data across all industries has increased dramatically over the past decade, AI has become more viable for even smaller organizations.

Think of all the searches done on Google.com and other search sites. There are billions *per day*. Now consider that Google does targeted advertising and customized search results based on users' searches. Every time you search for something, you are feeding data into a truly monumental system. This is only possible because of technologies that support the principles of big data.

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There is no single event that precipitated the rise of big data, but in 2008, several prominent computer science researchers from the Computing Community Consortium (CCC) published an influential white paper titled "Big-Data Computing: Creating revolutionary breakthroughs in commerce, science, and society." This paper outlined the rising phenomenon of companies collecting and storing volumes of data that were previously unheard of. It also offered predictions and advice for companies as they entered a big data world, including mention of AI applications.



Note: The three enabling technologies just mentioned are often used in combination; for example, a company might use GPUs hosted on a cloud service to process their big data collections.

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ACTIVITY 6-1

Discussing the History of AI

Scenario

Consider the following discussion questions to help review your understanding of AI's history and what has led it to the hot topic it is today.

1. Identify at least two events, people, or systems since the end of the second AI winter that have had a direct impact on the upward trajectory of AI.

2. Identify at least two relatively recent technologies that have enabled AI to flourish in the business world, and briefly explain how they have done so.

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TOPIC B

AI Concepts

Now that you know some of the history behind AI, you can start learning its major concepts. Although these concepts are technical in nature, you only need to grasp them from a high level—enough for you to be able to effectively communicate with your colleagues and clients.

General AI and Narrow AI

There are many ways to categorize AI. One way is to define it in terms of its scope and overall ability. When doing so, AI is typically categorized as either being *general* or *narrow*.

Artificial general intelligence (AGI), also called *general AI* or *strong AI*, refers to machines that have a similar level of cognitive abilities as humans. This means the machine has the ability to use reason—not just to solve one or a few problems, but to extrapolate those reasoning abilities to solve many novel problems. A machine with AGI would be able to fool a human observer into thinking that the machine itself is human. At the moment, AGI is far from being a reality. It is debatable as to whether or not it will ever *be* a reality. Nevertheless, much philosophical thought and scientific research is being done on the subject, including whether or not a general AI system would truly possess consciousness or sapience.

Narrow AI, also called *weak AI*, is AI that is able to solve one or a few specific problems that are assigned to it by humans. It can approximate certain human traits, including reason, but it does not truly possess those traits. Instead, a weak AI system is limited in scope and cannot be applied to a general set of tasks like AGI. Still, even systems with limited scope can be very powerful. This is where AI is at now and will likely be for the foreseeable future. So, when people talk about implementing AI in business, they are referring to narrow AI, not the thinking machines of science fiction.

Example

Consider the example of an AI-powered machine cleaning a house. There are autonomous vacuum cleaners on the market today that can navigate a room by themselves, avoiding obstacles and detecting dirty spots on the floor that need cleaning. These AI systems excel at determining how best to go about cleaning a floor. This is an example of narrow AI, because the vacuum robot is good at the one thing it was designed to do. It cannot do anything else, at least not very well.

Now consider a robot that has never vacuumed before. It is able to enter a house (after having been given a key), find the vacuum cleaner in the closet, clean out the dust container if it's full, replace the filter if the vacuum has one and it's old, begin cleaning the floors room by room, empty out the dust that accumulated, put the vacuum back in the closet, then leave the house and lock the door. That robot would possess AGI, because it would be able to perform multiple distinct tasks, similar to how an adult human would vacuum a house if they had never vacuumed it before.

Machine Learning

Machine learning is an AI discipline that makes predictions and other decisions about an environment based on a given set of data. These predictions are made without the aid of instructions programmed directly into the machine itself. Instead, the machine is able to process input data using one or more ***algorithms***—sets of rules for carrying out problem-solving operations—that can make intelligent decisions about that data and how it pertains to an environment.

The result of the algorithm's calculations on that data is a ***model***. A machine learning model "learns" by measuring how effective (or ineffective) it is at predicting its assigned task, then improves its own predictive abilities based on these performance measurements. As the name

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implies, machine learning ensures that not only can a machine make intelligent decisions, it can actually get better at making those decisions with time and experience.

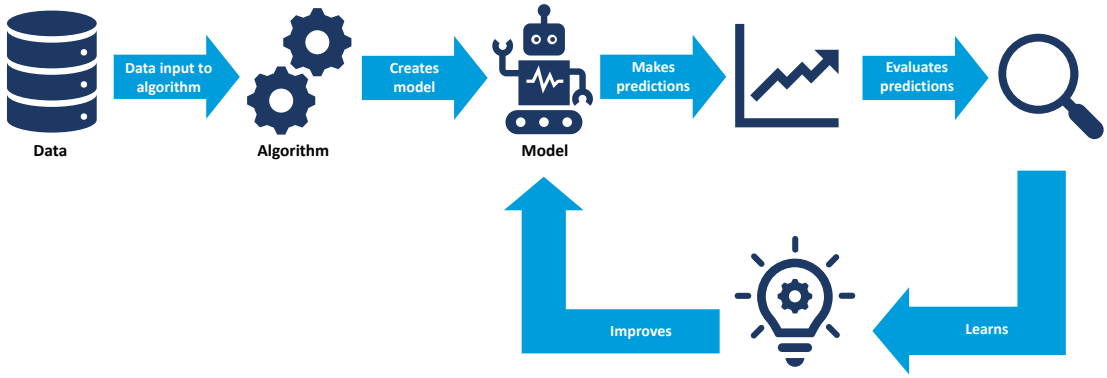


Figure 6-1: The general machine learning process.

Machine learning is the predominant form that AI takes in the business world. Most AI projects in an organization will incorporate some type of machine learning, as a machine learning model is the mechanism by which an AI system attempts to solve its assigned problem set.

There are many types of algorithms that implement machine learning, and each one might be more appropriate than others in certain situations. This course won't go into the technical details, but there are a few more aspects of machine learning worth diving into.

Note: Machine learning is occasionally abbreviated as ML.

Machine Learning Datasets

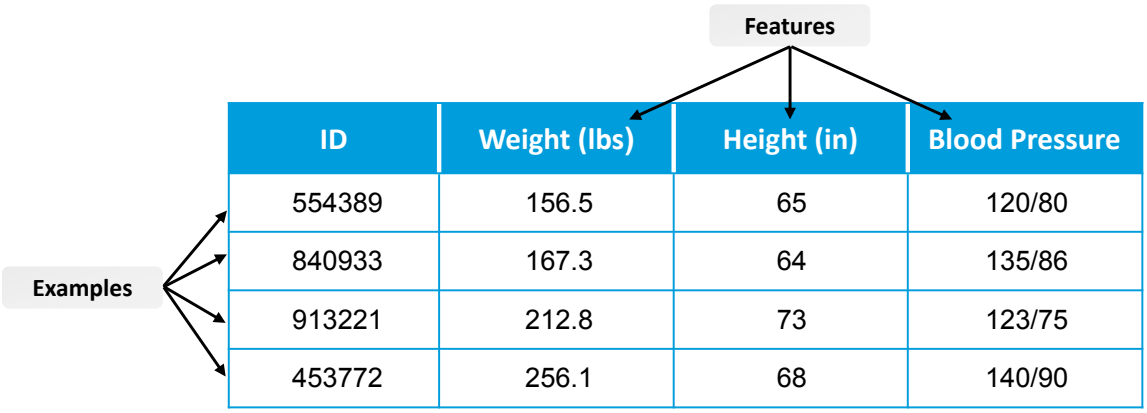
As mentioned, machine learning models are created by putting input data through an algorithm. But what exactly is being "learned" by the machine? Consider the following example: You have a spreadsheet of health data for individuals who participated in a study. The data is in a table, with each person as a row and various characteristics of each person as columns (e.g., weight, height, blood pressure level, etc.). There are two dimensions to this data:

- ***Example***—the individual instance of data. This corresponds to rows in a table, and in the health records scenario, refers to individual people.
- ***Feature***—each measurable property or characteristic of an example. This corresponds to columns in a table, and in the health records scenario, refers to a person's body weight, height, blood pressure level, etc.

Put together, this dataset is fed into the algorithm as input. The resulting model identifies the patterns in the features for each example, and uses these patterns to determine how to make predictions on new data. For example, the model might learn how a person's body weight relates to their blood pressure level—not just for a specific person, but in general. That way, the model can make informed predictions about a new patient based on their body weight and blood pressure level.

The process by which a model learns from input data is called ***training***.

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ID	Weight (lbs)	Height (in)	Blood Pressure
554389	156.5	65	120/80
840933	167.3	64	135/86
913221	212.8	73	123/75
453772	256.1	68	140/90

Figure 6-2: A dataset used to train a machine learning model.

Deep Learning

Deep learning is a type of machine learning that makes decisions using multiple layers of information. These layers are arranged as a hierarchy of decisions or other tasks, where the conclusions drawn from a lower layer can influence the conclusions drawn at higher layers. This makes a deep learning model more complex than a traditional machine learning model.

Consider the example of a model that processes an image of a human face. It will organize aspects of a face into different layers, starting with simple shapes like lines and curves and moving on from there. Eventually, at the highest layers, it will start identifying more recognizable features: eyes, lips, ears, a nose, etc. The deep learning model eventually is able to determine whether or not an image contains a human face based on what it has learned.

Deep learning moves past tabular datasets like the health records shown earlier. Instead, deep learning datasets usually consist of unstructured data—images, video, audio, text, and other data that can't fit neatly into a spreadsheet.

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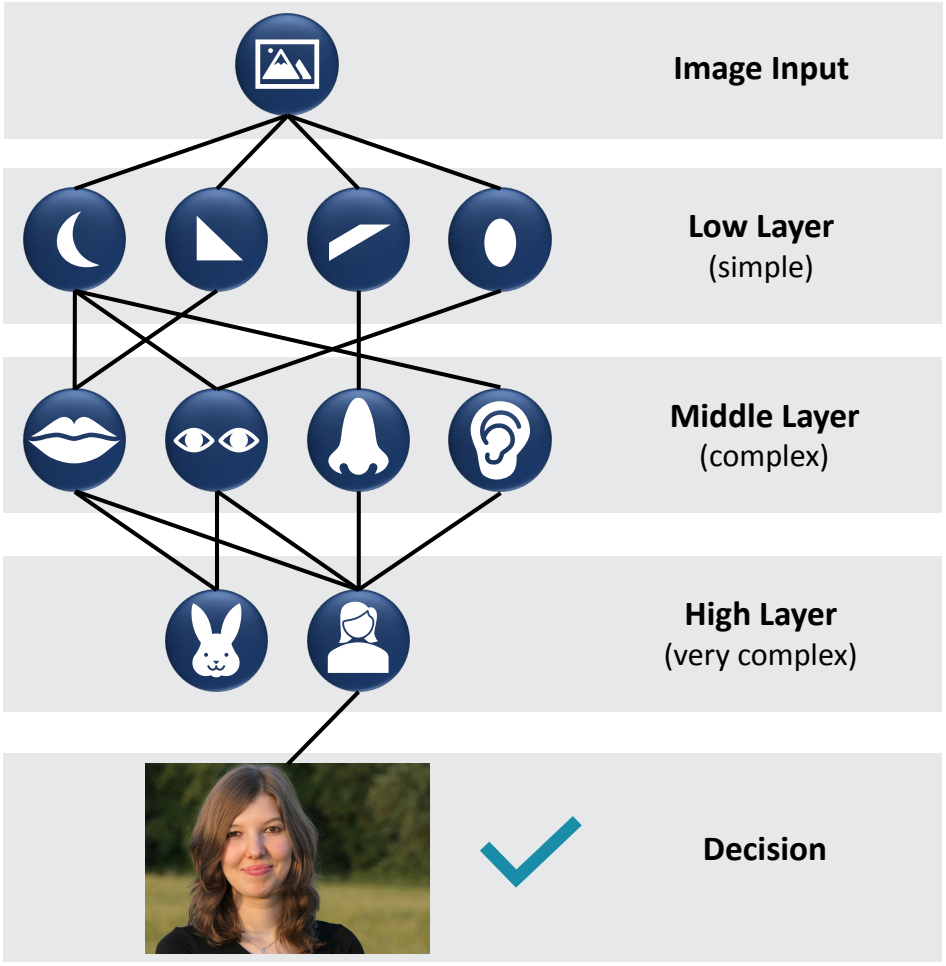


Figure 6-3: A deep learning model builds its understanding of an image through layers.

Note: The "deep" in deep learning refers to the many layers involved in the learning process.

Deep Learning Applications

There are many ways to apply deep learning to solving problems, but most applications of deep learning center on the following disciplines:

- ***Natural language processing (NLP)***: The discipline concerned with how computers analyze and apply human languages. There are actually many sub-disciplines that make up the field of NLP, including text analysis, speech recognition, natural language understanding, natural language generation, and natural language translation.
- ***Computer vision***: The discipline that involves computers understanding images and video at a high level. Like NLP, computer vision is a discipline that drives many more fields and technologies, including object recognition, object classification, image generation, motion detection, trajectory estimation, video tracking, and navigation.
- ***Robotics***: The discipline that involves studying, designing, and operating robots. There is no universally agreed-upon definition of "robot," however. In general, it can be said that a robot is a machine that can act in the physical world with some degree of autonomy (i.e., independent of human control).

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Generative AI

One particularly popular application of deep learning in recent years is ***generative AI***. Generative AI refers to deep learning models that are designed to generate any kind of content—text, programming code, audio, images, video, and so on. GPT-3, DALL·E, and ChatGPT are examples of generative AI because they can create lifelike examples of content.

Most generative models are trained on massive amounts of data that correspond to whatever the model is designed to generate. A model like GPT-3 that generates text is trained on textual sources like scientific papers, textbooks, novels, social media posts, chat logs, and potentially much more. A model like DALL·E that generates images is trained on photographs, hand drawings, computer-generated images (CGI), and more. Either one could use NLP, computer vision, or a mix of the two.

The power of generative AI is that any user, regardless of technical ability, can "ask" the model to create whatever it is the user wants. This is typically done by providing the model an input prompt, usually a short textual description of the thing that you'd like to generate. The model translates the words and phrases you've provided in the prompt into some type of content that (ideally) embodies those words and phrases. You might prompt the model to write a science fiction short story about explorers in space who discover an alien civilization, or you might prompt a different model to create a realistic photograph of a monkey with a small hat typing at a typewriter in an office. You're only limited by how effective the model is at understanding your prompt and translating it into useful content.

At first it might seem like generative AI is more of a toy or novelty, but it actually holds a great deal of potential in business applications. Being able to quickly generate images, such as logos or stock images, can greatly reduce the time and money that it would require for a human to create them. The same is true for text, as being able to quickly generate marketing copy, press releases, or responses to customers is also valuable to many organizations.



Prompt: **a photograph of a monkey typing at a typewriter, wearing a small hat, in an office, realistic, highly detailed**

Figure 6-4: The following images were generated using an open-source model similar to DALL·E called Stable Diffusion. The prompt used to generate the images is also shown.

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Note: Generative models don't provide perfect results every time, and there is a sort of art to writing effective prompts.

Large Language Models (LLM)

A related concept to generative AI is the ***large language model (LLM)***. LLMs are specifically designed to analyze patterns in very large bodies of text in order to determine how best to predict some kind of textual output. LLMs like GPT-3 and ChatGPT are able to determine what word should come next in a sentence based on various factors, including the user's input prompt and the context of the other words that have already been generated.

Machine Learning vs. Deep Learning

The most notable difference between traditional machine learning and deep learning has to do with features. Remember, in traditional machine learning, a model is given a set of features and learns their patterns. In deep learning, you don't explicitly give the model a defined set of features. Instead, the deep learning model itself learns the features inherent in the data. It is able to extract the features from the raw data it is given without human guidance.

Recall the example of a deep learning model determining if an image contains a human face. How would you describe someone's face in an image? You could focus on one aspect at a time, like eyes. You might measure each component of the eye—iris, retina, sclera, etc. You might also record the eye's color. But what if someone is facing a certain direction in one image but a different direction in another? What if the person's eyes are closed? What if the lighting is changing the color of the eye? There are so many variables involved in an image that it's just not feasible for you to do all this yourself.

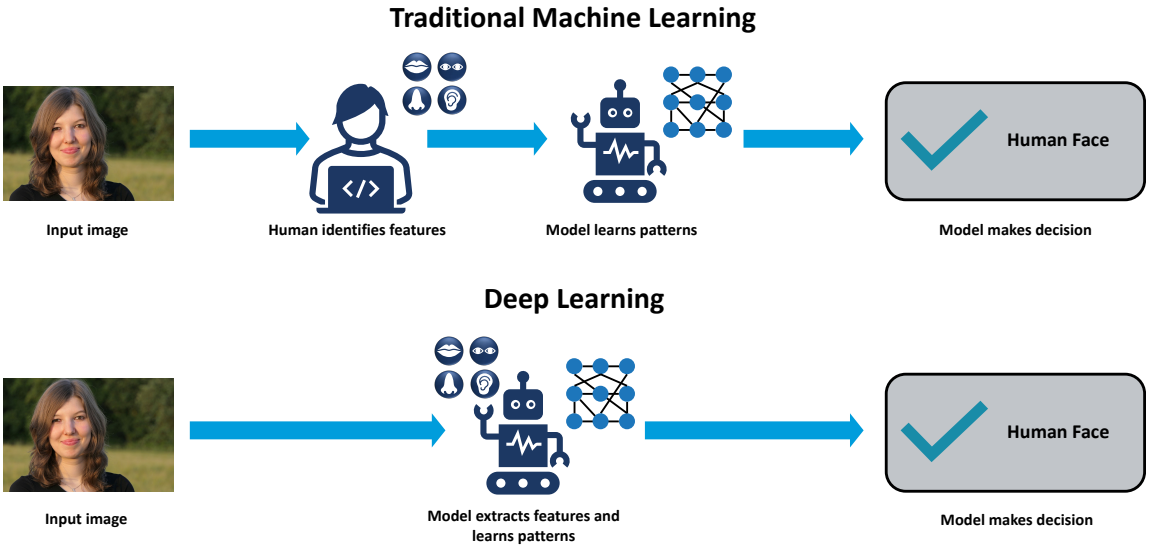


Figure 6-5: Traditional machine learning vs. deep learning. Deep learning is much more appropriate in this scenario, as manually identifying features in an image is usually not feasible.

This is where deep learning shines. The deep learning model has to decide for itself what constitutes a "feature" of a human face. Without a human operator telling it how, the deep learning model has learned what a human face looks like.

The Choice Between Machine Learning and Deep Learning

The choice between traditional machine learning and deep learning is highly dependent on the data you're working with and what you want to do with that data.

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Traditional machine learning is more commonly applied to smaller datasets in which an expert has already determined the features relevant to solving the problem. For example, customer survey responses, patient health information, and bank account activity are some common applications for traditional machine learning because you'd typically know what characteristics are useful for predicting customer preferences, health issues, and fraud.

On the other hand, deep learning is a better choice in situations where it's difficult to determine the characteristics that contribute to a prediction. For example, object recognition in images and video is a common application of deep learning because you don't necessarily know how to precisely describe a complex object. Likewise, deep learning is also used to help machines understand and generate human speech, as language is also a very complex concept. In addition, deep learning is more desirable when you have massive amounts of data, time, and resources at your disposal.

How AI Concepts Relate

You might still be wondering how these three concepts relate to one another. You can think of them as a hierarchy:

- **Artificial intelligence** is the umbrella term that encompasses many different techniques.
  - **Machine learning** is one subset of AI, though a fairly prominent one in today's world.
    - **Deep learning** is a type of machine learning that can solve more complex problems.

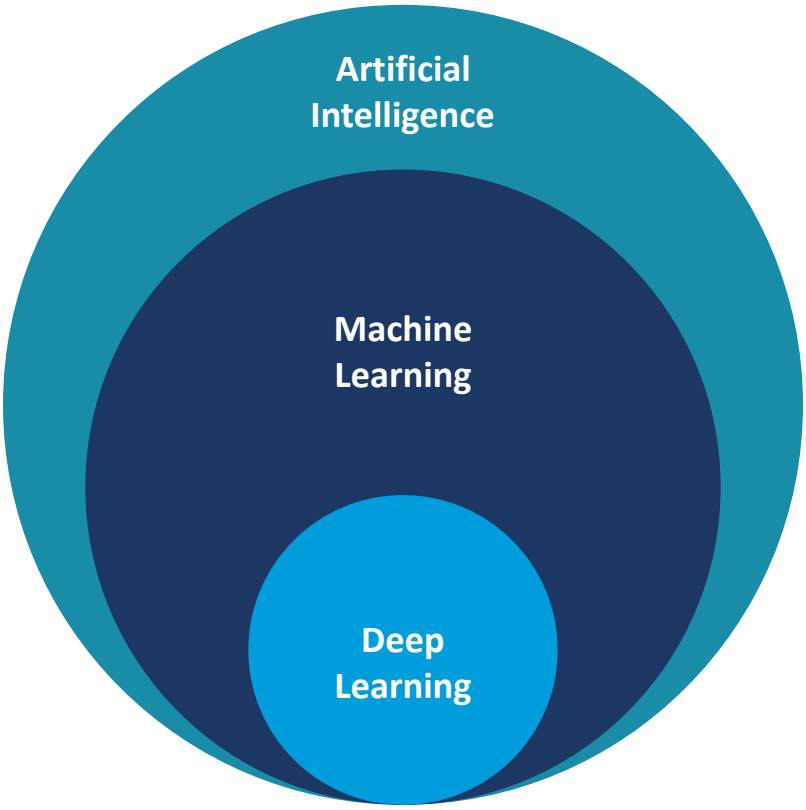


Figure 6-6: How AI, machine learning, and deep learning relate.

Note: In some cases, people use the term "machine learning" to mean the overall discipline, whereas other times they may use "machine learning" to mean traditional machine learning (i.e., not deep learning). Don't be afraid to ask for clarification when you're speaking with someone.

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Machine Learning Approaches

Regardless of whether you're using traditional machine learning or deep learning, you will need to take one of three approaches to building a model:

- Supervised learning
- Unsupervised learning
- Reinforcement learning

It's necessary to understand these approaches from a high level, as they directly affect the kind of problems your AI systems can solve.

Supervised Machine Learning

Supervised learning is an approach to machine learning that involves providing known values of the variable you are trying to predict as part of the input data. Recall the health records example: each data example had several features like weight, height, and blood pressure. You want your model to predict whether or not a new patient has heart disease given these features. In the data you use as training, you have a final column that records a "yes" or "no" as to whether or not this person has heart disease. This is called the ***label***, or the ***ground truth***, and it is a requirement of supervised learning.

By giving the model labels for each data example, it is able to identify the patterns not just among the features, but the among the features as they relate to the very thing the model is meant to predict. The model is able to learn how well (or how poorly) it is able to predict the label, gradually improving as a result.

There are two primary applications of supervised learning you should be aware of:

- ***Regression***, in which a supervised model is attempting to predict some numerical value. Examples include predicting tomorrow's temperature, predicting how many customers will patronize a restaurant on a given night, predicting how much product inventory will be available in the coming months, and so on.
- ***Classification***, in which a supervised model is attempting to place a data example in a class or category. Examples include classifying a patient as having heart disease or not, determining what type of clothing appears in a product image, determining whether or not a job applicant is worth considering for the interview process, and so on.

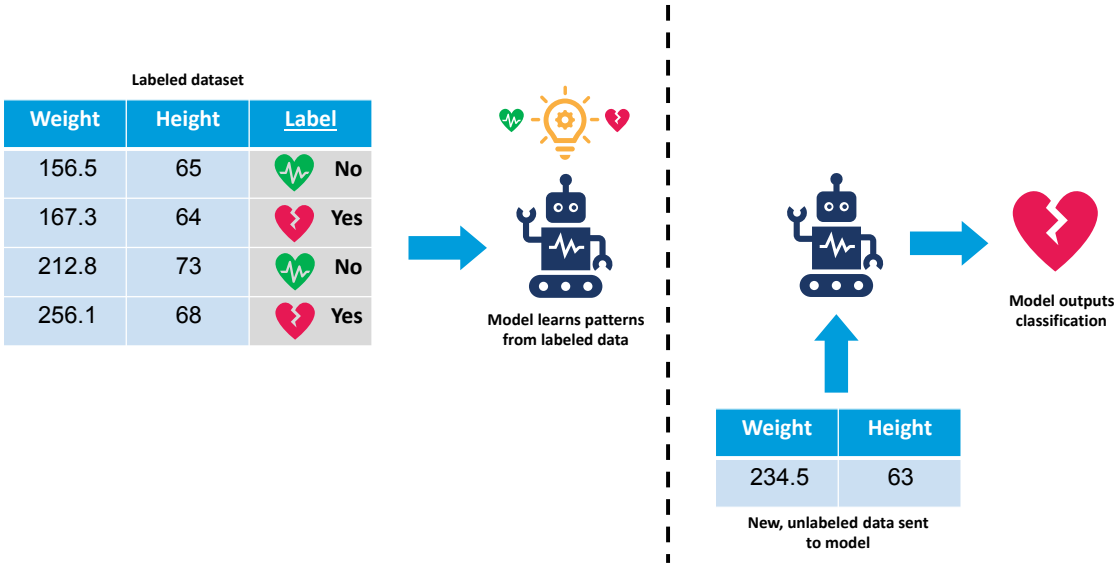


Figure 6-7: Training a supervised learning model to classify a patient as having heart disease or not having heart disease.

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Unsupervised Machine Learning

Unsupervised learning is a machine learning approach in which the model is *not* given label values. In other words, the model isn't explicitly told what it needs to predict, as this information is not provided as part of the training data. The model is still given the examples and a set of features, but no label. There are many reasons why the label might be absent—maybe it wasn't collected with the rest of the data, or maybe labeling the data is infeasible. In any case, a dataset without ground truth cannot be applied to regression or classification problems.

Without a set of labels to learn from, an unsupervised model must instead learn from the patterns in the features. The main application of unsupervised learning is called ***clustering***, and it identifies commonalities by grouping like data together into clusters. The model can also consider how these commonalities exist in new sets of data in order to group them into their relevant clusters.

Unsupervised learning is useful in areas such as customer segmentation. There isn't necessarily a specific thing that the business wants to predict; they just want to place customers into distinct groups so they can tailor their customer service and marketing approach to customers based on their attributes and behaviors.

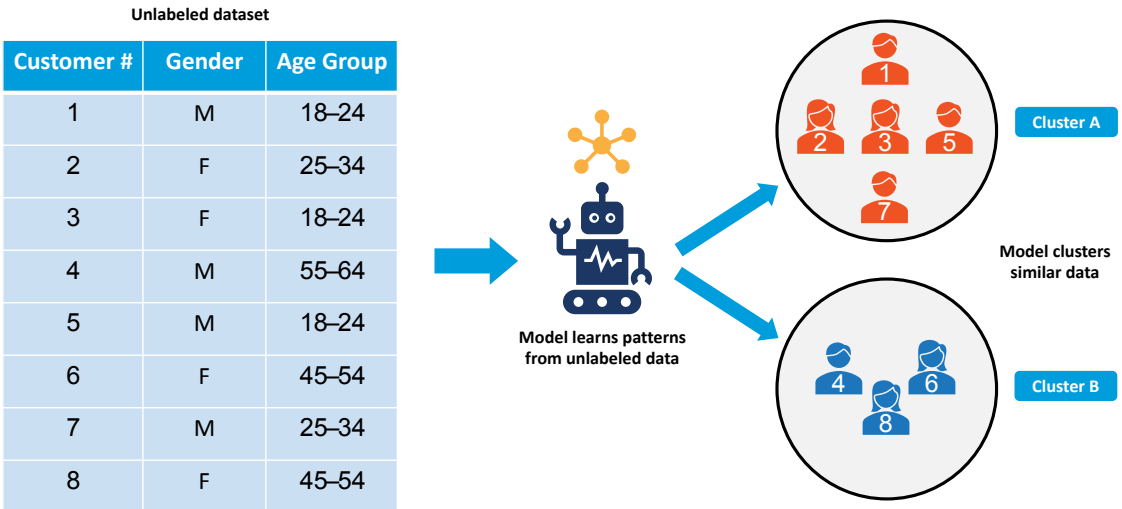


Figure 6-8: Training an unsupervised learning model to cluster customers.

Reinforcement Learning

Reinforcement learning, the third of the main machine learning approaches, is the process by which a software agent acts in an environment in order to obtain a reward. This involves the following steps:

1. The agent selects an action from an available set of actions.
2. The agent takes the action in the environment.
3. The environment changes state in some way.
4. The agent is rewarded or not, depending on how its action affected the state of the environment.
5. The process repeats.

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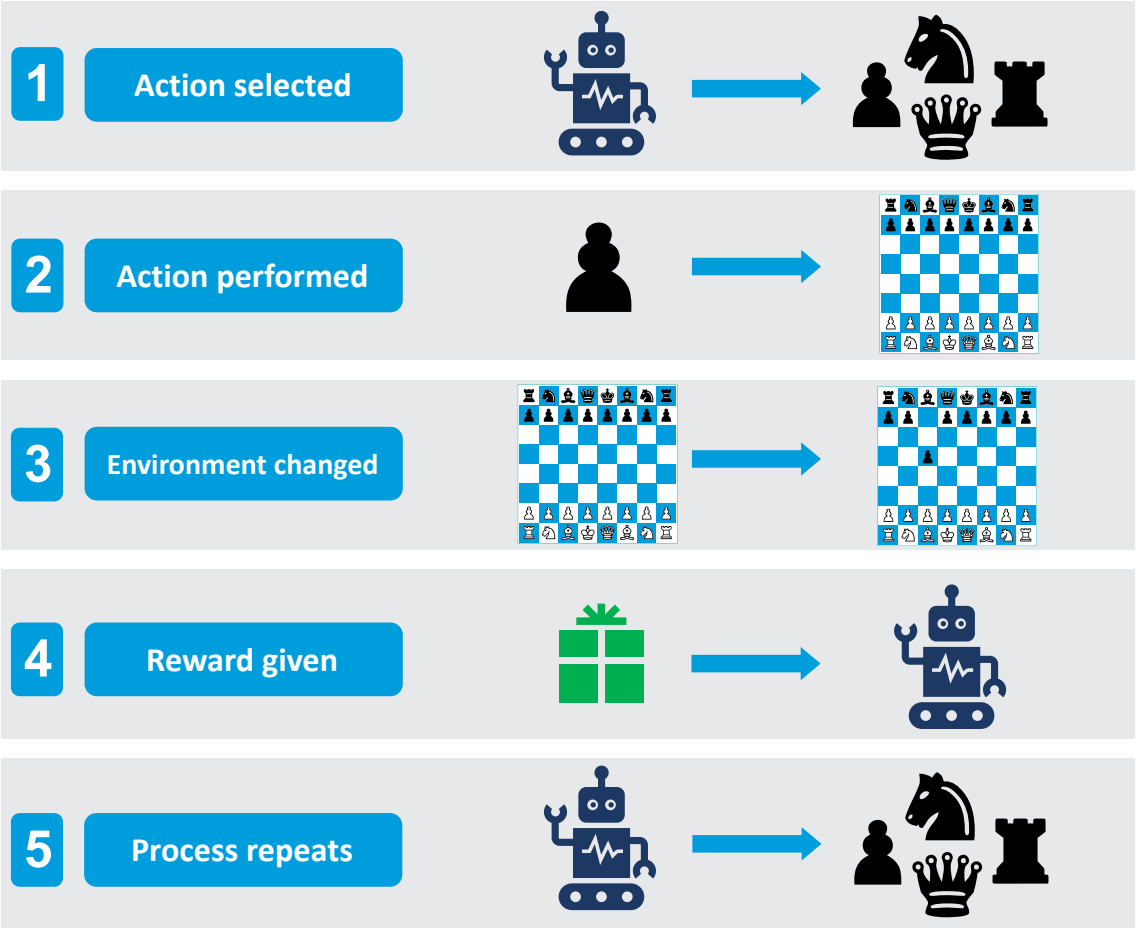


Figure 6-9: A machine being trained to play chess using reinforcement learning.

Ultimately, the goal of the agent is to maximize the amount of rewards it accumulates. So, it will learn which actions generate the most rewards, and thereby become better at interacting with the given environment. A reward is basically a function that indicates to the agent how it should act—i.e., what actions are desirable and worth repeating in the future.

Reinforcement learning is optimal in situations where the machine can learn effectively through trial and error. Robots that move or perform other independent actions in the physical world often use reinforcement learning, as do computers that play games like chess, checkers, and Go. While less popular than other forms of machine learning in the business world, reinforcement learning is still used in areas like personalizing customer experiences, planning optimal travel routes, and optimizing financial investment and trading decisions.

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ACTIVITY 6-2
Discussing AI Concepts

Scenario
Consider the following discussion questions to help review your understanding of foundational AI concepts.

1. Your marketing department started a campaign over a year ago in which customers were encouraged to take pictures of themselves with your company's products and post those pictures on social media. Now you have a very large collection of images, and you want to develop an AI model to categorize each different type of product in those images. That way, you can identify which products are the most popular among your highly engaged customers.

   Which technique, traditional machine learning or deep learning, would be most suitable to achieving this business objective? Why did you make this choice and not the other?

2. Over the years, you've acquired various data that describes your customers. The data includes their demographics and their shopping habits. There is no particular aspect that you're trying to predict; you just want to put your customers into groups to make it easier to market to them. Someone on your team could do this manually, but you think an AI model would be smarter and quicker at doing it.

   Is this an example of supervised, unsupervised, or reinforcement learning? Explain why.

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Summary

In this lesson, you identified the fundamental concepts that comprise artificial intelligence, including the prominent disciplines known as machine learning and deep learning. With this foundational knowledge, you'll be better prepared to facilitate AI projects and other initiatives in your organization.

What sort of additional research, if any, do you plan on doing to learn more about the concepts behind AI?

Does your organization currently use, or plan to use, machine learning or deep learning techniques? How? What about you personally—how are you affected by AI in your daily life?



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7 Functions of AI in Business

Lesson Time: 1 hour, 30 minutes

Lesson Introduction

Artificial intelligence (AI) has many applications across a wide variety of industries and sectors. In this lesson, you'll explore several of the most common ways in which AI enables new business functions or enhances existing functions. This will get you thinking about how AI can fulfill your own business needs and goals.

Lesson Objectives

In this lesson, you will:

- Identify how AI can be used to improve the user experience.
- Identify how AI can help segment audiences.
- Identify how AI can help secure organizational assets.
- Identify how AI can help optimize business processes.

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TOPIC A

Improve User Experiences

The first set of business functions you'll look at are those that improve the user experience by leveraging AI. These types of improvement are great at increasing customer engagement and promoting brand loyalty.

Natural Language Processing

Recall that one of the applications of deep learning is natural language processing (NLP), a discipline concerned with how computers analyze and apply human languages. Some of the sub-disciplines that make up the field of NLP include:

- Text analysis
- Speech recognition
- Natural language understanding
- Natural language generation
- Natural language translation

If you've ever used a text-to-speech or speech-to-text program, you've leveraged technology that is made possible by one or more of the previous NLP techniques. Computers can attempt to recognize language whether it is written or spoken, then analyze that language for familiar linguistic traits like syntax, semantics, phonetics, grammar, spelling, and more. In some applications, computers can use this analysis to generate new speech or written text that approximates how humans actually talk or write.

NLP seeks to bridge the gap between how computers communicate and how humans communicate. It can enable people to more easily understand machines, and vice versa. NLP also helps people with disabilities communicate with other people through the use of a computer.

Chatbots

A ***chatbot*** is an NLP system that can simulate real-time communication with users over a text-based chat. In the past, rudimentary automated chat systems would just give canned responses to certain questions; on the other hand, AI-driven chatbots actually exhibit a degree of intelligence in how they interpret and respond to user messages. Even though the same basic idea can be communicated in many different ways, a chatbot will be able to "understand" the core idea behind a user's message and then decide which response is most suitable.

There are many reasons why you might implement one or more chatbots in your organization. Some major examples include:

- **Customer service**: Chatbots can alleviate the burden of your customer service team by being the first line of contact between the customer and the organization. The user might be directed to a contact form on your website or app, which then runs the chatbot to interact with the user. The chatbot then interprets the customer's question or concern and provides the appropriate response, such as a troubleshooting tip or who to contact for more information. In many cases, businesses use customer service chatbots to filter out customers who have common issues; only the most unique or difficult questions are escalated to a human representative.
- **Appointment scheduling**: Businesses that require their customers to book appointments, such as in the healthcare industry, can implement chatbots to streamline the process. This might be a more attractive option than having the customer select an available time from a list or calendar, as the chatbot can be more helpful at offering alternative times or responding to queries.
- **Human resources management**: There are some tasks that employees regularly engage HR with that can be automated through the use of a chatbot. For example, employees could schedule

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vacation time or sick time with the chatbot. Again, this might be preferable to filling out a form, since the chatbot can be trained to identify issues with the request, such as someone trying to take an entire month off in a busy part of the year.

- **Information research**: Some industries require quick access to accurate information, such as an automotive engineer researching information about parts that will be used in next year's model. An organization can build a chatbot to simulate expertise in one or more fields by taking a user's query and determining how best to extract the desired information from available data stores. This can greatly reduce the time an employee spends looking up information that is hard to find.

ChatGPT

In November 2022, OpenAI released ChatGPT. ChatGPT is a generative large language model (LLM) based on GPT-3 (specifically, version 3.5) that represents a major leap forward in chatbot systems. It is able to take user input and produce responses that are more realistic and useful than any chatbot before it. Part of the reason why is because ChatGPT is able to "remember" earlier parts of a conversation, enabling it to maintain important context, whereas most other chatbots consider each user message independently when forming a response.

ChatGPT has many use cases in business beyond the chatbot use cases previously mentioned. It can also be used to write essays, articles, technical documentation, and even working code. However, ChatGPT is not without its flaws. If the user asks a factual question, the model may return an answer that sounds very plausible, but is incorrect. Still, it has been shown to exhibit an impressive ability to mimic human communication. OpenAI plans to continue work on the model and improve it in the years to come.



Note: For more information on ChatGPT and to try it out, see **https://openai.com/blog/chatgpt/**.

Agent Assistance

Agent assistance is automation technology that helps human agents in a call center do their jobs more efficiently and effectively. This can take many forms, including:

- **Call context**: Calls in a call center are often transferred between agents or departments, and customers can become frustrated when the next person they talk to doesn't have all of the relevant information. An agent assistance system can provide the proper context of a call to any agent the call gets transferred to, including who the customer is, what their general issue is, and any other pertinent information—all of which changes dynamically as the customer is routed from agent to agent.
- **Immediate responses**: Just like a chatbot can address simple or common customer service requests, an agent assistance system can automatically send the proper response to a customer—whether it's greeting them, saying goodbye, apologizing for any inconvenience, prompting them for more information, and so on. Not only does this make the human agent's job easier, it means the call goes quicker and is less likely to waste the customer's time.
- **Suggested responses**: Instead of just replying automatically using a pre-recorded message, the AI system can suggest to the human agent how they should respond to the customer. This suggestion can take the customer's query into account while also looking up useful information in a centralized repository. That way, the human agent doesn't waste time digging through a database for the proper information.
- **Live transcription and summarization**: Using various speech recognition techniques, the system can transcribe the call to make later reference easier for both the system and a human agent. Alternatively, the system can create a summary of the call, including the key points of what was discussed and whether or not the issue was resolved. That way, the agent can quickly reference a past call if the same customer calls again.

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Sentiment Analysis

Sentiment analysis is the technique of determining the attitudes and feelings someone has toward something. In the world of business, this usually means assessing how a customer feels about the business or its products and services. This can be everything from a simple positive or negative sentiment—the customer does or does not like their experience—to more complex classifications like displeasure, disgust, ambivalence, satisfaction, happiness, and so on.

Most sentiment analysis incorporates NLP, as most consumers' opinions are expressed through direct communication—usually through social media or a review on a website. The NLP system can pick up on certain key words, as well as the context in which those words appear, to determine how the person feels. For example, a review that begins with the sentence, "I usually love this company's products, but . . ." would probably be classified as a negative sentiment if the AI is properly trained.

There are various reasons to apply sentiment analysis in an organization, including:

- **Customer retention**: Obviously, unhappy customers are less likely to continue to do business with you. You can attempt to address their misgivings in various ways, like offering them coupons for future purchases. However, customers don't always express these sentiments in ways that are easy for a human to assess, which is why NLP systems with sentiment analysis capabilities are an effective alternative.
- **Employee retention**: Just as customers express their satisfaction or dissatisfaction with a product, so too do employees express these sentiments about their jobs. You might use an AI system to assess the tone of a meeting transcript (or the live meeting itself) to determine how your workforce is generally feeling. You might then choose to assign resources for morale-boosting exercises if morale is low.
- **Product development strategy**: The public can influence the strategic direction your company takes with regard to product development. For example, if people are posting positive opinions on social media about certain types of products you sell, you may choose to focus on developing more versions of those products while reducing development effort on products that have not been as warmly received. Assessing the sentiment of thousands or even millions of social media posts is a job for an AI, not a human.
- **Recommendations**: Sentiment analysis is also used to recommend products and services to people. If a user posts a positive review of a movie on a streaming service, an AI system may be able to recommend similar movies to them. And vice versa, the system would take similar movies out of the recommended pool if the user posted a negative review.

Emotion AI

Emotion AI is very similar to sentiment analysis. Some experts see them as different terms referring to the same thing, and some see one as a subset of the other. However, many AI researchers are starting to draw a distinction: Sentiment analysis focuses on analyzing language through NLP, whereas emotion AI is mostly concerned with analyzing facial expressions and other facial features. Both, however, have the same purpose: to determine how someone feels about something.

When analyzing facial features, emotion AI applies computer vision techniques like object recognition and object classification. For example, it can detect the furrowing of a person's brow to interpret this to mean anger or some other show of displeasure. Emotion AI can also incorporate motion detection as part of its analysis. For example, it can detect the difference between a smile and a laugh by looking at the movement of the person's jaw rather than just focus on the shape of the mouth.

The applications of emotion AI in business are similar to those of standard, NLP-based sentiment analysis. However, it has proven more difficult to implement, not to mention controversial. After all, many people are uncomfortable with the idea that a business is taking a picture or video of their face and using it for marketing purposes. So, you may have less opportunity to use emotion AI for things like customer retention, recommendations, and so on.

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Note: Emotion AI can also incorporate the analysis of more subtle physical cues, like the tone of someone's voice.

Recommendation Systems

A ***recommendation system***, or recommender system, suggests items, services, and other things of interest to users. Recommendation systems powered by machine learning are able to more accurately determine what most interests the targeted user(s), or what recommendations users will respond most positively to. Ideally, this makes the system more useful to both the user (who is more likely to see something they like) and the business (who is more likely to generate sales or interest in their brand).

Recommendation systems typically work on the principle that, faced with an overwhelming selection to choose from, most users would rather have those selections filtered to show only the few that are most relevant to them. This can be everything from what product the user might want to buy next from a store, to what movie a user might want to watch next on a streaming platform, and much more.

As mentioned before, recommendation systems can rely on sentiment analysis, but they can also rely on other behavioral attributes. For example, if a user watches a movie on a streaming service for only a few minutes and then quits to watch something else, it could indicate to a recommender model what the user is or is not interested in seeing next.

Some examples of recommendation systems in business include:

- **Content recommendation**: The aforementioned example of a streaming service suggesting a movie to watch is a classic example of a recommendation engine. The business isn't necessarily generating more direct sales, but it is increasing engagement, which can lead to better customer retention.
- **Product recommendation**: Other than recommending content within an existing subscription, recommendation systems can be used to suggest any type of product or service to a customer whose attributes are known to the model—their demographics, how they interact with the business, etc. This usually leads to increased sales, as customers are more likely to buy the product or service relevant to them.
- **Social networking**: Recommendation engines can "match" people who have common interests or goals. In the business world, this can mean bringing people with relevant skills and talents together to work on a project or brainstorm ideas. Recommending the right people can lead to more effective results and reduce a lot of time spent searching for qualified personnel.
- **Research**: Academics, scientists, and other people in technical or information-oriented fields may spend a lot of time on research. Recommendation systems can find high-quality results that pertain to the person's area of research much faster than any human. The results themselves may also be of better quality than what a human could find.

Digital Personalization

Recommendation systems are just one way that businesses use AI to personalize the user experience. There are many other facets to digital personalization, including serving specific content (e.g., search results that are of specific interest to the user), hiding specific content (e.g., search results that are not of interest to the user), rearranging content on an app, changing the visual theme of an app, and so on. All of these changes can be made at a group level or even down to individual users.

AI systems determine how to personalize user experiences based on a number of features that build a user's profile. Aspects of that profile may include:

- The user's geolocation.
- The user's race, gender, and age demographics.
- The user's socioeconomic status.

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- The user's browsing or online shopping history.
- The user's general sentiment toward the business.

The system can tailor the user's experience in such a way that maximizes engagement with the app through metrics like click-through rate (CTR) or watch time. The model improves as it learns what experience is best suited to what users.

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ACTIVITY 7-1

Improving User Experiences Using AI

Scenario

Consider the following discussion questions to help review your understanding of how AI is used to improve user experiences.

1. Your organization's customer service team is constantly flooded with calls—so much so that some customers have to wait a very long time to speak to an agent, and some of the calls even go unanswered. The agents themselves are also overworked and have been making mistakes. Your customers have voiced their frustration to you and on social media.

   What AI system(s) could help solve this problem to improve the customer experience, and how?

2. Your online store currently gives recommendations to customers based on their past purchases. These recommendations are rather basic—for example, the customer orders a product in Category X, so the system recommends a bunch of other products in Category X. This is only so helpful, and your customers have complained that it's hard to find what they're looking for.

   How can an AI-driven recommendation system provide more effective product suggestions to customers?

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TOPIC B

Segment Audiences

Another business function enhanced by AI is audience segmentation, also known as customer segmentation or market segmentation. This can lead to better customer retention and higher sales, among other benefits.

Customer Segmentation

Businesses focus their marketing efforts on specific segments of their customer base. Traditionally, these segments have been created using overt customer attributes, particularly demographics like age and gender. The two downsides of this approach are: it ignores more subtle factors that might be useful in grouping customers; and it can only handle a small number of factors, reducing the complexity of the groups.

Both of these concerns are addressed by AI-powered systems that do customer segmentation. They are able to build more intricate and useful groups by considering hundreds, perhaps thousands of factors that go into what makes a "customer"—not just overt factors like age, but also factors like lifestyle, general worldview, sentiment toward the business, and so on.

There are several ways AI can intelligently segment customers, but perhaps the most common tactic is to cluster customers according to a set of their features. In this unsupervised approach, you can train a model to place customers into one of several groups, where customers in a group have similar values for *all* relevant features.

One interesting aspect of clustering is that you can engineer the clusters in different ways. You can set a hard number of clusters from the beginning—for example, if you only want to work with a total of nine groups, you can tell the model to assign a customer to one of nine clusters. Alternatively, machine learning practitioners can analyze a model using various metrics to determine the "optimal" number of clusters. You may go this route if you don't have a specific number of segments in mind, and want to make sure each segment is as distinct as possible.

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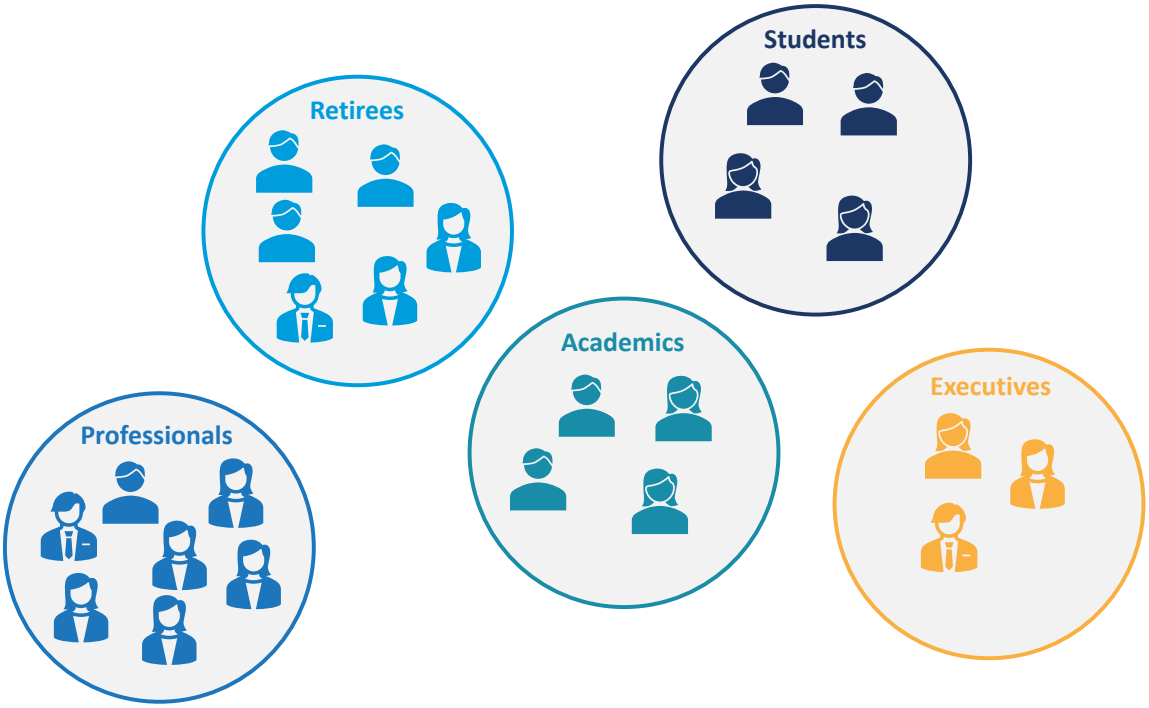


Figure 7-1: Customers segmented using a clustering model. Note that the names of each cluster are based on an analysis of the customers in each cluster.

Predictive Segmentation

Unsupervised clustering is not the only way AI can improve customer segmentation. Depending on your available dataset(s) and what you're trying to accomplish, you might also be able to use supervised techniques like classification and regression to generate customer segments based on some label that the machine learning model predicts.

For example, you have a dataset of many customer attributes, and one of those attributes you consider to be the label you're trying to predict. Perhaps it's "Did the customer return to make more purchases, yes or no?" You want to be able to predict whether future customers will return to purchase more items. So, this is a classification problem. But, you can also parlay the results into an opportunity for customer segmentation. Customers the model predicted will return could go in one segment, whereas customers the model predicted will not return could go in another. Or, you might incorporate the predictions into a clustering model to combine both unsupervised and supervised learning. In either case, the predictive system has improved your ability to segment customers.

Targeted Campaigns and Customer Profiles

Targeted campaigns are built from customer segments. One segment receives one campaign, whereas another segment receives a different campaign. But AI can be involved in more than just the direct segmentation aspect—it can also be useful in shaping the content of the campaigns according to the characteristics of each segment they target.

Consider a scenario where you use unsupervised learning to cluster your customer base. Each customer is put into, for example, one of nine clusters. But, clustering by itself does not label each cluster in a meaningful way. It simply does the grouping. So, how will you shape your marketing campaigns to accommodate each cluster?

Cluster analysis can help you learn more about the common attributes of the customers in each cluster. What age group do the customers fall into? What is the most common gender in the cluster? What general location are the customers from? A practitioner can perform statistical analyses on these factors and any others that are included in the input data to help generate a customer profile

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or persona. The profile/persona describes each segment using words rather than just raw numbers. It is not a depiction of a real person, but a depiction of a fictitious person who embodies the traits common to people in that segment. Using profiles and personas, you can start to get a better idea of what language and content might be most appropriate for a certain campaign.

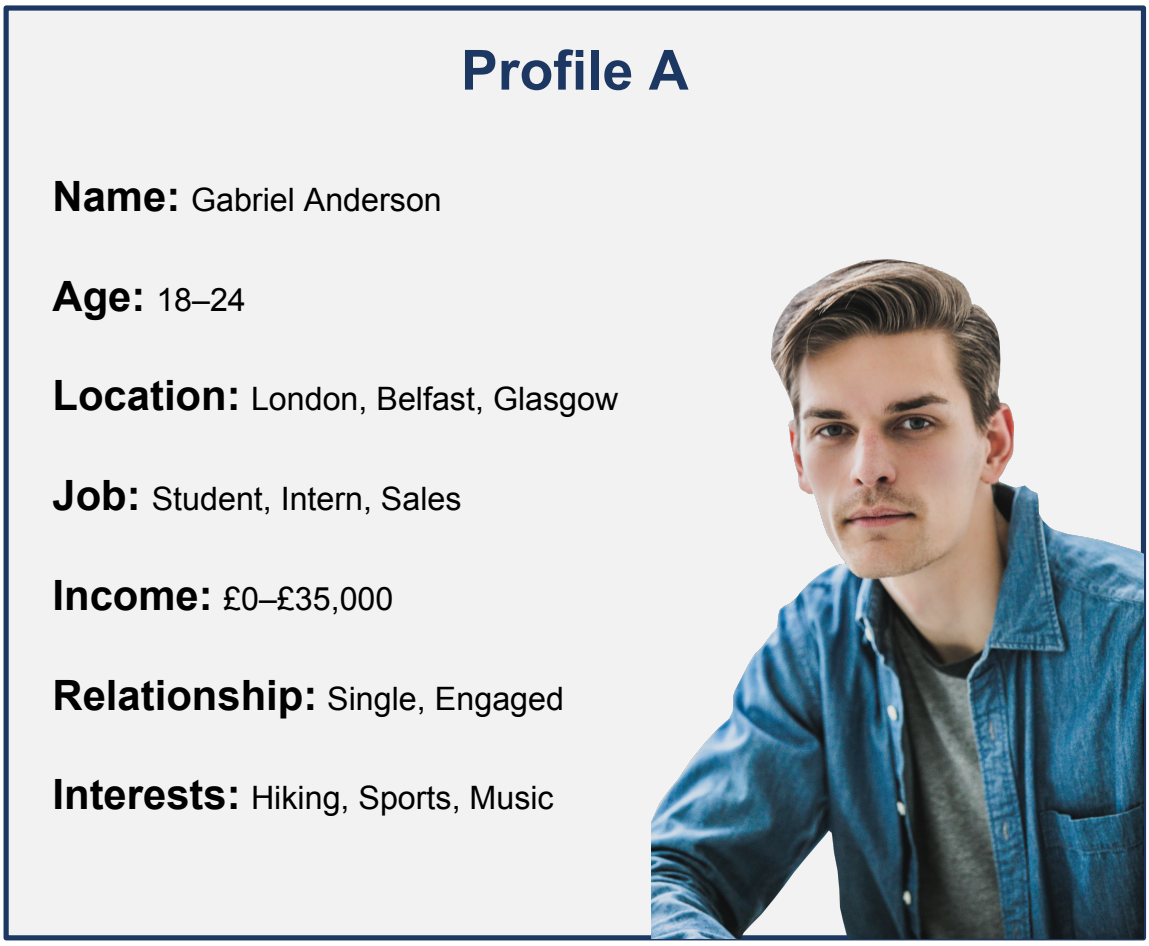


Figure 7-2: A customer profile built from traits common to a specific market segment.

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ACTIVITY 7-2

Segmenting Audiences Using AI

Scenario

Consider the following discussion questions to help review your understanding of how AI is used to segment audiences.

1. What machine learning technique can help you place customers into different groups for the purposes of market segmentation? Provide a basic explanation for how the technique works.

2. How can a machine learning model that makes predictions be used to enhance customer segmentation?

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TOPIC C

Secure Assets

Another major application of AI in business centers on security. Intelligent security systems can protect key assets against attack and exploitation, preventing the business from losing money, time, and its foothold in the market.

Risk Management

Risk management is the cyclical process of identifying, assessing, analyzing, and responding to risks. It is one of the most fundamental processes in an organization's cybersecurity operations, as risk permeates all aspects of business.

Examples of AI applied to risk management include:

- **Risk detection**: The first step in addressing risk is to actually identify what is a risk and where it's coming from. An AI system can assess various dimensions of the organization's operations to identify risks that have a historical basis. It can also continuously monitor the organization for new risks as they appear.
- **Risk analysis**: Revealing the attributes of a risk can be difficult because they are often complex and subtle. However, AI systems that have been trained on historical risk attributes may be able to categorize or cluster certain behaviors or entities in ways that make it easier to understand the risk. For example, an AI-powered analysis might be able to show how features like system vulnerabilities, equipment failure rates, network instability, etc., can lead to risk.
- **Risk scoring**: There are several methods for scoring risk, or evaluating its likelihood and impact in one calculation. AI systems can accurately assess the factors that contribute the most to likelihood and impact, such as the information revealed through a risk analysis. The output of the model might be a literal numeric score, or it might be more qualitative in nature (e.g., low risk vs. medium risk vs. high risk).
- **Risk response**: Knowing about risk is one thing, but knowing how to respond is just as important. The organization can avoid, transfer, mitigate, or accept a risk. Choosing which approach is not always obvious, but a classification model may be able to suggest the optimal approach with more confidence than a human analyst.

Threat Analysis

Threat analysis, also called *threat modeling*, is the study of people, events, and technologies that can harm an organization's assets—its data, networks, systems, employees, etc. Modern cybersecurity threats are growing more and more complex; many organizations dedicate time and resources to uncovering information about those threats so that the organization can better prepare its defenses.

Threat analysis has been a staple of organizational cybersecurity for many years, but only recently have security experts begun to leverage AI in this area. AI-based threat analysis systems can not only enhance existing capabilities, they can also deploy novel approaches to detecting and assessing threats.

Some example applications include:

- **Threat classification**: Using a supervised classification model, AI can help to categorize threats in many different ways depending on the data used as input. Common features that describe a threat might be its vector (i.e., the path it takes during an attack), its level of severity, its direct and indirect target(s), the projected damage it will inflict (e.g., in lost revenue), its point of origin (whether physical or virtual), and many more. The simplest classification result would be to mark the entity or event as either "is a threat" or "is not a threat," but more sophisticated systems can classify threats in more qualitative ways. For example, the system might label an event as a web-

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based attack, a password cracking attack, a data exfiltration attack, a backdoor attack, and so on. This can help you prioritize certain threats over others, or provide guidance on how to deal with specific types of threats.

- **Threat ranking**: An alternative approach would be to rank a threat instead of classifying it using qualitative terms. A threat can be ranked by its severity or its likelihood of success, and the cybersecurity team can triage threats according to these rankings. For example, a labeled dataset might assign a severity "score" to each threat, which a regression model can use to calculate the scores for new threats.
- **Threat segmentation**: Obtaining or assigning a meaningful label to cybersecurity threats can be difficult, which is why unsupervised clustering approaches are also commonly used to place threats in groups. An analyst can extract insights from the nature of each threat according to how it was assigned to a cluster, which can in turn prompt the cybersecurity team to take certain preventative or responsive actions.
- **Threat research**: Like other forms of data extraction and research, a machine learning model can greatly assist human cybersecurity personnel in retrieving relevant information about threats much more quickly and effectively than they would be able to do on their own. This keeps the personnel focused on proactively addressing threats rather than wasting time combing through voluminous threat databases.

IT Asset Management (ITAM)

IT asset management (ITAM) is the process of identifying, inventorying, tracking, and maintaining information technology assets as they are used across the organization. The purpose is to ensure that these assets are being used effectively across their entire lifecycle and to reduce the incidence of lost, damaged, or inoperable equipment.

Depending on the size of the organization and its available IT personnel, traditional ITAM processes may be inefficient, tedious, or even non-existent. AI can enhance ITAM in several ways, including:

- **User requests**: Users put in requests for new and upgraded hardware and software, and it can be difficult to prioritize (or deny) certain requests as compared to others. An AI system can consider numerous features, such as the user's job role, their daily duties, their history with IT assets, the number of available assets relevant to the request, etc., to either rank requests or classify them in different ways. That way, IT personnel aren't spending a lot of time deciding who should get what and when.
- **Fault prediction**: Some AI systems can continuously or periodically scan assigned devices to perform a "health check" of sorts. For example, the system might keep track of a laptop's lifetime service, the number of reads/writes to the storage drive, the presence of malicious software, the score it receives on certain computing benchmarks, and so on, to predict the likelihood that the laptop will need to be replaced soon. That way, the IT team doesn't have to wait until the user informs them that they need a replacement, avoiding a situation where the user is stuck without a functioning device waiting for a replacement to arrive.
- **Asset procurement**: When ordering a new asset or a replacement, IT personnel can struggle to choose the right product or the right vendor to purchase it from. AI systems could be trained to identify the most suitable way to procure an asset based on the recipient's features, the features of any asset it is replacing, or features that describe the daily tasks that the asset will need to perform.
- **Compliance issue identification**: Most organizations have acceptable usage policies that they expect employees to follow while using assets like equipment and software. An AI system could be trained to spot behavior that violates these rules and either block the behavior or notify a member of the IT team. If the source of compliance is a law or regulation instead of just organizational policy, having an AI-powered auditing process becomes even more attractive.

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Fraud Identification

Fraud is a major problem in certain industries and sectors, particularly banking and other financial institutions. Poring through thousands or millions of transactions to look for instances of fraud is just not feasible for most organizations, so AI systems are put in place to streamline the process.

Some examples of fraud identification using AI include:

- **Transaction assessment**: Banks use fraud detection systems to identify transactions that are likely to indicate malicious actions taken against users or the bank itself. The AI system can draw from historical records of accounts or transactions that are known to engage in fraud, which it uses to classify activity as being legitimate or fraudulent. Or, it can rank activity on a sliding scale, where the activity has some *likelihood* of being fraudulent.
- **Outlier detection**: In the absence of labeled fraud data, AI-powered fraud detection systems can also learn to divide accounts or transactions into groups that can indicate suspicious activity. For example, most transactions might be placed into one of several well-represented groups, whereas a few outliers are in one or more small groups. These outliers might suggest fraudulent behavior since they deviate from the norm so significantly.
- **Signature analysis**: AI systems can be trained with computer vision capabilities to identify fraudulent signatures on official documents like checks, contracts, tax forms, and ballots. A deep learning model can pick up on visual indicators of a forged signature that are too subtle for human eyes to detect.
- **Identity theft detection**: Some AI systems build profiles of typical behavior for their users, and any pattern of behavior that deviates from this profile can indicate identity theft. For example, a bank might cluster users according to their purchase history. If the model abruptly reassigns the user to a different cluster due to some change in their pattern of behavior, it could indicate that a malicious user has hijacked their account.

Security Notifications and Alerts

Security professionals and other IT personnel need to stay well informed about potential issues in their IT environments, particularly when it comes to behavior that could indicate a breach. However, the challenge comes in finding the balance between too few alerts (leading to missed problems) and too many alerts (leading to time wasted on false alarms). Thankfully, AI can provide support.

Some ways in which AI can improve security notifications and alerts are as follows:

- **Alert sorting and filtering**: AI system can sort and filter alerts to minimize false positives and maximize true positives. The nature of the alert and how such alerts are usually addressed by the security team might go into its decision to suppress or emphasize an alert.
- **Security dashboards**: Getting random alerts out of context will slow down personnel's response to an event and potentially cost the organization precious time in an emergency. However, an AI system that can quickly and effectively summarize an issue in a dashboard will be helpful to personnel who review the alerts. The system can determine which components and metrics are most helpful in displaying on the dashboard, and how (e.g., through a visual graph or as raw numbers/words).
- **Notification dissemination**: Sending an alert to every member of the security team, or to all of the devices and accounts the recipients use, is not a good use of an alert system. It can waste people's time and lead to communication issues. An AI system can be trained to categorize an alert based on several features, including its severity, which systems it affects, and so on, to determine which individuals are the most important to contact, and where they are most likely to need to see the alert (e.g., on their phone vs. on their laptop).
- **Remediation suggestions**: Security personnel might be satisfied with being informed about a potential issue, but an AI system can go one step further and suggest how to remediate the issue. Based on the cause of the alert or the behavior it describes, there may be one or a few common solutions. The security team can take these into consideration when they receive the alert, enabling them to act more quickly.

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Cloud Security and Virtualization

Cloud security is another business function that can be enhanced by AI. It includes everything from controlling access to a cloud service's management interface, to hardening the virtual servers that run in the cloud, to protecting the network that underlies an organization's cloud operations, to preventing unauthorized access or modification to data stored by and transmitted through the cloud, and much more.

In the application of cloud security, AI can improve event sorting and filtering, anomalous activity detection, application programming interface (API) protection, and data and privacy protection.

Aside from security, AI can enhance cloud virtualization as well. ***Virtualization*** is the process of creating an abstract representation of a computing environment, and it is the fundamental technology underlying the cloud. AI can work with cloud virtualization to enhance provisioning and deprovisioning, resource allocation, and server and application management.

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ACTIVITY 7-3

Securing Assets Using AI

Scenario

Consider the following discussion questions to help review your understanding of how AI is used to secure assets.

1. The cybersecurity team in your organization has been having a hectic week. Their security monitoring systems have been flooded with alerts, and most of them are false positives that end up wasting the team's time. As a result, they haven't been able to respond adequately to the real threats.

   How might AI help the security team focus on the real incidents that threaten the organization?

2. There have been a few incidents lately where employees' laptops have died and the employees have been left without working devices. Your IT team doesn't have any adequate backup laptops, so they need to spend time researching and purchasing new ones. This is leading to lost productivity, which is costing the organization money.

   How can AI improve your IT asset management process so that these kinds of incidents are avoided as much as possible?

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TOPIC D

Optimize Processes

AI is also sought after for its ability to optimize many different types of business processes through automation. This can greatly reduce costs and open up new and profitable business opportunities that wouldn't otherwise be available.

Industrial Robots

Industrial robots and other warehouse machines are a common application of machine learning, particularly reinforcement learning. The machine can perform a task in a physical environment while learning which tasks are most desirable. Many of these robots use computer vision to "see" the world around them so that they can move without hitting objects or people.

Some example applications of industrial robots include:

- **Welding**: Robots can weld components together without a human remotely controlling the machine. Because welding involves very high heat, it can be dangerous to a human worker. The robot can therefore minimize safety issues. Robots are also more efficient at welding than any human, which can increase productivity.
- **Assembly and disassembly**: In manufacturing plants, humans are traditionally needed to assemble parts on an assembly line. However, industrial robots with robotic "arms" can articulate those arms faster than a human, making them much more productive at assembling and disassembling parts. The robot scans the parts as they come down the line, then determines how to assemble or disassemble them based on their reinforcement learning reward system.
- **Inspection and testing**: Regardless of whether a product is made on an assembly line, it is a common practice to inspect products—either all of them or a random selection—for defects. Sometimes, those defects are obvious, but in other cases, they can be very difficult for a human to perceive. Inspection robots may be able to scan a product in much more detail and identify subtle cracks or deformities. They are also effective at testing products in extreme ways, such as blasting them with high heat to see how they withstand a fire.
- **Packaging and labeling**: Once a product is ready to be packaged (perhaps after having been assembled from smaller parts and then tested), a robot on an assembly line can then package and label the product for shipping out to a vendor. Again, a properly trained robot can do this much more quickly than a human and is less likely to make mistakes.

Autonomous Vehicles

An ***autonomous vehicle*** is any vehicle that can drive without human input. They are also called self-driving vehicles. Using techniques like reinforcement learning and computer vision, an autonomous vehicle is able to sense the road or ground beneath it, as well as the surrounding environment, and react accordingly. It can speed up and slow down as necessary to obey speed limits, take turns, avoid other cars, or stop short of obstacles. It can also adapt to weather conditions like slick roads caused by recent rainfall. Autonomous vehicles are also designed with safety in mind, particularly when it comes to minimizing the harm of pedestrians and passengers.

Some common applications of autonomous vehicles in business include:

- **Product transport and freight**: Human truckers may be overworked and more prone to making mistakes, whereas an automated truck does not have these problems. It can operate without rest and therefore can minimize accidents while also saving time.
- **Delivery**: On a smaller scale (as far as distance traveled), many businesses rely on cars to deliver products locally to their customers. Think of food or grocery delivery services. Automating the

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trip from business to home and vice versa can greatly increase the efficiency of travel, leading to more satisfied customers and a more productive business.

- **Public transportation**: It's becoming more common for cities to commission automated buses and subway systems for public transport. These systems can pick up and drop off passengers at designated stops while also having the safety capabilities that any other autonomous vehicle has.
- **Tourism**: A hotel, resort, or other tourism company might employ a fleet of buses that drive tourists around an area to show them the sights or drop them off at notable spots. By automating their tours, a business can save money and get people where they want to go more efficiently.

Robotic Process Automation

Robotic process automation (RPA) is a method by which software agents, also called bots, automate the performance of one or more tasks. The bots are not strictly robots in the sense that they act in a physical world, but rather they act with some degree of autonomy in a digital world.

In traditional RPA, the bot "watches" a user perform a task on their computer—for example, entering data in a database. The bot can then repeat this task much more quickly and efficiently, so the user's input is no longer needed. This does not necessarily exhibit AI since the bot is just repeating the same exact steps. However, RPA systems that *do* exhibit AI are able to learn from a user's input and then extrapolate that input into similar, but novel scenarios. For example, the bot may be able to enter data into a different database. Or, it is able to optimize the steps it takes to achieve the same goal. For example, the bot may have found a quicker way to connect to the database it needs to enter data into.

Some examples of AI-enabled RPA are as follows:

- **Data entry and extraction**: Data entry can be a tedious process, and humans are likely to make mistakes. RPA systems can quickly and accurately take raw data and transfer it to a more structured database. Likewise, they can extract data from a database that a data analyst needs, without that analyst needing to write complex querying code.
- **Software testing**: Testing is a crucial component of the software development lifecycle. Developers must ensure their code is robust and minimizes errors. However, there are many different kinds of tests, and running them manually can be slow and burdensome. RPA can start a number of test harnesses and respond to the results with some degree of intelligence. For example, the bot might run the test again for verification purposes, or it might tweak the parameters of the test slightly to see if the results are different.
- **Customer service**: Customer service reps need to carefully balance customer needs and business needs. It can be difficult to know what the appropriate response is, and when/should they offer a discount? A refund? Should they transfer the customer to another rep? In many cases, customer service reps struggle to follow a flow chart of responses. With RPA, a bot can determine the proper course of action based on various factors.
- **Financial processes**: The amount of transactions that banks and other financial institutions need to process is staggering. Some involve a degree of judgment or complexity, such as knowing whom it's appropriate to lend money to and how much. AI-enabled RPA systems can make these decisions with greater accuracy than a human. The bot doesn't necessarily replace a human entirely; a human likely reviews the bot's decisions and does or does not approve them.

Predictive Analytics for AI

Predictive analytics is a form of data analytics that uses data to forecast future behaviors. It provides estimates of the likelihood of a future outcome based on existing data, statistical models and algorithms, and probability theory to interpolate missing information or to extrapolate what might happen in the future. You can think of predictive analytics as providing the answer to the question, "What could happen?"

When it comes to using predictive analytics to optimize business processes, some examples include:

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- **Inventory levels**: If an organization can predict how much inventory it will need in the future for one or more of its products, it will be able to increase or decrease current levels as needed to meet customer demand and business processing capability.
- **Purchasing patterns**: Just like maintaining the optimal inventory, it can be difficult to know the purchasing patterns to engage in for the business to run optimally. Predictive analytics can help you determine when it's the best time to buy something, whether due to pricing factors or external marketing factors. It can also help you pinpoint the optimal vendors to purchase from.
- **Staffing needs**: Some businesses need to plan and schedule their human workforce around certain events or other market factors. For example, a retail store might want to know what the optimal number of backroom workers is during a holiday or a sale so that the store can stay properly stocked and the workers can remain busy. Rather than guess, you can use predictive systems to anticipate the optimal level of staffing.
- **Customer behavior**: Customers, whether as individuals or as a collective, don't always appear to behave consistently. They may take advantage of a sale one day, but barely show up the next day. Of course, there are many factors that affect customer behavior that a human analyst can't always perceive. A predictive system, on the other hand, may be able to better anticipate customer behavior, giving you the chance to adapt your business strategy to optimize sales.

Supplier Relationship Management (SRM)

Supplier relationship management (SRM) is a process in which a business evaluates and attempts to strengthen the relationships it has with its suppliers, as well as better align those relationships with the business's overall goals and strategy. The purpose of SRM is to identify areas of improvement in a supplier relationship, whether that means finding new areas of value, minimizing cost, minimizing risk, etc. It is similar to custom relationship management (CRM), but focused on the role that suppliers play in the business.

AI can greatly improve on common SRM tasks, and even enable new approaches to building and maintaining supplier relationships. Some example applications include:

- **Supplier segmentation**: To make it easier to communicate with each supplier and manage other aspects of your relationships, you would typically segment suppliers according to one or more aspects. It might be based on the kind of product they supply, or where they are located, or your general attitude toward them. Using a technique like clustering, you may be able to more intelligently segment your suppliers according to multiple features instead of just a few obvious ones.
- **Supplier evaluation**: One major task of SRM is to evaluate your relationship with one or more suppliers. You need to make sure that such relationships are strong and still providing value to your business. An evaluation can include everything from "scoring" a supplier according to some metric, or describing them in more qualitative terms. In any case, an AI system might be able to score or classify each supplier more effectively than traditional tools.
- **Supplier collaboration**: It can be difficult to collaborate with suppliers that cross cultural, national, or language boundaries. Even suppliers in different sectors may operate in a way that is foreign to your business. AI can help your collaboration efforts. For example, you may be able to cluster your suppliers in such a way that you communicate differently to each cluster. Maybe one cluster prefers terse, results-driven discussions, whereas another cluster engages in open-ended discussions.
- **Risk management**: Risk is an important consideration in SRM just as it is in any business relationship. The business and the supplier both take on some degree of risk and must work together to minimize that risk. As discussed earlier, AI has many applications in risk management, and they apply to SRM as well. For example, an AI system may be able to identify areas of risk in the supplier and provide suggestions for how to address that risk in your own business.

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Route Optimization

Route optimization is the process of finding the shortest path between two points. Traditional algorithms are able to identify this path in simple scenarios. But, it takes AI to account for the many factors that could impact how one gets from place to place, including weather conditions, road construction, hazards, traffic, rest stops, and so on.

Some examples of route optimization enhanced by AI include:

- **Delivery**: Whether a delivery vehicle is self-driving or not, an onboard navigation system powered by AI can quickly determine the fastest route from the business to the customer's home in order to get the product to the customer as fast as possible. It can also maximize fuel economy by having the vehicle avoid traffic-congested areas.
- **Tourism**: Tourism agencies can use AI-powered route optimization to help tourists get to their destinations faster. So, the tourists may have more time to see more sights.
- **Warehouse navigation**: Some industrial robots are able to move around an environment rather than just move a limb while stuck in place. These mobile robots may need to get from place to place within a warehouse or factory in order to retrieve or drop off parts or products. They'll need to be able to navigate various obstacles and get to their destination quickly; route optimization using AI can make this a reality.
- **Video games and virtual worlds**: Video games have been using rudimentary forms of AI for many years. Modern AI techniques can help developers create intelligent non-player characters that can move through the virtual world more efficiently, making them seem more lifelike and thus increasing the player's immersion.

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ACTIVITY 7-4

Optimizing Processes Using AI

Scenario

Consider the following discussion questions to help review your understanding of how AI is used to optimize processes.

1. A recent internal review of your business processes confirms that your product inventory levels are not being managed efficiently. At any given time, there is either not enough inventory to meet customer demand, leading to customers being frustrated at out-of-stock notices; or, there is too much inventory taking up limited warehouse space, increasing storage costs and reducing cash flow from unsold product.

   How might AI help address these inventory management challenges?

2. Another internal review of your business processes suggests that the assembly line in your warehouse is not running optimally. The human workers are not able to assemble your products quickly enough to meet business goals, no matter how skilled they are. There are frequent stoppages on the line, and there have been several safety issues as well.

   How might AI help address assembly line challenges?

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Summary

In this lesson, you explored many different facets of AI as applied to business functions. This includes enhancing the user experience, segmenting audiences, securing assets, optimizing processes, and more. By identifying the many ways in which AI is used, you can start to zero in on how AI can be useful to you and your organization in particular.

What function(s) of your own business are you most interested in applying AI to?

What other business functions not discussed in this lesson do you think AI could improve upon in your organization?



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8 Implementing Business Requirements for AI

Lesson Time: 1 hour, 15 minutes

Lesson Introduction

Artificial intelligence (AI) is not a magic wand that you can wave to solve all of your problems. There are real costs involved, and many aspects you need to consider and plan for. In this lesson, you'll take a look at some of the requirements and challenges that AI poses, as well as how to develop strategies to address them.

Lesson Objectives

In this lesson, you will:

- Identify design requirements for implementing AI.
- Identify data requirements for implementing AI.
- Identify risk related to implementing AI.
- Develop a strategy for implementing AI.

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TOPIC A

Identify Design Requirements

Like any project, you should begin an AI project by identifying design requirements that will affect the rest of your team's efforts.

Use Case Identification

AI is transformative. Numerous organizations have already implemented AI solutions and are reporting their successes with it. It is easy to get tempted by the hype, focusing on the interesting possibilities of AI and looking for ways to force them into your business process, when instead you should focus on specific problems your organization needs to address, and then consider how AI might help you address the issues you have identified.

You'll need to identify use cases for AI that are relevant to your business. Some important factors to consider when doing this include:

- The problem(s) in your organization that need to be solved.
- Which problem(s) may or may not be solvable through traditional, non-AI technologies and processes.
- Existing processes that may be in place to address the problem(s).
- The overall goals of fixing the problem(s) (e.g., make operations more efficient and effective).
- Data that you have or can obtain that may pertain to the problem(s).
- Similar problem(s) that other organizations have faced and how they have solved them using AI.

In many cases, to address these factors, you'll need to consult with a wide variety of stakeholders. Such stakeholders can include technical personnel like AI practitioners who know more about the implementation of AI than you; but can also include non-technical audiences like customers who will be directly impacted by an AI solution.

The Value Proposition of AI

You also need to consider that AI is often an expensive business proposition. It can cost your organization a great deal of time, money, and effort to bring an AI solution to fruition. That's why you should try to quantify the value of any proposed AI solution.

For example, if you believe that a recommendation system will help lead to greater customer retention for your online store, how much more money in sales do you think you'll generate from these customers? How much money will you *need* to generate to justify the cost of designing, developing, testing, and implementing the recommendation system into production? Are there other ways the system could provide value to the business besides direct monetary increases (e.g., increased customer loyalty)? These are important questions to consider in any AI venture.

Prototype Assessment

In fields like AI, it is common to create a prototype to demonstrate to stakeholders what the solution will generally look like. So, if you plan on developing a system that can identify fraud in transactions, your team may build a preliminary classification model based on a sample of data, deploy that model into an ad hoc software environment, and then test the overall system by sending it some example transactions.

Presumably, you're one of the stakeholders they're demonstrating this prototype to. You're obviously not responsible for evaluating the technical details and inner workings of the prototype,

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but you should be involved in assessing its design. You need to ask yourself (and your team) questions like:

- Is the overall design efficiently and effectively serving the needs of the target users?
- Will this system be scalable and able to support an increase in user demand?
- How much does the design rely on on-premises resources (e.g., servers) vs. cloud-based resources?
- Can the model be easily decoupled from the deployment environment in case it needs to be ported to another environment (e.g., as part of an upgrade)?
- Is the system designed in such a way that the model can be upgraded or improved over time using new training data?
- What will the maintenance effort and costs be for this system?



Note: Before a prototype, the business may also create a *proof of concept (POC)* to first determine whether or not the idea is even feasible.

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ACTIVITY 8-1
Identifying Design Requirements for AI

Scenario
Consider the following discussion questions to help review your understanding of how to identify design requirements for AI.

1. You've heard about AI and how it's been a boon to many businesses in your industry. You want to begin brainstorming how AI can help your own organization.

   What are some important factors you should consider when identifying use cases for AI in your organization?

2. You decide that it's best for the team to create a prototype of the entire AI solution before development begins on a production-ready version. The solution will take the form of an integrated system, in which a mobile app will retrieve a prediction from a deep learning model stored on a cloud server. The prototype includes a rudimentary version of the mobile app, the model trained on sample data, and a small-scale cloud server environment.

   What questions should you ask yourself or your team to help you assess the design of this prototype?

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TOPIC B

Identify Data Requirements

Because AI is so data driven, the data itself becomes a crucial asset worthy of your special attention. In this topic, you'll address issues related to data that could impact your AI project.

Data Quality and AI

The quality of data can refer to a few things:

- **Accuracy**: How well it describes what it is meant to describe. Data is often recorded incorrectly due to human error, but is sometimes deliberately altered to be incorrect as part of an attack.
- **Recentness**: How up to date the data is. The data may have been perfectly accurate at one time, but if what it's describing undergoes significant change over time, then the data will be out of date.
- **Completeness**: Whether or not any is missing. Enough missing data can lead to low-quality AI models.
- **Formatting**: Whether or not it is in a format that requires a significant amount of processing. Data from different sources, each of which might have used different formatting standards, can be very time consuming to reconcile.
- **Volume**: The overall amount, whether this is measured by examples, features, or both. Generally, the more data you have, the better your AI models will be.
- **Usability**: How relevant it is to the problem at hand. Irrelevant data can be mixed in with relevant data, and it sometimes takes a skillful data analyst to spot the problem.

A deficiency in any one of these areas is going to negatively affect the data your team has to work with, which in turn will affect the quality of the models they create, which then has an effect on the overall AI solution the models are integrated into.

It's up to you and your team to determine which of these aspects is at greatest risk, as well as how to address that risk. For example, volume might not be as much of a concern if you're not developing a deep learning solution. Or, maybe completeness isn't as much of a factor if your team thinks it can ignore the missing data and still produce an effective model.

You might also take the opposite approach, where you identify which aspects of data quality are most important, then prioritize efforts to sustain them. For example, an AI system that makes life-and-death medical decisions should be trained from highly accurate data. You might double or triple check the data you use in an AI system against canonical data sources to ensure the data is kept accurate throughout the project.

Data Quality Models

There are many ways to represent the aspects of data quality beyond those just mentioned. You may want to consider reviewing an industry-recognized data quality model, such as ISO/IEC 25012: *Software engineering — Software product Quality Requirements and Evaluation (SQuaRE) — Data quality model*. This model includes 15 characteristics of data quality. You can learn more at **https://iso25000.com/index.php/en/iso-25000-standards/iso-25012**.

There are also other data quality models out there that you may want to research on your own time. Some examples include the AHIMA Data Quality Management Model, the eCQI Quality Data Model, the UK Government Data Quality Framework, and so on.

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Data Control and AI

Data in the organization's possession must be tightly controlled, particularly if that data is of a sensitive or private nature. Some common data control techniques include:

- **Acquisition**: How is data collected and procured? Only certain sources or types of data may be allowed for use in an AI project. Untrusted, irrelevant, or sensitive data is either not acquired or is filtered out during acquisition.
- **Storage**: How and where is data stored? Data that is private in nature may, for legal reasons, need to be stored in certain locations and not others. Or, the organization may decide to segment the data for security or data management reasons, creating borders between physical or virtual storage locations. In either case, the organization is able to more tightly control data transfers across these borders.
- **Ownership**: Who "owns" data? In other words, who is ultimately responsible for the data. The data owner makes high-level decisions about how the data can be collected and used in an AI project, and is also accountable for the data should it be used improperly.
- **Access permissions**: What people or systems are able to read and use the data? Most AI practitioners and data scientists will need access, as do servers that ingest and process data for model building and prediction. Other stakeholders may also need to work with the data.
- **Modification permissions**: Aside from accessing the data, what people or systems need to be able to alter the data (e.g., to correct an inaccurate record)? You would typically restrict this level of access to only a few database administrators.
- **Sharing permissions**: What outside parties should have access to the data? Perhaps an external contractor is helping out with some facet of the AI project and they need access to the data. Their access should be controlled differently than internal personnel.
- **Disposal**: When and how should the data be disposed of? In many organizations, data that has served its purpose is disposed of to reduce the risk of it being compromised. You may also be legally obligated to dispose of certain data.

There may already be IT processes in place for generally controlling data in the organization that can be repurposed for data used in AI applications. However, because AI is such a heavily data-driven field, the use cases and needs regarding that data often require specialized planning. For example, an AI training system may need to continuously access a stream of user data, whereas in more traditional applications, the system may only need to access user data in an ad hoc fashion.

Data Security and AI

Security is a paramount concern when it comes to data of any sensitive or personal nature. This problem is compounded in a field like AI, where the organization is expected to consume and process massive amounts of sensitive and personal data. The more data you work with, the more you put at risk. Data is at risk of being:

- Copied by and disseminated to unauthorized users.
- Modified in unauthorized ways.
- Deleted without authorization.
- Made unavailable for use.

There are many ways to secure data. Controlling access to data, as mentioned earlier, is one major way. One guiding principle is the ***principle of least privilege***, which states that a user should only be given enough permissions to be able to do their job, and no more than that. That way, if the user's account is compromised, the damage that account could do is minimal compared to if they had full access.

Another major data protection technique is ***encryption***, which encodes data in such a way that only those in possession of the decryption key can read its actual contents. If an unauthorized user gains access to an encrypted database, they won't be able to do much with it if they don't have the key. Encrypting data is important not only when it is stored in databases (*at rest*), but also when it is *in transit* over a network or currently *in use* by a server or other computer processing that data.

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Ultimately, you must work closely with your IT team to develop a reasonable plan for securing the data that goes into your AI project.

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ACTIVITY 8-2

Identifying Data Requirements for AI

Scenario

Consider the following discussion questions to help review your understanding of how to identify data requirements for AI.

1. Your data analysis team has been pulling in data from many different sources (some internal, some external) over the past few weeks. The data describes different features of your customers, including their demographics, their purchasing behavior, and so on. Once combined, you'll use this data to build an AI solution that can segment your customer base for targeted marketing.

   What are some data quality concerns that you have regarding this project?

2. As part of the development and testing phases of your AI project, your team members grab copies of the master dataset over your company VPN and then use those copies on their work laptops at home or in other remote locations.

   What are some security risks that could affect the data being used in this project?

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TOPIC C

Identify Risks in Implementing AI

All business ventures, AI included, are subject to risk. However, AI has its own unique set of risks that you must identify and address as early in the project's lifecycle as you can.

Impact on Employees

One of the most pressing issues of introducing AI in the business is its effect on human labor. In particular, AI may displace jobs traditionally performed by humans on an unprecedented scale. This can include blue-collar jobs as well as white-collar jobs. The jobs most at risk are those that incorporate high amounts of routine and predictable behavior, especially behavior that a machine learning model can quickly learn with minimal amounts of training.

Naturally, your employees may feel anxiety (sometimes called ***technostress***) as some of their job functions become more and more automated. This can lead to symptoms such as frustration, irritability, insomnia, headache, and more. Any one of these symptoms can negatively affect employees' productivity—for example, they may be more likely to commit errors or lose focus on the task at hand. They may also seek employment elsewhere if they feel their health is suffering.

Even if employees don't fear being replaced by AI, they may still struggle to adapt to it. They likely are used to following a specific set of processes that require their full input and perhaps some degree of creativity. AI can disrupt this workflow, making it seem it's "getting in the way" of the employee doing their job. Over time, employees will likely adapt to the AI as they would with any organizational change, but there is still a risk of lost productivity during that transitional period.

Operational Risks

AI is a disruptive force to the business—in a good way. But, just like employees need to adapt to AI, so too does the business. By implementing AI solutions, you necessarily take on a new set of risks, including the risk that the AI solution will:

- Outright fail.
- Not bring as much value to the business as the solution(s) it's replacing.
- Not be cost beneficial.
- Not be well integrated into other areas of the business.
- Not be easily updated and maintained.

Of course, not every implementation of an AI solution is going to be disruptive. But the more your business starts relying on AI to support its operations, the greater the risk. Your organization must therefore meet a certain level of readiness before you begin incorporating disruptive AI solutions. A change management approach can ensure that any disruptive AI solutions are incorporated into business operations in an efficient and effective way, minimizing the risks just mentioned and better preparing the organization to harness the power of AI.

Ethical Considerations

The ethics of AI involve thinking about how:

- Intelligent machines treat human beings.
- Intelligent machines act as moral agents.
- AI may cause harm to humans, whether physical, psychological, financial, or other types of harm.

As a business decision maker, you won't necessarily be asked to take on the deep philosophical issues involved with AI. However, you are still responsible for ensuring that your AI solutions

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follow general ethical principles. In some organizations, ethics have actually become a standard part of corporate board reporting.

The following table describes five general principles in emerging and data-driven technologies, like AI, that your business should strive to uphold.

Ethical Principle	*Description*
Privacy	Privacy focuses on protecting sensitive data about people from unauthorized or unwanted access. AI systems often collect an incredibly vast amount of information, and any business or individual in control of these AI systems effectively has control over the information. Users may not even be aware of the vast amount of data that is being collected about them and fed into AI systems. Likewise, the entities controlling this data may not be completely aware, either—especially if the AI systems are not closely monitored and managed.
Accountability	Accountability is all about ensuring people are held responsible for their actions. AI can make accountability a challenge, since decisions of great consequence are made by machines and not people. If a machine learning model makes a false cancer diagnosis, and the patient goes through an ordeal as a result, who is responsible for the mistake? The practitioners who trained the model? The project manager for the AI solution? The doctors who relied on the model's diagnosis? These questions are not always easy to answer, and unfortunately, many businesses do not even attempt to address them.
Transparency and explainability	Transparency is about enabling people to see into the inner workings of the technology and its effects on people and society. Explainability is about being able to truly understand and interpret how the technology works and why. The complex nature of AI can make supporting these principles a challenge. This is especially true when it comes to the complexities of deep learning. Some deep learning algorithms are considered ***black boxes***, meaning it is difficult (or impossible) for a human practitioner to determine why the model made a certain decision. So, it is unreasonable to "trust" the model. If the decision the model makes is a bad one, then fixing the problem may not be plausible.
Fairness and non-discrimination	These principles ensure people are not treated improperly because of who they are. They are primarily undermined by bias, which is a major issue in AI. Machines process the data that is given to them, just as we expect them to. These machines might be very good at learning the data and drawing conclusions from it, but what of the data itself? The data is typically fed from a human, or is at least the product of some prior human guidance. This means the data has the potential to exhibit bias, or undue positive or negative attitudes applied to individuals, groups, or ideas. So, if the model is trained on biased data, it will produce biased results. This can lead to significant harm.

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Ethical Principle	*Description*
Safety and security	Security, as you've seen, is all about protecting the organization's assets. Just as AI can be used to enhance security countermeasures, so too can it be used to enhance attacks. Safety is related in that it is about protecting the human asset. In other words, minimizing the chance that people are physically harmed and that property (whether physical or virtual) is compromised. AI-powered industrial machinery is commonly thought of as a source of safety risk, but even virtual systems can put people at risk. For example, AI-powered traffic management systems can increase the rate of car crashes if they make poor decisions.

Operationalizing Ethical Principles

Recognizing ethical principles is one thing, but actually putting them into practice in the organization is another. There are many policies, procedures, tools, metrics, and other techniques that you can use to operationalize these principles. Applying encryption techniques to private user data is an example of operationalizing privacy. Using software tools that can identify what features an AI model considers most important to its predictions is an example of operationalizing explainability. There are many such examples.

AI Risk Management Framework

A valuable resource in addressing operational and ethical risks incurred by AI is the AI Risk Management Framework (RMF) published by the National Institute of Standards and Technology (NIST). The AI RMF outlines a process by which an organization can manage such risks. It does this by describing characteristics of trustworthy AI, such as AI that is valid, reliable, safe, fair, secure, and so on.

The AI RMF also offers four functions (Map, Measure, Manage, and Govern) that comprise the AI RMF Core, where each function enables certain outcomes and actions that are favorable toward managing AI risk (e.g., identifying risk is an outcome of the Measure function).

Lastly, the AI RMF encourages organizations to create profiles, which describe how the aforementioned characteristics, functions, and outcomes are applied to specific use cases, sectors, or industries. For example, a fair hiring profile could help an organization address fairness risks in an AI system that assists the hiring process.



Note: Learn more about the AI Risk Management Framework by visiting **https://www.nist.gov/itl/ai-risk-management-framework**.

Legal and Regulatory Considerations

Given how new AI is in the world of business, not to mention how complex it is from both a technical and ethical standpoint, there are not yet many laws and regulations that govern its usage. However, as the technology matures and becomes more commonplace in people's lives, you will undoubtedly need to become more proactive about identifying what your business must comply with to stay in good legal standing.

There are a few examples of documents that may have some legal implications regarding the usage of AI. The following table lists four.

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Document	Overview
DoD AI Principles	In February 2020, the U.S. Department of Defense (DoD) adopted a set of principles after consulting with various AI experts across multiple industries and sectors. These principles are expected to be upheld in all facets of the military, including combat and non-combat roles. The principles require that usage of AI is: • Responsible • Equitable • Traceable • Reliable • Governable For more information, see **https://www.defense.gov/News/Releases/Release/Article/2091996/dod-adopts-ethical-principles-for-artificial-intelligence/**.
U.S. AI Bill of Rights	In October 2022, President Joe Biden unveiled a blueprint for a new AI Bill of Rights. This Bill of Rights is meant to enumerate five core principles that are useful in upholding the rights, opportunities, and needs of American citizens as they pertain to AI. The principles involve: • Safe and effective systems • Algorithmic discrimination protections • Data privacy • Notice and explanation • Human alternatives, consideration, and fallback For more information, see **https://www.whitehouse.gov/ostp/ai-bill-of-rights/**.
EU AI Act	In 2021, the European Union proposed the Artificial Intelligence Act as a way to regulate the use of AI in the EU. As of December 2022, it has not yet been made into a law, but if it is, it will be the first law governing AI by any major regulatory body. Unlike the AI Bill of Rights, this act is not focused on the rights of individuals, but as a way to regulate the use of AI systems. The regulation recognizes three major categories of AI risk: • Unacceptable risk (e.g., government-run social credit scoring) • High-risk systems (e.g., job applicant ranking) • Low-risk systems (unregulated) For more information, see **https://artificialintelligenceact.eu/**.
National Strategy for Artificial Intelligence	In 2018, India published its National Strategy for Artificial Intelligence, which provides guidance for how the country could adopt AI in a way that promotes social and economic growth as well as serve the greater good. The strategy outlines three distinct components for AI adoption in India: • Economic opportunities • Social development and inclusive growth • India as an "AI playground" or "garage" in which organizations can develop solutions for use in other economies For more information, see **https://indiaai.gov.in/research-reports/national-strategy-for-artificial-intelligence**.

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Note: You may have the opportunity to consult with regulatory bodies as new AI regulations are developed. It's important to take a measured approach to regulation so that it will not have a significant negative impact on innovation and competitiveness in the AI space.

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ACTIVITY 8-3

Identifying Risks in Implementing AI

Scenario

Consider the following discussion questions to help review your understanding of how to identify risks in implementing AI.

1. Up until now, your HR department has been manually screening job applicants. However, this takes up a lot of the team's time. You want to implement an AI solution that will automatically perform preliminary job applicant screening, quickly and efficiently selecting a pool of candidates most qualified to continue on to the first wave of interviews.

   What are some of the business risks involved in adopting this AI solution, from both an employee and an operational standpoint?

2. Consider the same scenario where you're implementing an AI-powered job applicant screening solution.

   What are some ethical concerns you would have regarding this solution?

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TOPIC D

Develop an AI Strategy

Unless you have a strategy for incorporating AI into your business operations, you'll be unable to truly harness the value of your work. In this last topic, you'll ensure that your AI projects fit within the overall mission of the business.

Alignment to Organizational Initiatives

As part of its overall strategy, your organization likely has one or more initiatives that are currently driving the business. The AI solutions you develop and implement must be considered alongside these initiatives, otherwise they risk being stranded without adequate support from decision makers. AI solutions that are isolated from strategic initiatives also risk being a hindrance to the business rather than a benefit.

Consider an initiative like ***digital transformation***, in which the organization seeks major change through the use of digital technologies like cloud services, mobile devices, Internet of Things (IoT), and, of course, AI. The goals of such an initiative usually center on promoting innovation in the marketplace or improving the efficiency of business processes. AI is obviously an attractive option for accomplishing this. The organization should develop an AI solution not just as a way of solving a specific problem, but as a way of achieving digital transformation.

There are many other types of initiatives that hold relevance for AI. For example, your organization might have a data strategy that seeks to drive innovation by focusing on the possibilities that can be unlocked through data. As a data-driven technology, AI is a great candidate for supporting this initiative. Or, perhaps your organization has a general business growth strategy, where it wants to grow the business by some dollar amount over some period of time. You can fit an AI solution into this strategy by aligning it with those specific growth objectives, e.g., how can the AI solution help generate sales or reduce costs over the time period?

It's also important to recognize that AI *must* align with one or more existing strategies rather than being its own initiative disconnected from the rest of the business. This is particularly true regarding integration with IT strategies. For example, an autopilot AI system developed for an aircraft must be interoperable with other aircraft systems and must also have undergone rigorous safety and security tests in alignment with existing strategies, otherwise, it could prove disastrous.

Alignment to Environmental, Social, and Governance (ESG) Initiatives

Environmental, social, and governance (ESG) is a framework by which stakeholders can evaluate an organization's risks and opportunities regarding these three areas of concern. AI solutions have the potential to impact these areas in significant ways, so it is a good idea to align them with ESG initiatives whenever feasible.

There are several things to consider with each area of ESG as it pertains to AI.

- **Environmental**: The "E" is focused on how the organization impacts the environment in various ways, and how it is committed to sustainability. When it comes to AI, one of the most overarching environmental issues is power consumption. The hardware that AI solutions run on requires massive amounts of power. So, the organization could demonstrate to stakeholders that it is taking steps toward improving the efficiency of its power consumption.
- **Social**: The "S" is focused on how the organization has an effect on various social issues, and how it is committed to treating its employees, customers, partners, and communities fairly. As discussed earlier, AI solutions are subject to bias, which can lead to social harm. In addition to promoting fairness and non-discrimination, the organization could demonstrate to stakeholders

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that an AI solution is supporting the ideas of diversity, equity, and inclusion. When these ideas are considered during AI development and deployment, the solution is more likely to positively affect at-risk communities, achieve fair and just results, and generate participation from a variety of viewpoints.

- **Governance**: The "G" is focused on the organization's leadership and management, and how it is committed to the viability of the business. There are various concerns in governance, including upholding the rights of shareholders, conducting internal audits, and reviewing executive pay structures. One concern that is relevant to AI is the implementation of internal controls—in particular, data protection processes. The organization could demonstrate that leadership is taking user privacy and company security issues seriously during the development and deployment of AI solutions.

Organizational Resources

An AI project can be a significant undertaking for the organization, and will likely require resources that span many different departments and areas of expertise.

- **Talent resources**: As a field, AI is complex and technical. It requires skilled practitioners to successfully implement in any organization. You may wish to engage third-party contractors who specialize in AI, but don't discount your existing pool of talent. With the right training and hands-on experience, personnel like data analysts and software developers can become effective members of your AI development team. You also need to consider that AI solutions are about more than just direct development; you'll need IT personnel who are familiar with the intersection of development and operations (DevOps) to actually deploy an AI solution that can be accessed over a network.
- **Data resources**: As you're well aware by now, data is the main driver of any AI solution. It is perhaps the most important resource for the project. You need to account for how you'll acquire, process, and store data that may be useful to the project. Some data is open source and easily acquirable, whereas other data is closed source and requires licensing agreements and possibly fees. Other data may be internal to your organization, whether it's already available or waiting to be generated. Whatever the case may be, your strategy must incorporate data handling and management processes.
- **Computing resources**: Even if you plan on running every part of an AI project in the cloud, you must still be aware of what underlying computing resources are necessary to do so. This includes the CPU and GPU clusters used in training models; the storage devices used to maintain sources, backups, and copies of datasets; and the network bandwidth used in delivering predictions and other decisions to hundreds, thousands, or perhaps millions of consumers. You must consider these resources as flexible, able to scale up and scale down as demand rises and falls.
- **Budget resources**: AI projects can vary greatly in size and scope, and so can their costs. There are several common sources of cost that you should consider focusing on, including: the cost of acquiring software licenses used in developing AI solutions; the cost of deploying AI solutions as a service, whether in the cloud or using on-premises servers; and the cost of forward-looking activities like maintaining, improving, and scaling up solutions over their lifecycles. Also, the resources described previously (talent, data, and computing) naturally have costs associated with them. You must consider all of these factors as you budget for your AI projects.
- **Stakeholders**: As mentioned before, stakeholders can provide valuable assistance and unique perspectives when it comes to an AI project. Don't hesitate to solicit their opinions and requirements before, during, and at the end of a project. After all, stakeholders like users and employees will be the ones most directly affected by your AI solutions.

Competitive Considerations

AI can give an organization a great competitive edge in the marketplace. Products, services, and processes that are rooted in traditional technologies and practices are starting to feel outdated now, and in many cases, will soon become obsolete. Organizations that don't start seriously considering

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how they can adopt AI could lose business to those that have been embracing AI for years. So, part of your business strategy should be to position AI as a force that drives your business to industry leadership, or, at the least, maintains your current position as an industry leader.

Some examples of how AI can boost an organization's competitiveness include:

- Improving the efficacy or usefulness of existing products and services that are common to the industry.
- Fostering innovation in a stagnant industry.
- Streamlining business processes so that an organization's operational margins are much better than its peers.
- Generating renewed interest from an industry's customer base, or attracting new customers to that base.
- Attracting new talent to the organization so that it can grow more rapidly.

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ACTIVITY 8-4

Developing an AI Strategy

Scenario

Consider the following discussion questions to help review your understanding of how to develop an AI strategy.

1. Consider the example of the AI system that screens job applicants. You want to be able to align this system with an environmental, social, and governance (ESG) framework.

   How might the AI job applicant screener impact each of the three aspects of the ESG framework? How might you address these impacts in your AI strategy?

2. Consider the same job applicant screener example.

   What are some organizational resources that you need to consider as part of your AI strategy? Why are these resources relevant to the job applicant screener?

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Summary

In this lesson, you ensured that your AI projects were bringing value to the business by identifying design and data requirements, addressing inherent AI-based risks, and aligning your efforts to business strategy. You're now better prepared to adopt a leadership role in an AI project that has the power to transform the business.

When it comes to implementing AI in your organization, what risks most concern you, and why?

What organizational resources have the most bearing on your AI strategy, and why?



Note: Check your CHOICE Course screen for opportunities to interact with your classmates, peers, and the larger CHOICE online community about the topics covered in this course or other topics you are interested in. From the Course screen you can also access available resources for a more continuous learning experience.

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Course Follow-Up

Congratulations! You have completed the *ETBIZ (Exam ETZ-210)* course. You have gained foundational knowledge and insight to help guide business decisions for implementing and managing IoT, data science, and AI projects.

You've also gained knowledge that will prepare you for the CertNexus® ETBIZ (ETZ-210) credential. If you combine this class experience with review, private study, and hands-on experience, you will be prepared to demonstrate your emerging-technology management expertise both through professional certification and with solid competence on the job.

What's Next?

If you want to dive deeper into the ethics of data-driven technologies like AI, consider taking the CertNexus *DEBIZ™ (Exam DEB-110): Data Ethics for Business Professionals* course. If the team members you manage want to obtain hands-on and practical skills in implementing emerging technologies, or if you yourself want to expand your practical skillset, the following CertNexus courses can help you do just that:

- *Certified Internet of Things (IoT) Practitioner (Exam ITP-110)*
- *Certified Data Science Practitioner (CDSP) (Exam DSP-110)*
- *Certified Artificial Intelligence (AI) Practitioner (Exam AIP-210)*

You are encouraged to explore emerging technologies further by actively participating in any of the social media forums set up by your instructor or training administrator through the **Social Media** tile on the CHOICE Course screen.



Note: The **Files** tile includes a document called **AI Project Checklist for Participants.docx** that you can use to ensure you are following best practices when participating in an AI project, regardless of your role or level of technical knowledge.

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Solutions

ACTIVITY 1-1: Understanding Value, Insights, and Data

1. **A) Identify new/incremental value: What elements of value would be important to produce year-round, high-quality fresh flowers for Greene Organix's distributors?**

   A: Answers may vary, but should include: cost, real-time availability, consistency and quality of product, freshness, unique varieties, year-round availability, longevity, and others.

2. **B) Understand insights needed to support value: What information may be needed to drive down production costs so you can continue to deliver best-in-class flowers at improved competitive prices?**

   A: Answers will vary, but should focus on specific ways to drive down operational costs. Information required for this goal might include: investigating the feasibility of introducing additional automation into the operational process.

3. **C) Data: What data do you need to collect to support these insights?**

   A: A local area network could be implemented through cabled connections, such as Ethernet, although a wireless network would make it easier to relocate sensors as needed.

4. **Sensors: What useful information might an IoT device measure in this environment?**

   A: Answers may vary, although a greenhouse operator might be interested in CO_2 levels, humidity, temperature, light exposure, soil moisture, nutrients concentration, airflow, plant imaging, and liquid-level detection.

   Actuators: What actions might an IoT device need to perform in this environment?

   A: Actuators in a greenhouse environment might include CO_2 generators, air blowers and vents, irrigation pumps and valves, plenum or shade actuators, artificial lighting controls, heaters, and air conditioners.

   Other sources: What external data might be of use to support the insights?

   A: Answers may vary, but might include: weather data, material-supplier data, logistics data, inventory data, and commodities-market pricing information.

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ACTIVITY 1-2: Selecting an IoT Infrastructure

1. **A) What would be a typical IoT infrastructure setup for our greenhouses?**

   A: Answers will vary, but should include: Sensors, actuators, gateways, and connectivity between them; remote connectivity for our gateway to communicate to the cloud; a process to configure (add, delete, and move) sensors and actuators within our solution (provisioning and tracking); the ability to supply power to sensors, actuators, and gateways; a remote-user connectivity application; a cloud IoT platform to ingest all the data and a streaming analytics process to analyze the data; and access to external data sources (weather data, market commodities data, etc.) to merge with our collected data.

2. **B) What are some constraints we may be dealing with within our greenhouse operation?**

   A: Answers will vary, but should include: The cost to implement and maintain (the incremental cost must be less than the incremental value created); security; remote locations of warehouses and extreme conditions where internal equipment may not be reliable; greenhouse sites may be too remote for Wi-Fi-based communications from the hub to the cloud (the solution might need to be cellular or satellite based); safety-related parameters that must be considered first, and possibly an extra layer of security for this data and control; any data that must be analyzed and processed locally to ensure rapid response and no reliance on remote network connection issues; and whether or not to limit the amount of data sent to the cloud (more data means more cost, both for transmission and for storage).

ACTIVITY 1-3: Identifying Potential Challenges of IoT

1. **A) Challenges: What challenges might your organization encounter in implementing an IoT solution?**

   A: The answer depends on the nature of the proposed implementation and unique challenges within the organization. However, several themes are common, such as cost and possible disruption to current operations. Because IoT implementations may lead to automation and changes in job responsibilities, there may be challenges related to change management, such as getting buy-in, retraining staff, updating documentation, and so on.

2. **B) Countermeasures and remediations: How might you address the challenges you identified?**

   A: Identifying what might go wrong is an important part of the design process. If you identify potential problems at the start of the project, you can design a solution that will prevent them from occurring. Various countermeasures and remediations to common IoT challenges are described in the next lesson.

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ACTIVITY 2-1: Using IoT to Solve Problems

1. **How might IoT help Best Elevators increase customer loyalty?**

   A: If IoT can provide improvements in all of the other problem areas, then presumably customer satisfaction will increase, and customers will be less likely to explore competitive options. Also, implementing a *connected services* strategy will improve the customer experience, and will provide customers with a sense that Best Elevators is a trusted partner in their daily operations, which should also contribute to increased customer loyalty.

2. **How might IoT help Best Elevators improve service maintenance and predictive maintenance?**

   A: Data collection from sensors combined with analytics can be used to inform Best Elevators and their customers when maintenance is required. Early-warning data and remote diagnostics would enable the organization to dispatch technicians promptly, ensuring they are equipped with the necessary information, tools, and parts to restore the equipment to safe operating conditions quickly.

3. **How might IoT help Best Elevators manage spare-parts inventory more efficiently?**

   A: IoT can be very helpful for keeping spare-part inventories in warehouses and service vehicles stocked appropriately, which is normally a notoriously difficult task to manage cost-effectively. On the supply side, inventory can be tracked using typical IoT devices such as RFID tags. On the demand side, elevator equipment can be monitored and analyzed using data analytics and Best Elevators' own domain expertise to predict future needs. Combining historical data of part failure rates with analytic algorithms, Best Elevators can optimize their part stocks in the warehouse and technicians' service vehicles, reducing inventory costs and improving repair time.

4. **How might IoT help Best Elevators improve product design and technician training?**

   A: Once IoT systems are in place, the system will be able to collect and analyze usage patterns and maintenance requirements over time. This information can be used to improve product design and technician training.

5. **How might IoT help Best Elevators improve uptime and field service efficiency?**

   A: The solutions described in the previous questions will help improve uptime and field service efficiency.

6. **How might IoT help Best Elevators allocate scarce service technicians more efficiently?**

   A: If product design, elevator uptime, and field service efficiency are improved (as described in the previous two answers), then this will reduce the demand on service technicians. Improvements in technician training should make technicians more efficient and effective.

7. **How might IoT help Best Elevators communicate more effectively with suppliers?**

   A: Parts distributors and value-added resellers will also see additional opportunities to leverage the insight garnered from the IoT connected-services solution. Best Elevator can share its experience and knowledge with its equipment manufacturers, creating a value-based competitive advantage in a market that is highly cost driven with pricing volatility.

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ACTIVITY 2-2: Preparing for an IoT Initiative

1. **What would need to be done next in this project?**

   A: Answers will vary depending on the development methodologies your organization follows. You've already initiated the project and developed a system concept. There may be more planning to do, and requirements have not been completely identified yet. So it seems that the project is in the late planning stage. The next phase would be identifying requirements. Typically, before delving into design and development, the team would translate the general concept into formal requirements so all project stakeholders can agree on specifically what the project should accomplish.

2. **What sorts of change-management issues might arise as the company shifts from primarily selling physical goods to selling services?**

   A: Best Elevators is currently focused on manufacturing and maintenance operations. The new system will establish a new branch of operations focusing on software services. This will have some impact on staffing and organization structure. The payment model may change, as the company's orientation shifts from primarily selling physical goods to selling services. This will have numerous effects on the organization. You will also have some change-management issues with customers. While many customers will welcome the improvements Best Elevators is proposing, some will be suspicious and may resist becoming more dependent on Best Elevators. You'll need to ensure that customers understand how the improvements will benefit them and their tenants, and may even save them money on maintenance costs. Planning the technical aspects of an IoT project is complex enough, but to ensure the success of the project you should also carefully consider the impact on personnel, contractors, customers, business operations, and the financial bottom line.

ACTIVITY 3-1: Discussing Data Science and Related Technologies

1. **Susan begins by asking you how you would define data science.**

   A: Data science is a discipline that involves accumulating and analyzing data so that you can extract value from the data. It also involves presenting the value of the data in a meaningful way that answers a business question or solves a problem.

2. **You then spend a little time discussing the parts of data science. How would you describe the skills and knowledge required to perform data science?**

   A: Data science is a combination of mathematical skills, computing skills, and domain knowledge.

3. **You mention to Susan that part of your duties at the Esker Road store includes reviewing shrinkage reports to determine the percentage of loss due to theft each month. What type of analytics is this activity?**

   A: This activity consists of reviewing data to determine past behavior, so this is an example of descriptive analytics.

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4. **Susan asks if you are involved in setting up employee schedules for your store, and you agree that this is part of your responsibilities. What type of analytics is this most closely related to?**

   A: Scheduling staff includes an estimation of how many associates will be needed on certain days. In other words, you are predicting how busy the store will be and how many associates are needed to ensure that customers can be assisted promptly. So, this activity is a form of predictive analytics.

5. **As you wrap up your session with Susan, she suggests that if GCE is truly going to embrace a data-driven culture, other technologies will probably be needed. What types of technology is she referring to?**

   A: Related concepts and technologies include artificial intelligence (AI), generative AI, machine learning (ML), deep learning (DL), Python or R programming, application programming interfaces (APIs), web scraping, and business intelligence (BI).

ACTIVITY 3-2: Identifying Data Science Opportunities from a Dataset (Optional)

2. **What kind of information about each transaction is recorded in the dataset?**

   A: The dataset tracks the unique invoice ID of the transaction; the date and time of the transaction; the name, gender, and age of the customer; whether the customer is a member of the store's rewards club; the branch location the transaction took place at; and what category of item was purchased.

3. **How might you use descriptive analytics with this dataset?**

   A: Answers may vary, but some examples include: drawing conclusions about the overall shopping behavior of customers; creating visualizations that compare shopping behaviors between the different branches; calculating statistics concerning different genders and age groups, like the age group that purchased the most clothing; and so on.

4. **How might you use diagnostic analytics with this dataset?**

   A: Answers may vary, but some examples include: identifying why certain times of day or days of the week see a decline in sales; determining why a gender or age group is less likely to be a member of the rewards club; understanding why a certain branch location is lagging behind in sales of a specific product line; and so on.

5. **How might you use predictive analytics with this dataset?**

   A: Answers may vary, but some examples include: determining what time of day someone of a certain age is likely to do their shopping; predicting what type of product someone will purchase based on their age, gender, and membership status; predicting days or weeks in which sales are likely to decline for future months; and so on.

6. **How might you use prescriptive analytics with this dataset?**

   A: Answers may vary, but some examples include: proposing suggestions for changing store hours to better accommodate customer preferences; encouraging a different approach to advertising certain product lines, such as increasing the visibility of deals and promotions for those product lines; creating new membership-club perks to entice a certain gender or age group into joining; and so on.

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ACTIVITY 3-3: Discussing Types of Data Used in Data Science Projects

1. Store inventory is kept in a database consisting of a series of linked tables. What type of data is this?
   - (•) Structured
   - ( ) Unstructured
   - ( ) Semi-structured
   - ( ) Big data

2. GCE has a web presence, with each store location having a page to highlight local employees and events. It also contains a link for customers to provide feedback via a web form that includes clickable fields for identifying age ranges and departments and free-form text fields for the actual feedback. What type of data is the customer feedback?
   - ( ) Structured
   - ( ) Unstructured
   - (•) Semi-structured
   - ( ) Transactional

3. What type of repository is most likely used to store the customer feedback?

   A: An operational data store.

4. Your manager said that when they had the soft opening for the store, they used census data to select certain potential customers for incentive savings offers. What type of data is census data?
   - ( ) Big data
   - (•) Open data
   - ( ) Proprietary data
   - ( ) Unstructured data

5. What type of repository is most likely used to store the census data?

   A: A data warehouse.

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ACTIVITY 3-4: Discussing the Data Science Lifecycle

1. In which stage of the lifecycle is data wrangling performed?
   - ○ Problem Identification
   - ○ Data Collection
   - ◉ Preprocessing
   - ○ EDA
   - ○ Modeling

2. In which stage of the lifecycle are ML algorithms tested?
   - ○ EDA
   - ◉ Modeling
   - ○ Model Deployment
   - ○ Communication of Results

3. Which stages of the data science lifecycle would you (as a manager/business professional) expect to have the most influence over? (Choose two.)
   - ☑ Problem Identification
   - ☐ Preprocessing
   - ☐ Model Deployment
   - ☑ Communication of Results

4. Why did you select the stages you did in the last step?

   A: Responses will vary, but might include: As a manager or other business professional, you might possess specific domain knowledge that can help identify business problems. Also, if you have participated in problem identification, it is likely that you would want to be informed of the results of the project.

ACTIVITY 4-1: Improving Customer Experience

1. How could the company address this issue?

   A: Responses will vary, but might include implementing a self-service support chatbot. The chatbot could be trained to deal with basic support questions, and to forward more complex issues to a human agent. After implementation of the chatbot, the company could perform sentiment analysis on the online reviews and use the results to evaluate whether or not CX has been sufficiently improved.

2. How can you tell how effective these displays are?

   A: Responses will vary, but might include: gathering and analyzing sales data for the displayed items, sending email to loyalty customers to ask for their reactions, and creating a blog post on the store's web page that shows the displays and includes a poll or other mechanism for customers to vote for their favorite displays.

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ACTIVITY 4-2: Improving Marketing Efforts

1. **How can the use of data science help alleviate their concerns?**

   A: Responses will vary, but might include: Using audience segmentation to identify and refine target audiences for each of the new product lines. For example, people who live in apartments may not be interested in lawn and patio furniture, and retirees may not be interested in beach volleyball sets. Once the target audience segments are identified, GCE can implement targeted advertising to those groups who are more likely to be interested in the new product lines. Data science comes into play because audience segmentation relies on the analysis of customer information to identify patterns and categories according to demographic, geographic, psychographic, and behavioral data. Data science also supports targeted advertising by collecting and analyzing data related to website visits, location, and device usage.

2. **How can data science help address this issue?**

   A: Responses will vary, but might include: Using data from past campaigns and the current campaign to perform marketing measurement analysis. While campaign optimization focuses on a particular campaign, marketing measurement analysis can compare the effectiveness of different campaigns.

ACTIVITY 4-3: Optimizing Organizational and Transactional Security

1. **How would you explain possible uses of data science to prevent fraudulent activity in the online store?**

   A: Responses will vary, but might include: using data analytics to identify suspicious activity such as an excessive number of failed login attempts or numerous anomalous purchases from existing customers.

2. **What security issues might this be an example of, and how would you suggest that the issues be addressed?**

   A: Responses will vary, but might include: IP theft and fraudulent activity. Although the counterfeit goods do not constitute IP theft from GCE itself, the detected actions still have an adverse effect on the business. Proposing that data regarding the identified counterfeit items be analyzed to identify potential patterns, such as which brands are most likely to be affected or demographic information about the people returning the false items, may help employees identify any questionable transactions in the future.

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ACTIVITY 4-4: Enhancing Operational Practices

1. **What do you think the issue is? What advice would you offer?**

   A: Responses will vary, but might include: This appears to be a possible example of customer churn. You might suggest that the store manager gather data about what types of customers have historically shopped at peak and off-peak hours, maybe by using a customer satisfaction survey. This data, along with other demographic and behavioral data, can be analyzed to identify the characteristics of typical peak and off-peak customers. You might also suggest that the manager take into consideration any factors that might be affecting the density of early and late shoppers, such as reduced staff during off-peak hours or the opening of a big-box store three blocks away. After analysis, identify the top two or three reasons for customers no longer shopping during off-peak times, and identify strategies to increase the satisfaction and loyalty of off-peak customers, possibly by offering early-bird and night-owl specials.

2. **How can TA analytics help streamline the recruitment and hiring process?**

   A: Responses will vary, but might include: automated candidate sourcing and tracking, résumé parsing, and email communications.

ACTIVITY 5-1: Discussing the Development of Data-Centric Organizations

1. **What recommendations would you make to ensure the success of the endeavor?**

   A: Responses will vary, but might include: ensuring that the data strategy aligns with organizational objectives and culture, performing a resource assessment to identify the people, equipment, and processes required for transforming your company to a data-centric organization, and identifying potential contributors.

2. **Which staffing option or options would you recommend the company use to optimize costs?**

   A: Responses will vary, but might include: Wherever possible, use existing personnel, general training, and upskilling to fill the open positions. This approach takes advantage of existing domain knowledge. Other options include combining new job roles with existing ones and using third-party talent.

ACTIVITY 5-2: Developing an Implementation Strategy

1. **What are some determining factors you would use to prioritize the project ideas?**

   A: Responses will vary, but might include: relative benefit to the entire company, estimated cost of implementation, and estimated time frame for completion. Because the initial request includes a time frame of two quarters, it would probably be most beneficial to use the estimated time frame for completion as the primary factor in prioritization.

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2. What is one issue that you would suggest the implementation team consider before beginning data collection?

   A: Responses will vary, but might include: verifying that the identified problem is explained clearly to support the collection of relevant data. Other responses include identifying preferred methods of and tools for collecting customer data and determining the optimal data storage locations and methods.

ACTIVITY 5-3: Discussing the Impact of Data Science on Business

1. Of the issues discussed (impact on reputation, impact on talent pool, and legal/regulatory compliance), which do you think carries the most weight? Why?

   A: Responses will vary, but might include: reputational impact, because loss of reputation is one of the hardest things for a company to overcome. In addition, loss of reputation also affects the ability to optimize the talent pool.

2. What side of the discussion would you take?

   A: Responses will vary, but might include: If cost and time to implement are the primary issues, you might agree to investigate the use of third-party services, provided that someone in the company performs a thorough review of the data and the terms of the vendor agreement, and ensures that any applicable compliance requirements are met.

ACTIVITY 5-4: Discussing Governance Measures

1. Would you consider this a cause for concern? Why or why not?

   A: Responses will vary but might include: Yes, this could be evidence of biased data, because the dispersion of the location data suggests that customers who do not live in the identified areas may be under-represented in the dataset.

2. Of the laws, regulations, and guidelines discussed in this topic, which do you think has the largest reach?

   A: Responses will vary but might include: the GDPR, because it provides data privacy protection to every resident of the EU and regulates business entities that operate within the EU or that collect data from EU citizens.

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ACTIVITY 6-1: Discussing the History of AI

1. **Identify at least two events, people, or systems since the end of the second AI winter that have had a direct impact on the upward trajectory of AI.**

   A: Answers may vary. Some examples include: In 1997, Deep Blue defeated chess grandmaster Garry Kasparov in one of the first demonstrations of creative problem solving in AI. In 2004, the DARPA Grand Challenge kicked off, putting self-driving cars in the spotlight. In 2011, IBM's Watson defeated Ken Jennings and Brad Rutter at *Jeopardy!*, and Apple's Siri became the first virtual assistant with AI capabilities. In 2016, Google's DeepMind project demonstrated significant advances in general-purpose deep learning capabilities by defeating champion Lee Sedol at Go. Since 2020, systems like GPT-3 and DALL·E have given end users the ability to generate convincingly realistic text and images.

2. **Identify at least two relatively recent technologies that have enabled AI to flourish in the business world, and briefly explain how they have done so.**

   A: Answers may vary, but there are essentially three recent advances that have paved the way for AI in business: cloud computing, general-purpose GPUs, and big data. Cloud computing has made it easier for even small businesses to access and manage the high infrastructure costs of AI systems without having to operate equipment locally. General-purpose GPUs have greatly accelerated the performance of AI systems due to their high memory bandwidth and ability to efficiently perform complex linear algebra calculations. Big data has increased the value proposition of AI systems since more data (in terms of volume, variety, etc.) tends to lead to better AI.

ACTIVITY 6-2: Discussing AI Concepts

1. **Which technique, traditional machine learning or deep learning, would be most suitable to achieving this business objective? Why did you make this choice and not the other?**

   A: Answers may vary, but deep learning is almost certainly the better option. Deep learning models excel at processing large amounts of data, especially unstructured data like images. Deep learning models can extract the features in those images without anyone from your team needing to extract them manually, which would be extremely difficult if not impossible. Traditional machine learning would be better suited to simpler, more structured datasets where extracting features is either easy or has already been done for you.

2. **Is this an example of supervised, unsupervised, or reinforcement learning? Explain why.**

   A: This is an example of unsupervised learning because there are no labels for the model to learn from. The model must instead determine common attributes from the customers' features and then place each customer into a cluster with other, similar customers.

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ACTIVITY 7-1: Improving User Experiences Using AI

1. **What AI system(s) could help solve this problem to improve the customer experience, and how?**

   A: Answers may vary, but there are two types of systems that could actually help: chatbots and agent assistance systems. Chatbots could help by funneling any customer service requests that come in through a website or app into an automated system that can act as a first point of contact, answering basic and common questions by processing customers' text-based messages. This will free up agents to address more complex requests. On the other hand, agent assistance systems can work alongside call center agents to more efficiently address customer issues. For example, the assistant can quickly search and retrieve troubleshooting steps for a specific problem faster than the human agent. This will reduce time spent on each call and will make the call more productive, leading to more satisfied customers.

2. **How can an AI-driven recommendation system provide more effective product suggestions to customers?**

   A: Answers may vary. An AI-driven recommendation engine can learn many different factors that can help it make an informed decision about what to suggest. For example, it might build a profile of the customer based on their location, age, gender, spending habits, etc., then use that profile to discover what other people with similar profiles tend to buy. The AI system can also make more intelligent decisions based on the nature of the products themselves, such as recommending a product that "matches" the product the customer purchased—like recommending a monitor to a person who just bought a desktop computer.

ACTIVITY 7-2: Segmenting Audiences Using AI

1. **What machine learning technique can help you place customers into different groups for the purposes of market segmentation? Provide a basic explanation for how the technique works.**

   A: Answers may vary, but unsupervised learning, specifically clustering, is the most prominent method for segmenting customers using machine learning. It finds commonalities in the attributes of each customer and then places each customer alongside other customers that they most closely resemble.

2. **How can a machine learning model that makes predictions be used to enhance customer segmentation?**

   A: Answers may vary. A prediction system can use techniques like classification or regression to predict some aspect of each customer that isn't already known. This aspect may play an important role in determining which segment to assign the customer to. The assignment might be directly based on the prediction result, or it might be incorporated into a clustering model that considers the prediction result alongside many other features when segmenting customers.

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ACTIVITY 7-3: Securing Assets Using AI

1. **How might AI help the security team focus on the real incidents that threaten the organization?**

   A: Answers may vary, but AI-powered systems can help the security team classify, rank, and prioritize threats and alerts so that only the most pressing matters are brought to their attention. This will reduce the chance that the team acts on false positives, enabling them to spend more time addressing the important issues.

2. **How can AI improve your IT asset management process so that these kinds of incidents are avoided as much as possible?**

   A: Answers may vary. First, an AI system that can receive monitoring data from the device, particularly factors that indicate the device's health, can be used to predict the failure of that device. That way, the IT team can order replacements ahead of the actual failure, ensuring there is no downtime when it actually happens. Second, instead of having a team member spend time researching devices, an AI system can help them select a proper replacement to narrow down their choices, making the procurement process quicker and easier.

ACTIVITY 7-4: Optimizing Processes Using AI

1. **How might AI help address these inventory management challenges?**

   A: Answers may vary, but predictive analytics can help the business estimate how much inventory will be needed to meet customer demand in the future, along with considering factors like storage costs and a reduction in cash flow. The business can use these predictions to determine whether they need to increase or decrease their current inventory levels. This will lead to optimal inventory levels, keeping costs low and customers happy.

2. **How might AI help address assembly line challenges?**

   A: Answers may vary, but industrial robots were created to address issues like this. They can assemble and disassemble parts much more quickly and accurately than any human. They also help minimize safety issues, as some assembly line work can be dangerous if not done properly.

ACTIVITY 8-1: Identifying Design Requirements for AI

1. **What are some important factors you should consider when identifying use cases for AI in your organization?**

   A: Answers will vary. Perhaps the first and most important step is to start by identifying the problems you need to solve, then determining how AI can solve them. This is better than the opposite approach, in which you just try to force AI into your business without good reason. You should also consider what your overall goals are, and how AI can help you meet those goals. Another important factor is data—how can existing data be used, or how can new data be obtained, that is relevant to the problem? You should also consider stakeholders and what input they have that could guide your decisions.

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2. **What questions should you ask yourself or your team to help you assess the design of this prototype?**

   A: Answers will vary. One important question is, will the cloud servers be scalable? In other words, if demand for predictions increases significantly, will the cloud environment be able to meet this demand? Another question is, will the AI engineers have the ability to decouple the model from the cloud environment in case some major change needs to be made? Likewise, is the model able to be easily trained with new data so that its effectiveness can be improved over time? There are many such questions and concerns that are important to assessing the design of a prototype.

ACTIVITY 8-2: Identifying Data Requirements for AI

1. **What are some data quality concerns that you have regarding this project?**

   A: Answers will vary. One concern is how long or how many resources it will take to actually combine all of the data. If the data is from disparate sources, then consolidating all of it into a single, consistent format may prove very challenging. Another potential concern is the recentness of the data. Could the data about the customers' demographics or purchasing activity be outdated? It might no longer align with current information if enough time has passed.

2. **What are some security risks that could affect the data being used in this project?**

   A: Answers will vary. The data could be at risk of being intercepted when in transit over the VPN, though it likely has some encryption in place. However, once the data is on the employees' laptops at a remote location, it is open to many more risks, including theft of the data if the user's home network isn't secured, or if the laptop is stolen from a public location. The security of the master dataset within the enterprise network should also be considered. Is it properly encrypted? If not, then the security of the data is at risk from the very beginning. Is it protected against availability issues? If not, then the employees won't be able to work on the data remotely.

ACTIVITY 8-3: Identifying Risks in Implementing AI

1. **What are some of the business risks involved in adopting this AI solution, from both an employee and an operational standpoint?**

   A: Answers will vary. From an employee standpoint, the HR team members may be pleased that they no longer need to spend time screening applicants, but they also may be wary of their jobs being totally replaced by automation. Their job performance may suffer as a result, or they may seek other employment. They may also need time to learn how to use or interpret the results of the job applicant solution, which can lead to lost productivity in the short term. From an operational standpoint, there is the risk that the job applicant solution will do no better than the HR personnel at filtering applicants—or it may perform worse. Even if it's faster, it may not be worth the cost of development and maintenance. Speaking of maintenance, you have to consider the risk of the solution becoming outdated as the job landscape changes, and without a clear plan for updating the solution, it could end up being more of a hindrance than a benefit in the long term.

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2. **What are some ethical concerns you would have regarding this solution?**

   A: Answers will vary. The model behind the solution may need to be trained on large volumes of personal data about job applicants. It will also likely need to take even more of that data as input when it's making its filtering decisions. If the personal data is compromised at any point in the process, it will lead to major privacy issues (and possibly legal action). The model might also be found in violation of fairness and non-discrimination principles by exhibiting bias in its filtering practices. It might filter out perfectly qualified applicants based on protected attributes or attributes not relevant to the job, like race, gender, nationality, religion, etc.

ACTIVITY 8-4: Developing an AI Strategy

1. **How might the AI job applicant screener impact each of the three aspects of the ESG framework? How might you address these impacts in your AI strategy?**

   A: Answers will vary. The AI system doesn't seem like it'd have much of a direct impact on the environment. However, the resources used to create, operate, and maintain that solution likely will. So, you might impress upon stakeholders your commitment to running the system as efficiently as possible. The social impact can be significant since the screener is directly impacting people's lives. Its fairness may be called into question, so, as part of the AI strategy, you might make a commitment to ensuring it is free of major bias. On the governance side of things, the organization's leadership must commit to implementing user privacy and data protection controls. Otherwise, job applicants' personal data could be leaked or otherwise mishandled.

2. **What are some organizational resources that you need to consider as part of your AI strategy? Why are these resources relevant to the job applicant screener?**

   A: Answers will vary. In any AI project, talent is a big resource concern. The organization must employ or contract skilled AI practitioners who know how to turn data into an effective AI model. Likewise, data itself is a major resource that the organization needs to consider. Where will it be getting data that it can use to train the job applicant screener? Are there historical datasets of job applicants that the model can be trained on? Are there enough examples, or enough features? Are the datasets labeled? These are questions that the AI strategy needs to answer. Another resource to consider has to do with stakeholders. There are potentially many stakeholders in a project like this, but of particular note are HR personnel and the department managers who perform interviews of candidates screened by the AI system. They likely have valuable input as to what constitutes an acceptable job candidate at each stage of the process.

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Glossary

agent assistance
Automation technology that helps human agents in a call center do their jobs more efficiently and effectively.

aggregation
The process of gathering and summarizing data in a summarized report.

AGI
(artificial general intelligence) A speculative type of AI in which machines have a similar level of cognitive abilities as humans. Also called *general AI* or *strong AI*.

AI
(artificial intelligence) The ability of machines to exhibit human-like intelligence, as well as the scientific discipline concerned with this idea.

algorithm
A mathematical procedure or technique that, when converted into code, enables a computer to identify patterns, make predictions, or perform tasks without explicit instructions.

API
(application programming interface) A component that connects disparate pieces of software together.

audience segmentation
The process of dividing individuals into groups based on common characteristics.

autonomous vehicle
A vehicle with self-driving capabilities.

BI
(business intelligence) Strategies and technologies used by enterprises for the analysis of business data, which transforms that data into useful information.

big data
Collections of data that are so large and complex that they require advanced tools to process and analyze them.

black box
The property by which a machine learning model's decisions are difficult to understand or explain.

C2
(command and control) A set of organizational and technical attributes and processes that use human, physical, and information resources to solve problems and accomplish missions to achieve the goals of an organization or enterprise.

causation
A relationship between two variables where a change in one variable causes a change in the other variable.

chatbot
A type of AI system that uses natural language processing (NLP) to engage in real-time, text-based chatting with a human user.

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classification
A type of supervised learning in which a model attempts to place a data example into a class or category.

cloud computing
An approach to providing computing services and other resources over a network of distributed servers and other infrastructure.

cloud security
The application of cybersecurity techniques and principles to cloud platforms, the infrastructures they are built on, and the resources they use.

clustering
A type of unsupervised learning in which a model groups data examples together that exhibit similar attributes.

COM
(Component Object Model) A standard for software components introduced by Microsoft in 1993 that provides a language-neutral way of implementing objects that can be used in environments different from the one in which they were created, even across machine boundaries, and that provides the basis for several other Microsoft technologies and frameworks.

computational statistical inference
See ***statistical inference***.

computer vision
The discipline that involves computers understanding images and video at a high level.

connected services
A service relationship between a customer and a product vendor made possible through network connections provided in the vendor's product.

constrained device
A product that contains a processor whose capabilities are limited by low power-usage requirements, low memory, low processor speed, compact size, and so forth.

correlation
A statistical association between two or more variables.

CPAP device
(continuous positive airway pressure device) A machine that facilitates a patient's healthy breathing while sleeping.

CPS
(cyber-physical system) A system in which software (cyber) technologies model and control physical production equipment and processes.

crowdsourcing
The act of outsourcing work and services to a group of people, such as an online community, who aren't internal employees of the organization.

customer churn
The tendency of customers to cancel subscriptions to a service, or otherwise cease their relationship with a business.

CX
(customer experience) The overall perception of a company based on customers' interactions with that company.

data analytics
The science of examining raw data to derive insights and conclusions that help businesses make better decisions or to verify or disprove existing models or theories.

data augmentation
The process of generating new data points from existing data.

data cleaning
The manipulation of data to remove inconsistencies and address missing data points.

data dashboard
A visualization tool that provides users with a way to view key performance indicators and status for an enterprise.

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data ethics
A collection of moral principles that govern how organizations gather, protect, and use data.

data governance
A set of policies and procedures that govern how data is collected, stored, and used.

data lake
A collection of raw data that is retained in its original format until it is needed.

data literacy
The ability to read, analyze, communicate, and reason with data.

data mining
The process of pulling actionable insight out of a set of data and putting it to good use.

data model
A description of the relationships among various types of information.

data munging
See ***data wrangling***.

data profiling
The process of examining, analyzing, and summarizing a dataset to determine the level of data quality of the dataset.

data sandbox
A scalable development platform used to enable collaboration and exploration of large datasets.

data science
The discipline that involves accumulating data, analyzing the data, extracting value from the data, and presenting the value of the data in a meaningful way.

data team
A group of individuals involved in the strategic development, implementation, and management of business analytics technologies, processes, and the overall success of business analytics at the organization.

data visualization
The practice of communicating meaningful data in a visual format, such as graphics, charts, and dashboards.

data warehouse
A collection of structured data that comes from one or more sources, optimized to support data aggregation.

data workbench
See ***data sandbox***.

data wrangling
The process of transforming raw data into a format that is better suited to analysis.

data-centric organization
An organization where management and employees not only believe that data analytics are an integral component of the overall business strategy, but also implements a framework in which data is stored independently of the applications and tools that rely on it.

DCS
(distributed control system) A device used in industrial control that can control multiple manufacturing processes and locations, perhaps performing the role of multiple PLCs and PCs.

descriptive analytics
A form of data analytics that uses data to summarize past behaviors.

diagnostic analytics
A form of data analytics that focuses on examining past data to identify the causes of specific events.

digital transformation
A business initiative in which the organization seeks major change through the use of digital technologies.

DL
(deep learning) A type of machine learning that makes complex decisions using multiple layers of information.

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dynamic pricing
The process of adjusting the price of goods or services according to customer behavior or changes in the market.

EDA
(exploratory data analysis) A form of analysis that examines datasets, summarizes their main characteristics, and provides the summary information in a visual format.

edge
Refers to the boundary where the public Internet (the cloud) meets local networks and devices owned by customers.

emotion AI
A type of sentiment analysis solution that focuses on the physical attributes of a person, particularly their facial expressions and features.

encryption
A security technique in which data is encoded so that it cannot be understood unless it is decrypted using a specific key.

energy management
The process of controlling how a device operates to conserve energy when it is not needed so it is available when it is needed—for example, putting a processor into sleep mode when there is no data for it to process.

energy trading
The buying, selling, and moving of bulk energy (electricity and natural gas) from where it is produced to where it is needed.

ESG
(environmental, social, and governance) A framework by which stakeholders can evaluate an organization's risks and opportunities regarding these three areas of concern.

ETL
(extract, transform, and load) A data-loading technique that pulls data from one or more sources, cleaning and formatting it before loading it into the sandbox.

example
In the field of machine learning, an individual instance of data in a training set.

feature
In the field of machine learning, a measurable property of an example in a training set.

feature engineering
The process of using domain knowledge, mathematics, and statistics to manipulate a dataset to facilitate the training of an ML model.

generative AI
(generative artificial intelligence) A deep learning application of AI that is used to generate content such as text, audio, images, and video, in a way that appears convincingly human like.

GPU
(graphics processing unit) A computer chip typically used to process and output visual data like graphics and images, but which also optimizes the performance of mathematical calculations used in AI.

ground truth
See ***label***.

hypothesis testing
A statistical inference process that enables you to prove (or disprove) an assumption.

IIoT
(Industrial Internet of Things) The use of IoT technologies to extend the traditional capabilities of industrial control systems. Also called Industry 4.0.

industrial robot
A robot used in warehouses and manufacturing facilities to automate dangerous or difficult processes.

IoT
(Internet of Things) A network of physical devices, vehicles, home appliances, and other items embedded with electronics, software, sensors, actuators, and connectivity that enables these things to connect and exchange data and integrate the physical world with

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computer-based systems that can use advanced techniques to process the data that is obtained, resulting in efficiency improvements, economic benefits, and reduced human exertions.

IoT client application
An application that enables end users to view and analyze data obtained through IoT devices, and control IoT devices.

IP
(intellectual property) A category of property that includes intangible creations of human intellect.

IT
(information technology) The use of computers to store, retrieve, transmit, and manipulate data or information.

ITAM
(information technology asset management) The process of identifying, inventorying, tracking, and maintaining information technology assets as they are used across the organization.

label
In the field of machine learning, the variable in a training set that you are trying to predict for new samples of data.

linear regression
A type of regression analysis in which the relationships between variables can be depicted with a straight line on a graph.

LLM
(large language model) A deep learning model that analyzes patterns in very large bodies of text in order to determine how best to predict some kind of textual output.

M2M
(machine to machine) Communication between computers, embedded devices, or other machines.

machine data
Data that is generated by industrial equipment, sensors in Internet of Things (IoT) devices, and even logs and counters that track visits to a web page.

machine learning model
The object that makes predictions and other decisions and is the result of putting data through a machine learning algorithm.

marketing measurement analysis
The process of collecting, analyzing, and interpreting data to measure the effectiveness of marketing activities.

ML
(machine learning) An AI discipline in which a machine is able to gradually improve its predictive capabilities without being given explicit instructions.

model target
A business variable that you want to predict by using data analysis.

model training
The process of feeding a small set of data (training data) into an ML algorithm, evaluating the results of the algorithm, and optimizing the algorithm for performance.

model validation
The process of feeding a set of testing data into an algorithm to validate that a model actually achieves the intended purpose.

narrow AI
(narrow artificial intelligence) A type of AI in which machines are able to solve one or a few specific problems that are assigned to them by humans. Also called *weak AI*.

NLP
(natural language processing) The discipline that involves computers analyzing and applying human languages.

NTA
(network traffic analysis) The process of capturing and analyzing network data packets to gain insights into the behavior, performance, and security of a computer network.

OLE
(Object Linking and Embedding) A proprietary technology developed by Microsoft that enables embedding and linking to documents and other objects.

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OPC
(Open Platform Communications) An industrial standard for interprocess communication initially based on Microsoft's standards for distributed computing.

open data
Data that is freely available for use and that can be shared or built on by anyone for any reason.

operationalization
The process of moving data and models out of a sandbox and into a live production environment.

OT
(operational technology) Hardware and software used to detect or cause changes in physical processes through direct monitoring and control of physical devices such as valves and pumps, typically used for industrial control.

PAC
(programmable automation controller) A type of industrial automation controller that combines attributes of a programmable logic controller and a personal computer.

PLC
(programmable logic controller) An industrial digital computer that has been ruggedized and adapted for the control of manufacturing processes, such as assembly lines, robotic devices, or any activity that requires high reliability control, ease of programming, and process fault diagnosis.

predictive analytics
A form of data analytics that uses data to forecast future behaviors.

predictive maintenance
Techniques that help determine the condition of in-service equipment in order to predict when maintenance should be performed, potentially saving costs over routine or time-based preventive maintenance because tasks are performed only when warranted.

preprocessing
A series of activities that improve the quality of data to facilitate in-depth analysis.

prescriptive analytics
A form of data analytics that uses data to predict future outcomes and then offer recommendations for actions based on those predictions.

prescriptive maintenance
An approach to predictive maintenance in which required maintenance is predicted by systems, and a course of action is prescribed. Essentially, it is preventive maintenance with built-in intelligence.

preventive maintenance
A statistical approach to periodic maintenance that requires maintenance at standard intervals, determined by the frequency with which parts have failed in the past.

principle of least privilege
A security principle that states a user should only have enough permissions to be able to do their job effectively, and no more than that.

proactive maintenance
Any approach that attempts to schedule maintenance before a problem occurs, hopefully preventing a run-to-failure (RTF) situation.

proprietary data
Data that is owned and controlled by an individual, organization, or group and that might be protected by copyright, patent, or trade-secret legislation.

recommendation system
A system that suggests items, services, and other things of interest to users.

recommender system
A form of machine learning that suggests potential items of interest based on a customer's profile and search history.

regression
A type of supervised learning in which a model attempts to predict some numerical value.

regression analysis
A statistical inference method that estimates the relationship between a dependent variable and one or more independent variables.

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reinforcement learning
A type of machine learning in which a software agent acts in an environment in order to obtain a reward.

risk management
The cyclical process of identifying, assessing, analyzing, and responding to risks.

robotics
The discipline that involves studying, designing, and operating robots.

root-cause analysis
See ***diagnostic analytics***.

route optimization
The process of finding the shortest path between two points.

RPA
(robotic process automation) A method by which software agents, also called bots, automate the performance of one or more tasks.

RTF
(run to failure) Waiting until a component fails before replacing it.

RTU
(remote terminal unit) A microprocessor-controlled electronic device used in industrial control systems to monitor and control a remote asset.

sampling
The process of studying a population through the gathering and analysis of information about the population.

SCADA
(supervisory control and data acquisition) A control-system architecture typically used in industrial, energy, and utility companies to support machine-to-machine communication.

SDLC
(system development lifecycle) A system development process that divides system development work into distinct phases to improve design, product management, and project management.

semi-structured data
Data that does not reside in a relational database but includes organization characteristics that support some sort of analysis.

sentiment analysis
The technique of determining the attitudes and feelings someone has toward something.

smart city
A municipal area that uses various types of electronic data-collection sensors and actuators to supply information used to manage assets and resources efficiently, and to control remote resources.

smart grid
An electrical grid that includes a variety of operational and energy measures including smart meters, smart appliances, renewable energy resources, and energy-efficient resources.

smart pig
An embedded device that can be sent into a pipeline to perform maintenance tasks such as cleaning or detecting leaks.

social data
Data that is garnered from social media (such as Twitter, Facebook, LinkedIn, and YouTube) or other websites.

SRM
(supplier relationship management) A process in which a business evaluates and attempts to strengthen the relationships it has with its suppliers, as well as better align those relationships with the business's overall goals and strategy.

stack
An implementation of a computer-networking protocol suite or protocol family.

statistical inference
The process of using data analysis to predict the properties of a set of data. Sometimes referred to as computational statistical inference.

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structured data
Data that contains elements that are easily accessible for effective analysis.

supervised learning
A type of machine learning in which known label values are provided as input so that a model can predict these values in future datasets.

synthetic data
A type of computer-generated data that mimics the characteristics of real-world data.

TA analytics
(talent acquisition analytics) The process of analyzing recruitment data and candidate data to optimize the recruiting and hiring process.

technostress
A negative psychological condition brought on by a person's inability to cope with the introduction of new technologies.

telematics
The use of telecommunications technology to monitor the location, movements, status, and behavior of a vehicle or fleet of vehicles.

telemetry
An automated communications process by which measurements and other data are collected at remote or inaccessible points and transmitted to receiving equipment for monitoring.

thing
In the context of IoT, a device that is network connected, includes an embedded processor, and may include sensors and actuators that interact with the physical world around the device.

threat analysis
The study of people, events, and technologies that can harm an organization's assets.

training
In the field of machine learning, the process by which a model learns from input data.

transactional data
Data that results from buying and selling products and services.

UEBA
(user and entity behavior analysis) A cybersecurity tool that uses algorithms and ML to detect anomalies in the behavior of the users in a corporate network, as well as the routers, servers, and endpoints in that network.

unstructured data
Data that is not organized in any established manner and does not conform to any data model.

unsupervised learning
A type of machine learning in which label values are not provided as input, so the model does not have an explicit variable that it is predicting.

upskilling
A form of training that gives employees the skills and knowledge to help them perform their jobs more effectively.

virtual assistant
A software-based agent designed to assist users with performing simple tasks.

virtualization
The process of creating an abstract representation of a computing environment.

workbench
See ***data sandbox***.

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